

# Strategic Managers with Noisy Beliefs: Evidence from Carbon Pricing\*

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## Abstract

We study how strategic interactions and noisy beliefs determine managers' expectations about the impacts of a given macroeconomic shock in a general equilibrium setting. We propose a novel identification strategy based on a survey instrument designed to recover primitives governing firm managers' strategic interactions and beliefs. We apply our approach to the context of carbon pricing reform in Switzerland and find that a CHF 100 per ton carbon tax increase is expected to reduce output by 1.18% over a short- to medium-run horizon; that firms hold overly pessimistic beliefs about the economy-wide impact of carbon pricing; and that quantity setting tends to exhibit strategic substitutabilities, dampening the impact of noisy beliefs. Information counterfactuals reveal a counter-intuitive implication for policy communication: because responses are strategic substitutes, publicly revealing the true (and considerably less pessimistic) aggregate impact of carbon pricing leaves firms more optimistic about the economy as a whole but more pessimistic about their own output, deepening the expected economy-wide output decline.

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# 1 Introduction

This paper examines how strategic firm interactions and managers’ noisy beliefs determine expectations about the impacts of a macroeconomic shock or a policy reform. The central premise is simple. A manager contemplating the impacts of a macroeconomic shock must form expectations not only about the direct impact on their own firm, but also about the general-equilibrium response of competitors and demand conditions. These general-equilibrium effects can be summarized as a strategic game in which firms’ optimal choices depend, in part, on beliefs about others’ choices. As a consequence, firms’ responses to a shock are jointly determined by (i) managers’ beliefs about their respective firm’s direct shock elasticity, (ii) beliefs about other firms’ elasticity, and (iii) the intensity and sign of strategic interactions. Beliefs about the economy-wide elasticity can influence firms’ beliefs about their own elasticity but also about other firms’ elasticity. In turn, strategic interactions shape how managers’ pessimistic or optimistic beliefs about the economy translate into firm-level expectations. Strategic interactions amplify or dampen noisy beliefs depending on whether equilibrium outcomes are strategic complements or substitutes, respectively.

Empirically assessing these mechanisms based solely on observational data poses several challenges. It typically requires strong identifying variation, rich firm-level data (often released with substantial lags), and structural assumptions about production technologies and market structure. As a result, it is often difficult, if not impossible, to provide timely evidence to policymakers about the expected impacts of an upcoming change in economic conditions or a policy reform. To overcome these limitations, we propose a survey-based approach to elicit managers’ expectations directly and recover firm-specific primitives that govern strategic interactions and noisy beliefs in a general-equilibrium setting. Our approach has three steps. We first build a general-equilibrium model that delivers a linear quantity-setting equation capturing the nature of strategic interactions in firms’ response to a specific policy reform. Noisy beliefs, which in our setting are the difference between what firm managers believe about the response of others and their actual response, also enter the equation. The model’s strategic interaction parameters are sufficient statistics that require minimal assumptions about the economy’s production structure. In a second step, we design a survey instrument specifically tailored to identify both the strategic interaction parameters and noisy beliefs. In a third and final step, we enrich the model with an information structure for which we can estimate the primitives. This, in turn, allows us to perform counterfactual policy simulations.

We apply our approach in the context of carbon pricing reform in Switzerland, a timely and highly debated topic. Switzerland was among the first countries to adopt a broad-based carbon pricing scheme, doing so in 2008. In recent years, several referendums have addressed alternative measures to meet Switzerland’s climate goals. Notably, a 2021 initiative to

expand the coverage and level of the carbon tax was narrowly defeated at the polls. Among analysts, there is now renewed interest in identifying ways to make such proposals more politically acceptable. A better understanding of firm managers' beliefs about carbon pricing policy could play an important role in informing this debate.

Our identification strategy to recover firm-specific primitives is built around three scenarios that vary the scope of a carbon tax reform. Namely, managers need to form expectations about the impacts of an economy-wide, sector-wide, and firm-specific carbon pricing scheme over a medium-run horizon—three years in our setting. In addition, we need to elicit managers' own planned production impact, and, for some but not all scenarios, expectations of sectoral and aggregate production impacts (with natural restrictions for the sectoral and firm-specific scenarios). Our design allows us to identify: (i) the relevant tax semi-elasticity (beliefs about the quantity impact of carbon pricing); (ii) strategic interaction parameters capturing how a firm's own quantity choices respond to expected sectoral and aggregate changes; and (iii) noisy beliefs about the policy's sectoral and aggregate production impacts.

Our main findings can be summarized as follows. First, the survey-based identification strategy is feasible in practice: managers respond meaningfully to the scenarios, allowing us to recover firm-specific primitives. Second, the subjective output costs of carbon taxation reported by managers are broadly in line with estimates existing estimates from observational studies: in our baseline scenario, a CHF 100/ton increase in the carbon tax is associated with an expected decline in aggregate output of around 1.18% after three years. Third, we find evidence of overly pessimistic beliefs about how the rest of the economy responds to the tax, compared to the average firm's beliefs about their own response. This evidence suggests that firms receive excessively negative signals about the aggregate cost elasticity. Last, but not least, we find strong strategic substitutabilities in quantity setting, especially at the sectoral level: managers lower their optimal production response when they expect stronger activity by other firms in their sector, consistent with competitive interactions within sectors. In turn, strategic substitutabilities dampen the impact of their pessimistic beliefs about carbon pricing on their own outcomes.

Motivated by this evidence, we enrich the model with an information structure where firms observe some private signal of their own elasticity, and public signals about the aggregate and sectoral elasticity that can be driven respectively by aggregate and sectoral noise. We show that firms' overly pessimistic beliefs about the aggregate effect of the carbon tax come from the public signal on the aggregate elasticity that is too pessimistic. This pessimistic public signal leads firms to be also excessively pessimistic about their own elasticity.

We then use our data to identify the parameters that characterize firms' prior beliefs. Based on these estimates, we study a public communication policy in which the government reveals the true aggregate impact of carbon pricing, which our estimates show to be markedly

less severe than the general public “sentiment” on the costs of decarbonizing. Counterintuitively, this good news deepens the expected fall in firms’ own output, from 1.24% to 1.37% on average. The reason lies in the predominance of higher-order beliefs (what firms think about other firms’ costs) over first-order beliefs (what firms think about their own costs). First, because we estimate that firms have precise private information on their own cost, their assessment of this cost does not budge. Second, because of strategic substitutabilities, learning that the economy at large will be less affected leads each firm to become more pessimistic about their own output. Correcting overly pessimistic beliefs about carbon pricing thus need not raise firms’ support for it, and may even lower it.

Our paper contributes to a growing literature that uses survey data to recover firms’ structural primitives that are otherwise difficult to identify from observational data. For instance, [Keiller et al. \(2024\)](#) demonstrate how carefully designed survey instruments can be used to estimate production-function parameters and adjustment frictions at the firm level. [Coibion et al. \(2021\)](#) show how firms’ expectations can be exploited to study strategic interactions in price setting, highlighting the role of higher-order beliefs in firms’ decisions. Closest to our approach, [Afrouzi \(2024\)](#) shows that survey scenarios can identify strategic complementarities in price setting by eliciting higher-order beliefs. We build on and extend this method by designing survey scenarios that vary the scope of a policy shock to jointly identify firm-specific cost elasticities and strategic interaction parameters.

Our methodology also provides results that relate to a large literature on strategic interactions in quantity setting. Standard models of Cournot competition imply strategic substitutability ([Bulow et al., 1985](#); [Vives, 1999](#)), while macroeconomic models featuring aggregate demand externalities or network production can generate complementarities ([Cooper and John, 1988](#); [Benhabib and Farmer, 1994](#); [Farmer, 1999](#)). These complementarities typically amplify the impact of noisy information ([Benhabib et al., 2015](#); [Angeletos and La’O, 2013](#)).<sup>1</sup> Existing work assumes the sign of strategic interactions. We provide direct survey-based firm-level evidence that quantity setting is characterized by strategic substitutability, especially at the sectoral and, to a lesser extent, at the aggregate level. We are the first to document this “zero-sum thinking” at the firm level. As a consequence, pessimistic beliefs about aggregate responses to carbon pricing dampen—rather than amplify—firms’ own pessimistic responses. This mechanism highlights a previously undocumented channel through which information frictions interact with competitive forces to shape firms’ expectations about policy reforms.

We also contribute to an emerging literature that shows that support for carbon pricing

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<sup>1</sup>Strategic interaction parameters also shape the use of information, with complementarities leading to an under-use of private signals and substitutabilities leading to an under-use of public information ([Morris and Shin, 2002](#); [Angeletos and Pavan, 2007](#)).

is shaped by (mis-)perceptions about its costs, distributional incidence, and macroeconomic effects, and that correcting these beliefs can substantially increase policy support (Carattini et al., 2018; Beiser-McGrath and Bernauer, 2019; Kallbekken and Sælen, 2011; Andre et al., 2024). While this work focuses primarily on households, much less is known about firms’ beliefs about carbon pricing and how these beliefs interact with strategic considerations. Our paper contributes by measuring firm-level expectations about carbon pricing and showing how aggregate belief distortions shape firms’ expected real responses.

Finally, our paper contributes to a growing literature that uses surveys of firm managers to study the economic and political economy implications of environmental policies. Martin et al. (2014) show that firms’ perceived relocation risks under carbon regulation shape their responses and policy positions. Hensel et al. (2024) demonstrate that carbon pricing affects firms’ and households’ inflation expectations, pointing to a potentially important macroeconomic expectations channel. More broadly, OECD (2016) has emphasized the value of firm surveys as a tool to study environmental policies when administrative or balance-sheet data are unavailable or released with long lags. Our contribution is to use survey-based scenarios to go beyond eliciting firm managers’ opinions and expectations about future environmental policies. Our approach allows us to estimate structural parameters that permit us to conduct counterfactual policy simulations about the effects of carbon pricing along multiple margins and at different levels of aggregation. This approach allows us not only to document firms’ perceived costs of carbon pricing but also to quantify how aggregate beliefs and strategic interactions shape firms’ intended responses, providing a bridge between micro-level survey evidence and general-equilibrium analyses of environmental policy.

## 2 The Model

We model the impact of carbon pricing in a one-period general equilibrium setting with information frictions. We introduce carbon policies as a carbon tax. Firms know the tax, but they must decide how much to produce before they know how the tax impacts their costs. Firms can reduce their carbon emissions through a carbon abatement technology or by reducing production (equivalently, labor). We show that the equilibrium decisions of the firms boil down to a game between firms with strategic interactions where beliefs about the production impacts on other firms matter.

### 2.1 Representative household

There is a representative household with utility

$$U = \frac{C^{1-\gamma}}{1-\gamma} - \phi^{-1} \frac{L^{1+\eta}}{1+\eta} \quad (1)$$

where  $C$  is a consumption bundle that depends on the consumption of imperfectly substitutable goods from different sectors:  $C = \left( \sum_n C_n^{\frac{1-\epsilon^c}{\epsilon^c}} \right)^{\frac{\epsilon^c}{1-\epsilon^c}}$ , where each sector-specific consumption  $C_n$  is itself a bundle of different consumption varieties  $C_{in}$  within the sector  $C_n = \left( \sum_i C_{in}^{\frac{1-\rho^c}{\rho^c}} \right)^{\frac{\rho^c}{1-\rho^c}}$ .  $L$  is a function of the labor supplied to the different sectors:  $L = \left( \sum_n L_n^{1+\chi} \right)^{\frac{1}{1+\chi}}$ , where each sector-specific labor supply  $L_n$  is a function of the labor supplied to the firms producing individual varieties:  $L_n = \left( \sum_i L_{in}^{1+\psi} \right)^{\frac{1}{1+\psi}}$ . The budget constraint faced by the household is

$$\sum_n \sum_i P_{in} C_{in} = \sum_n \sum_i W_{in} L_{in} \quad (2)$$

where  $P_{in}$  is the price of variety  $i$  in sector  $n$ ,  $W_{in}$  is the wage paid by the firm producing good  $i$  in sector  $n$ .

The household's optimal choices imply

$$\phi^{-1} C^\gamma L^\eta = \frac{W}{P} \quad (3)$$

$$C_n = C \left( \frac{P_n}{P} \right)^{-\epsilon^c} \quad (4)$$

$$C_{in} = C_n \left( \frac{P_{in}}{P_n} \right)^{-\rho^c} \quad (5)$$

$$L_n = L \left( \frac{W_n}{W} \right)^{1/\chi} \quad (6)$$

$$L_{in} = L_n \left( \frac{W_{in}}{W_n} \right)^{1/\psi} \quad (7)$$

$$(8)$$

where  $P$ ,  $P_n$ ,  $W$  and  $W_n$  are price aggregators defined as  $P = \left( \sum_n P_n^{1-\epsilon^c} \right)^{\frac{1}{1-\epsilon^c}}$ ,  $P_n = \left( \sum_i P_{in}^{1-\rho^c} \right)^{\frac{1}{1-\rho^c}}$ ,  $W = \left( \sum_n W_n^{1+1/\chi} \right)^{\frac{1}{1+1/\chi}}$ , and  $W_n = \left( \sum_i W_{in}^{1+1/\psi} \right)^{\frac{1}{1+1/\psi}}$ .

## 2.2 Firms

The economy comprises  $N$  sectors, indexed by  $n$ , and each sector  $n$  contains  $N^n$  firms, indexed by  $i$ . Consider firm  $i$  in sector  $n$ . Its production function is

$$Q_{in} = [(1 + \beta_{in}) L_{in}]^{\frac{1}{1+\beta_{in}}} \quad (9)$$

where  $\beta_{in} > 0$  is a firm-specific return-to-scale parameter.

We assume for simplicity that carbon emissions are proportional to output, so that

$$E_{in} = \phi_{in} Q_{in} \quad (10)$$

where  $\phi_{in}$  is the firm-specific carbon-intensity of production. One can think of this carbon-intensity as arising for instance from the firm-specific energy mix that combines dirty and green energy sources. This parameter is endogenous, as firms can lower their carbon intensity if they pay the abatement cost

$$\frac{\kappa_{in}}{2} \left(1 - \frac{\phi_{in}}{\bar{\phi}_{in}}\right)^2 Q_{in}.$$

$\bar{\phi}_{in}$  is the business-as-usual carbon intensity, that is, the carbon-intensity that would hold in the absence of abatement effort.  $\kappa_{in}$  is a parameter that reflects how costly it is for that firm to decarbonize its production. The abatement cost can be thought of for instance as the cost of switching from a brown energy source to a green energy source, which requires some investment from the firm. The cost is quadratic in the carbon intensity and is proportional to total production. Firms are imperfectly informed about both  $\bar{\phi}_{in}$  and  $\kappa_{in}$ , which means that they ignore both their business-as-usual carbon intensity and how much it would cost them to decrease it. The firm faces a tax  $tax_{in}$  per unit of carbon emission.

**Timing and information.** At the beginning of the period, the firm decides how much labor to hire (equivalently, how much to produce). At that stage, the firm knows its own tax  $tax_{in}$  and the distribution of the tax within its own sector and in the firms' population, as well as the distribution of the preference and technology parameters. However, the firm does not observe the business-as-usual carbon-intensity  $\bar{\phi}_{in}$  and the cost parameter  $\kappa_{in}$ , nor their distribution within their sector or in the firms' population. Prices and abatement efforts are set at the end of period, after the firm learns about  $\bar{\phi}_{in}$  and  $\kappa_{in}$  and their distribution.

**Firm's decisions.** The firm behaves monopolistically within its sector. Its expected profits are

$$E_{in}\Pi_{in} = E_{in} \left\{ P_{in}Q_{in} - W_{in}L_{in} - tax_{in}E_{in} - \frac{\kappa_{in}}{2} \left(1 - \frac{\phi_{in}}{\bar{\phi}_{in}}\right)^2 Q_{in} \right\} \quad (11)$$

where  $E_{in}(\cdot)$  is the firm expectation conditional on its beginning-of-period information.

For a given tax  $tax_{in}$ , the firm maximizes its profits (11) with respect to production  $Q_{in}$  and abatement effort  $\phi_{in}$ , subject to the demand for its product (5), its production technology (9) and its emission technology (10).

According to our timing assumption, firms know the parameters of the abatement technology when they decide their abatement effort  $\phi_{in}$ . As a result, the marginal abatement cost must be equal to the carbon tax:

$$tax_{in} = \frac{\kappa_{in}}{\bar{\phi}_{in}} \left(1 - \frac{\phi_{in}}{\bar{\phi}_{in}}\right). \quad (12)$$

When deciding their production  $Q_{in}$ , the firms do not know the parameters of the abatement technology yet. Taking into account the abatement efforts of the firm, the optimal level of production is then set so that the markup is constant in expectations:

$$E_{in}(P_{in}) = E_{in} \left\{ \frac{\rho^c}{\rho^c - 1} \left( W_{in} Q_{in}^{\beta_{in}} + tax_{in} \bar{\phi}_{in} \left( 1 - \frac{tax_{in} \bar{\phi}_{in}}{2\kappa_{in}} \right) \right) \right\} \quad (13)$$

## 2.3 Equilibrium

The consumption of variety  $i$  in sector  $n$  should be equal to the production of variety  $i$  in sector  $n$ :

$$C_{in} = Q_{in} \quad (14)$$

## 2.4 Strategic interactions in quantity setting

We have assumed that the firm has to make quantity decisions before the uncertainty on decarbonization parameters is resolved, while prices fully adjust after. This assumption will give us strategic interactions in quantity setting.

We log-linearize the model around the firm-specific baseline tax level  $tax_{in} = \overline{tax}_{in}$ , that we assume to be close to zero. We denote the log-deviations from the baseline by lower-case letters and denote by  $\tau_{in} = tax_{in} - \overline{tax}_{in}$  the deviation of the tax from its baseline level, in euros say.

The firm must then set its production so that its markup does not deviate from its optimal level in expectations, and the price follows (13). The firm behavior can be approximated as follows:

$$E_{in} \{ p_{in} - (w_{in} + \beta_{in} q_{in}) - \tilde{\sigma}_{in} \tau_{in} \} = 0 \quad (15)$$

where  $\tilde{\sigma}_{in} = \frac{\bar{\phi}_{in}}{\bar{W}_{in} \bar{Q}_{in}^{\beta_{in}}}$  is the semi-elasticity of the marginal cost to  $\tau_{in}$ , with  $\bar{W}_{in}$ ,  $\bar{Q}_{in}$  the baseline wage and output. Output increases the marginal cost through a decreasing-return-to-scale channel, which depends on the curvature of the production function parameter  $\beta_{in}$ . The carbon tax  $\tau_{in}$  increases the marginal cost because it works as a tax on production.

Given our timing assumptions, the firm makes production decisions before observing market prices and before observing abatement cost parameters. But, to make a decision on output, the firm must infer the potential drivers of the inverse of the product wage  $p_{in} - w_{in}$ , and the price tax-elasticity  $\tilde{\sigma}_{in}$ , as implied by Equation (16):

$$q_{in} = \frac{1}{\beta_{in}} \{ E_{in}(p_{in} - w_{in}) - E_{in}(\tilde{\sigma}_{in}) \tau_{in} \}. \quad (16)$$

The product wage can be decomposed as follows:

$$p_{in} - w_{in} = (p_{in} - p_n) + (p_n - p) - (w_{in} - w_n) - (w_n - w) - (w - p). \quad (17)$$

It depends on the variety's relative price, within its sector, and on the relative price of the sector within the aggregate economy. These will depend respectively on the supply of the variety relative to the sectoral supply, and on the supply of the sector relative to the aggregate supply. The product wage also depends on the wage, which has an individual, sectoral and aggregate component. These will depend respectively on the individual, sectoral and aggregate input demand.

We now derive the key equation that will guide our empirical investigation. Using the optimal output (16) and the product wage expression (17), and combining with the consumption demand and labor supply equations (3)-(7), the equilibrium relations (14), and the production function (9), we obtain the following log-linear approximation of the optimal individual output:

$$q_{in} = \alpha_{in}^a E_{in}(q) + \alpha_{in}^s E_{in}(q_n) - E_{in}(\sigma_{in})\tau_{in} + \omega \quad (18)$$

with  $\sigma_{in} = \tilde{\sigma}_{in}/[\psi(1+\beta_{in})+(\rho^c)^{-1}+\beta_{in}]$  and where  $\alpha_{in}^s$  and  $\alpha_{in}^a$  are the strategic complementarity parameters at the sectoral level  $\alpha_{in}^s = \frac{(\psi-\chi)(1+\beta_n)+(\rho^c)^{-1}-(\epsilon^c)^{-1}}{(\psi(1+\beta_n)+\beta_{in})+(\rho^c)^{-1}}$  and at the aggregate level  $\alpha_{in}^a = \frac{(\chi-\eta)(1+\beta)-\gamma+(\epsilon^c)^{-1}}{(\psi(1+\beta_{in})+\beta_{in})+(\rho^c)^{-1}}$ , where  $\beta_n$  and  $\beta$  are respectively the sectoral and economy-wide averages of  $\beta_{in}$ . The constant term  $\omega = [(\psi-\chi)Cov(\beta_{in}-\beta_n, q_{in}-q_n) + \chi Cov(\beta_n-\beta, q_n-q)]$  is negligible under some mild assumptions, as we will show later.

The crucial point about Equation (18) is that it summarizes the general equilibrium effects with two strategic interaction parameters:  $\alpha^a$  describes strategic interactions between sectors, whereas  $\alpha^s$  describes strategic interactions within a sector. These parameters are sufficient statistics that depend on the degree of competition in the product markets (governed by the elasticities of substitution  $\epsilon^c$  and  $\rho^c$ ) and in the input markets (governed by  $\gamma$ ,  $\eta$ ,  $\psi$  and  $\chi$ ). Moreover, the strength of the decreasing marginal returns (governed by  $\beta$ ) also matters. Since the elasticity of substitution between sectors  $\epsilon^c$  tends to be small ( $\epsilon^c < 1$ ),  $\alpha^a$  is typically positive, which means strategic complementarities between sectors are more likely. On the other hand, since the elasticity of substitution within sectors, i.e., across varieties, tends to be large ( $\rho^c > 1$ ),  $\alpha^s$  is typically negative, reflecting strategic substitutabilities between competing firms in the same sector.

## 2.5 Noisy beliefs and strategic interactions

We define “noisy beliefs” as a situation where firms have erroneous expectation about their own elasticities on average, that is,  $\bar{E}(\sigma_{in}) \neq \sigma^a$ , where  $\bar{E}(\sigma_{in})$  is the economy-wide average of  $E_{in}(\sigma_{in})$ . As we will see, this implies that firms are also, on average, mistaken about aggregate and sectoral output:  $\bar{E}(q) \neq q$  and  $\bar{E}(q_n) \neq q$ , that is, the average responses about the aggregate and sectoral expected output differs from the actual aggregate output, computed as the aggregate average of the firms' responses about their own production. We

show that this has consequences on the expected impact of the carbon tax, and that these consequences are mitigated or amplified by strategic interactions.

We can show how strategic interactions and noisy beliefs interact by simply rewriting Equation (18) as

$$q_{in} = \alpha_{in}^a [q + (E_{in}(q) - q)] + \alpha_{in}^s [q_n + (E_{in}(q_n) - q_n)] - [\sigma_{in} + (E_{in}(\sigma_{in}) - \sigma_{in})] \tau_{in}. \quad (19)$$

This is the second equation that guides our empirical investigation. By quantifying noisy beliefs at the aggregate and sector level, we can show how they impact firm managers' beliefs about the impact of carbon pricing on their own output.

Moreover, we can also use Equation (19) to show how an aggregate error on the elasticity  $\bar{E}(\sigma_{in}) - \sigma^a$ , but also on aggregate production  $\bar{E}(q) - q$ , or on sectoral production  $\bar{E}(q_n) - q_n$ , affects the economy-wide response to a carbon tax. This can readily be seen by taking the economy-wide averages where we obtain:

$$q = \frac{1}{1 - \alpha^a - \alpha^s} [\alpha^a (\bar{E}(q) - q) + \alpha^s (\bar{E}(q_n) - q_n) - (\bar{E}(\sigma_{in}) - \sigma^a) \tau - \sigma^a \tau] \quad (20)$$

where  $\tau$  is the average tax. Note that the impact of the different types of errors still depends on the strategic interaction parameters  $\alpha^a$  and  $\alpha^s$ .

Equation (20) can serve as a basis for information counterfactuals. If we were to change the nature of the signals impacting managers' beliefs, this would affect the different types of aggregate errors, and, in turn, the aggregate impact of a carbon tax. However, to determine how we can change managers' beliefs, we need to make further assumptions on the information structure, as we show in Section 5.1. Before doing so, we propose a model-based identification strategy of the key objects in Equation (19) with minimal assumptions about the information structure.

## 2.6 Identification Strategy

Here we discuss how to identify the firms' beliefs about the elasticities  $E_{in}(\sigma_{in})$  and the degree of strategic complementarities  $\alpha_{in}^a$  and  $\alpha_{in}^s$ , as well as the higher-order errors  $E_{in}(q_n) - q_n$  and  $E_{in}(q) - q$ .

**Equilibrium solution.** To do so, we must first characterize the equilibrium solution. We need to make some assumptions on the information structure. So far, we have simply assumed that the carbon taxes  $\tau_{in}$  and strategic interaction parameters  $\alpha_{in}^a$  and  $\alpha_{in}^s$  are known to all firms, but firms are uncertain about the tax-related elasticities  $\sigma_{in}$ . To characterize the equilibrium solution, we now formalize the information structure about  $\sigma_{in}$ .

**Assumption 1** (General information structure). *Let  $\sigma_{in}^f$ ,  $\sigma_n^s$  and  $\sigma^a$  be some firm-level, sectoral and aggregate stochastic components of  $\sigma_{in}$  such that:*

$$\sigma_{in} = \sigma^a + \sigma_n^s + \sigma_{in}^f. \quad (21)$$

*We assume that firms have some priors on  $\Sigma_{in} = (\sigma^a, \sigma_n^s, \sigma_{in}^f)$ , such that, conditional on this prior,  $\Sigma_{in} \sim \mathcal{N}(\Sigma_{in}^0, P^0)$ , and that firms receive a vector of  $m$  signals  $s_{in} = h\Sigma_{in} + v_{in}$ , where  $h$  is a  $m \times 3$  matrix and  $v_{in}$  is a vector of  $m$  noise shocks:  $v_{in} \sim \mathcal{N}(0, R)$ , with  $R$  a  $m^2$  positive diagonal matrix.*

The firm's expectations are then conditional on their priors and signals:  $E_{in}(\cdot) = E(\cdot | \Sigma_{in}^0, s_{in})$ . Some technical assumptions are also necessary to fully characterize the solution. First, we assume that the number of firms in each sector and the number of sectors are large. Because firms are of comparable size and the shocks they face are Gaussian, a law of large numbers then ensures that idiosyncratic firm-level shocks have no impact on sectoral or aggregate quantities and that sector-specific shocks have no impact on aggregate quantities. Second, we make an independence assumption, namely, that firm-specific shocks and taxes are independent from firm-specific parameters and that sector-specific shocks and taxes are independent from sector-specific parameters. Third, we assume that parameter heterogeneity is limited. Together, these assumptions ensure that any linear solution at the firm-level involving firm-level parameters can be linearly aggregated at the sector and aggregate level using sectoral and aggregate parameters. We also make some invertibility assumption that ensures the existence of a unique linear solution (see Assumptions 3, 4, 5 and 6 in [Appendix A.1](#)).

**Proposition 1.** *Under Assumptions 1, 3, 4, 5 and 6 (see [Appendix A.1](#) for the latter four), we can show that  $\omega = 0$ , and that the equilibrium solution for  $(q_{in}, E_{in}(q_n), E_{in}(q))$  is linear in  $(\tau, \tau_n - \tau, \tau_{in} - \tau_n)$ :*

$$\begin{pmatrix} q_{in} \\ E_{in}(q_n) \\ E_{in}(q) \end{pmatrix} = \begin{pmatrix} \mathcal{Q}^{f(1)}(\Theta_{in}, S_{in}) & \mathcal{Q}^{f(2)}(\Theta_{in}, S_{in}) & \mathcal{Q}^{f(3)}(\Theta_{in}, S_{in}) \\ \mathcal{Q}^{s(1)}(\Theta_{in}, S_{in}) & \mathcal{Q}^{s(2)}(\Theta_{in}, S_{in}) & 0 \\ \mathcal{Q}^{a(1)}(\Theta_{in}, S_{in}) & 0 & 0 \end{pmatrix} \begin{pmatrix} \tau \\ \tau_n - \tau \\ \tau_{in} - \tau_n \end{pmatrix} \quad (22)$$

where  $\tau$  and  $\tau_n$  are the economy-wide and sectoral averages of  $\tau_{in}$  and  $\Theta_{in} = \{\theta_{in}^\alpha, \theta_{in}^{info}\}$ , with  $\theta_{in}^\alpha = \{\alpha_{in}, \alpha_n, \alpha\}$  the set of strategic interaction parameters,  $\theta_{in}^{info} = \{h, P^0, R\}$  the set of information parameters, and  $S_{in} = (\Sigma_{in}^0, s_{in}')'$  the vector of priors and signals. More specifically, the solution is linear in  $S_{in}$ , as  $\mathcal{Q}^{k(j)}(\Theta_{in}, S_{in}) = A^{k(j)}(\Theta_{in})S_{in}$ , for  $(j, k) = (1, f), (2, f), (3, f), (1, s), (2, s), (1, a)$ .

See proof in [Appendix A.1](#). This proposition has two important implications that will be key for our identification strategy. First, conditional on the parameters and priors, the equilibrium output and expectations are linear function of the aggregate, sector-specific, and

firm-specific taxes. Second, the expected sectoral output is not affected by the firm-specific tax, and the expected aggregate output is not affected by the firm-specific and sector-specific taxes.

**Strategic Interactions and firm-level elasticities.** Suppose that we want to identify the firms' beliefs about the elasticities  $E_{in}(\sigma_{in})$  and the degree of strategic complementarities  $\alpha_{in}^a$  and  $\alpha_{in}^s$ , or in other words, the vector  $(\alpha_{in}^a, \alpha_{in}^s, E_{in}(\sigma_{in}))'$ . Equation (18) suggests that this is possible if we generate variations in policies, firm-level quantities, and firm-level expectations about aggregate and sectoral quantities, namely in the vector  $(q_{in}, E_{in}(q), E_{in}(q_n), \tau_{in})'$ . On the other hand, Proposition 1 suggests that, conditional on the parameters, priors, and signals, variations in  $(q_{in}, E_{in}(q), E_{in}(q_n))'$  are driven by variations in  $(\tau, \tau_n - \tau, \tau_{in} - \tau_n)'$ . We thus can produce observations of  $(q_{in}, E_{in}(q), E_{in}(q_n), \tau_{in})'$  through survey questions where firms are provided with different scenarios on  $(\tau, \tau_n - \tau, \tau_{in} - \tau_n)'$ .

Because there are three parameters to identify, it is enough to generate three independent observations of the vector  $(q_{in}, E_{in}(q), E_{in}(q_n), \tau_{in})'$ . In that case, we can write

$$\begin{pmatrix} q_{in,1} \\ q_{in,2} \\ q_{in,3} \end{pmatrix} = \underbrace{\begin{pmatrix} E_{in}(q^{sc.1}) & E_{in}(q_n^{sc.1}) & \tau_{in}^{sc.1} \\ E_{in}(q^{sc.2}) & E_{in}(q_n^{sc.2}) & \tau_{in}^{sc.2} \\ E_{in}(q^{sc.3}) & E_{in}(q_n^{sc.3}) & \tau_{in}^{sc.3} \end{pmatrix}}_{P_{in}} \cdot \begin{pmatrix} \alpha_{in}^a \\ \alpha_{in}^s \\ E_{in}(\sigma_{in}) \end{pmatrix}$$

where the 1, 2 and 3 subscripts correspond to three distinct scenarios. Provided that  $P_{in}$  is invertible, we can recover  $(\alpha_{in}^a, \alpha_{in}^s, E_{in}(\sigma_{in}))'$  as follows:

$$\begin{pmatrix} \alpha_{in}^a \\ \alpha_{in}^s \\ E_{in}(\sigma_{in}) \end{pmatrix} = P_{in}^{-1} \begin{pmatrix} q_{in,1} \\ q_{in,2} \\ q_{in,3} \end{pmatrix}$$

Therefore, the parameters are identified if and only if  $P_{in}$  is invertible.

We can ensure that  $P_{in}$  is invertible by presenting firms with the following three scenarios, that vary in *scope*: (1) an increase  $\epsilon$  in the economy-wide tax:  $\tau = \epsilon$  and  $\tau_n - \tau = \tau_{in} - \tau_n = 0$ ; (2) an increase  $\epsilon$  in the tax in the sector where the firm operates (but not in the rest of the economy):  $\tau_n - \tau = \epsilon$  and  $\tau = \tau_{in} - \tau_n = 0$ ; (3) an increase  $\epsilon$  in the tax that applies to the firm (but not to the rest of the economy, including the sector where the firm operates):  $\tau_{in} - \tau_n = \epsilon$  and  $\tau = \tau_n - \tau = 0$ . In each of these scenarios, we ask firms about their own production, the production in their sector, and aggregate production.

Because, as Proposition 1 shows, a firm-specific tax has no impact on the expectations on the aggregate and sectoral output, and a sector-specific tax has no impact on the expectation on aggregate output, with these scenarios, there are less responses to collect (only those that

are non-zero) and  $P_{in}$  is upper-triangular, with non-zero diagonal elements:

$$\begin{pmatrix} q_{in}^{sc.1} \\ q_{in}^{sc.2} \\ q_{in}^{sc.3} \end{pmatrix} = \underbrace{\begin{pmatrix} E_{in}(q^{sc.1}) & E_{in}(q_n^{sc.1}) & \epsilon \\ 0 & E_{in}(q_n^{sc.2}) & \epsilon \\ 0 & 0 & \epsilon \end{pmatrix}}_{P_{in}} \cdot \begin{pmatrix} \alpha_{in}^a \\ \alpha_{in}^s \\ E_{in}(\sigma_{in}) \end{pmatrix} \quad (23)$$

which ensures invertibility (see [Appendix B](#) for a full derivation). [Appendix C](#) shows that our identification strategy is robust to individual-specific multiplicative response biases, in the sense that such biases do not affect the estimated strategic interaction parameters  $\alpha_{in}^a$  and  $\alpha_{in}^s$ .

Our identification strategy is different from the one implemented by [Afrouzi \(2024\)](#), who uses survey questions to identify the degree of strategic complementarity in price setting. Contrary to us, Afrouzi would consider only one policy scenario and would generate variations by artificially shutting down some channels (e.g. shutting down changes in expectations). His approach is close to ours, but the advantage of our approach is that it is more natural and allows a better control of the environment. For instance, it is not clear what firms infer from the expectation assumptions that they are provided with.<sup>2</sup>

**Higher-order errors.** Scenarios 1 and 2 provide us with a set of individual responses and higher-order beliefs. In Scenario 1, we collect  $(q_{in}^{sc.1}, E_{in}(q_n^{sc.1}), E_{in}(q^{sc.1}))$ , for an aggregate tax  $\tau = \epsilon$ . In Scenario 2, we collect  $(q_{in}^{sc.2}, E_{in}(q_n^{sc.2}))$ , for a sector-specific tax  $\tau_n - \tau = \epsilon$ . We can compute the higher-order errors  $E_{in}(q^{sc.1}) - q^{sc.1}$ ,  $E_{in}(q_n^{sc.1}) - q_n^{sc.1}$  and  $E_{in}(q_n^{sc.2}) - q_n^{sc.2}$ , where  $q^{sc.1}$ ,  $q_n^{sc.1}$  and  $q_n^{sc.2}$  are the aggregate and sectoral responses implied from the individual responses. Aggregating these higher-order errors would provide the aggregate errors driving Equation (20).

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<sup>2</sup>Transposing to our firms' quantity decisions, his strategy would amount to identifying the parameters of a version of Equation (18) that abstract from the sectoral level, using the following question: "Suppose that you get news that the general level of the carbon policy went up by  $\epsilon$  in the economy: (a) By what percentage do you think aggregate production would change? (b) By what percentage would your firm change its production level? (c) By what percentage would your firm change its production level if aggregate production did not change at all in response to this news?" This generates variations in  $(q_{in}, E_{in}(q), \tau_{in})$  that help identify the parameters  $\bar{\alpha}_{in}^a$  and  $E_{in}(\sigma_{in})$ :

$$\begin{pmatrix} \bar{\alpha}_{in}^a \\ E_{in}(\sigma_{in}) \end{pmatrix} = \begin{pmatrix} E_{in}(q^{sc.1}) & \epsilon \\ 0 & \epsilon \end{pmatrix}^{-1} \begin{pmatrix} q_{in}^{sc.1} \\ q_{in}^{sc.2} \end{pmatrix}$$

where  $E_{in}(q^{sc.1})$  is the answer to (a),  $q_{in}^{sc.1}$  is the answer to (b) and  $q_{in}^{sc.2}$  is the answer to (c).

Afrouzi would consider only one policy scenario and would generate variations by shutting down some channels, here  $E_{in}(q^{sc.2}) = 0$ . However, it is not clear what firms infer from the assumption  $E_{in}(q^{sc.2}) = 0$ .

### 3 The Survey

We rely on a survey for firm managers tailored to our identification strategy. The survey modules are organized to achieve three objectives. First, we elicit firm characteristics, which allow us to provide a firm-specific cost estimate of a carbon tax and to set up the carbon tax scenarios. Second, we present scenarios, consistent with the identification strategy outlined above, for which we elicit managers' expected effects on production and on other margins. Finally, we collect additional information about firms' and managers' characteristics to explore heterogeneity in our main estimates. In this section, we describe in detail each survey component achieving these objectives, and then we turn to the data collection details.

#### 3.1 Survey main components

**Carbon tax cost estimate.** One key innovation of our survey is to elicit a few firm characteristics such that we can provide a *real-time* estimate of a firm's total CO<sub>2</sub> emissions from direct fossil fuel consumption (referred as scope 2 emissions). We do so by asking managers about total operating expenses, the share of expenses devoted to energy, and the composition of energy inputs (electricity, fossil fuels, and other sources such as waste, coal, or wood). This information is first used to estimate CO<sub>2</sub> emissions, which we then translate into a cost estimate. Specifically, each manager is shown how a carbon tax increase of CHF 100 (treatment group 1) or CHF 300 (treatment group 2) per ton of CO<sub>2</sub> translates into an additional annual cost for their firm, where managers are assigned randomly to one of the two treatment groups.<sup>3</sup> [Figure 1](#) shows the information presented to managers. [Appendix D](#), which has the complete questionnaire, shows the modules leading to the carbon cost calculator. Formally, one can think of this procedure as providing a private, firm-specific signal about the cost elasticity  $\sigma_{in}$  that we aim to estimate. We thus provide a noisy signal but with no systematic aggregate or sectoral bias, preserving the information structure required for identification. [Table E.1](#) in the appendix contains additional information on how the cost increase is estimated, together with the CO<sub>2</sub> coefficients used.

**Carbon tax scenarios.** At the core of the survey are the carbon tax scenarios, which vary the coverage scope of the carbon tax increase but are identical in terms of the amount of the increase and all other information. In accordance with our identification strategy ([Section 2.6](#)), we present three different scenarios:

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<sup>3</sup>The current Swiss carbon tax on fossil thermal fuels is 120 CHF per ton of CO<sub>2</sub>. Our CHF 300 tax increase treatment, which then corresponds to an overall tax of 420 CHF per ton of CO<sub>2</sub>, aligns with [Ecoplan and INFRAS \(2024\)](#) who recommended, in a report for the Swiss government, a tax of 430 CHF to internalize the social cost of carbon in certain sectors.

- Scenario 1 (national tax  $\tau$ ): The tax increase applies to all firms in Switzerland.
- Scenario 2 (sectoral tax  $\tau_n$ ): The tax increase applies only to firms in the same sector as the respondent’s firm.
- Scenario 3 (firm-specific tax  $\tau_{in}$ ): The tax increase applies only to the respondent’s own firm.

For each scenario, we ask managers to assess the expected impact of the carbon tax reform. We first ask about the medium-run (three years after the implementation of the tax increase) impact on production, which we measure as the percentage change in annual production volume of goods and services (measured at current sales prices), relative to the situation with no additional tax increase. Specifically, firm manager  $i$  of sector  $n$  has to report the impact of Scenario 1 on Switzerland’s total production, the quantity  $E_{in}(q_1)$ ; the impact of Scenarios 1 and 2 on competitors’ production, the quantities  $E_{in}(q_{n,1})$  and  $E_{in}(q_{n,2})$ , respectively; and the impact of Scenarios 1, 2, and 3 on their own firm’s production, the quantities  $q_{in,1}$ ,  $q_{in,2}$ , and  $q_{in,3}$ , respectively.

The elicitation is done with a slide scale, one for each scenario and outcome. The slide scale (see [Appendix D](#)) has a range from -20% to 20%, but allows respondents to indicate that they expect “less than -20%” or “more than +20%.”<sup>4</sup>

Assuming that firms are small relative to the rest of their sector, and that the aggregate effect of the sectoral tax scenario is negligible, the above elicited production effects form the system of equations presented in Equation (23), which can then be used to recover a firm-specific parameter vector  $\{\alpha_{in}^a, \alpha_{in}^s, E_{in}(\sigma_{in})\}$ , for each respondent.<sup>5</sup>

Although our main analysis focuses on production outcomes, we also elicit managers’ expectations about the effects of the scenarios on final prices at the economy-wide, sectoral, and firm level.<sup>6</sup> For the national tax scenario, we further elicit expected effects on own firms’ energy consumption, number of employees, and profits.

**Firm and manager characteristics.** We complement the scenarios with the elicitation of various characteristics of the responding managers and their firms. We collect information

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<sup>4</sup>While most firm managers selected values between -20% and 20%, some selected the lowest bound, <-20%, and very few the highest bound, >20%. In our estimation, we retain these censored values and perform various sensitivity tests that show they have little impact on the main results.

<sup>5</sup>Note that for some firm manager the system of equations might not yield a solution if managers do not adjust some or all of their responses. We will thus report summary statistics for the subset of non-missing observations we recover for each parameter.

<sup>6</sup>Our model relies on assumptions about the information structure, in particular the timing of information shocks ([Section 2.2](#)), which are tailored to production decisions. Equation (18) applies to prices only if the information-timing assumptions also hold for pricing behavior.

on firms’ sector of activity, geographic location, number of employees, turnover, export share in the previous year, and a parsimonious set of manager characteristics: tenure at the firm, current position, age, gender, and education level. Managers are also asked to report perceived competition intensity in their sector (7-point ordinal scale), and the geographic location of their main competitors (five categories ranging from *all in Switzerland* to *all abroad*). We use these latter variables to corroborate, in part, our strategic parameter estimates.

We also ask managers about their beliefs about their firm’s CO<sub>2</sub> intensity relative to their sector, and their sector’s intensity relative to the economy as a whole, using an ordinal scale with categories *lower*, *equal*, and *higher*. In addition, managers have to report whether their firm is exempt from the Swiss carbon tax, participates in the Swiss Emission Trading System, and participates in the federal program for voluntary CO<sub>2</sub> reductions.

Finally, we ask managers about their view on the ideal carbon price for Switzerland under different scenarios. In one scenario, managers have to report what they would consider as the ideal carbon tax for Switzerland (in terms of CHF/ton) if the current tax were to be replaced by a new regime covering all sectors and CO<sub>2</sub> emissions in Switzerland. In a second scenario, they have to report the ideal carbon tax under a selective exemption where only the respondent’s own firm and its competitors were to be exempted from the new tax regime.

### 3.2 Data collection

**Contact sample.** We drew the firm contact sample from three sources: the Orbis firm database ([Moody’s Analytics, 2024](#)), the firm panel of the KOF Institute ([KOF Institute, 2025](#)), and an address panel from a 2023 survey ([Benhima et al., 2024](#)). The sample includes 26,900 firm contacts and covers the firm population in the Swiss Business and Enterprise Register (BER) at the sector–size stratum level, with two coverage exclusions: sectors outside the market economy and firms located in the Italian- and Romansh-speaking regions, which account for 6 percent of the BER. Rather than applying purely proportional stratified sampling, we oversampled larger firms and sectors expected to exhibit above-average responses to carbon tax increases or higher within-stratum heterogeneity, to ensure sufficient precision in strata that contribute disproportionately to aggregate outcomes. [Appendix F](#) provides additional details on the construction of the contact sample and the oversampling procedure. In the empirical analysis, we account for the resulting unequal sampling probabilities by reweighting firms to match population statistics, as discussed below.

**Survey implementation.** We invited firm contacts via e-mail to participate in the online survey on June 17, 2025, in German, French, or English, depending on the firm’s language region or previously indicated preference. The invitation, written in a neutral tone, provided

a brief explanation of the survey’s purpose, a secure direct link to the online survey, and, as an alternative, access via a login page using a password included in the invitation; it also offered exclusive access to a summary report of the results. Invitations were signed by the authors of this study and included links to their university websites. Two reminders were sent, and the survey was closed on August 31, 2025.

**Response sample statistics.** 4,116 firm managers started the survey (response rate of 15.3%) and 1,408 proceeded until the survey end (completion rate of 5.2%). Among the latter, 1,145 completely answered the scenario questions on the impact of a carbon tax on production. Among those, we exclude managers (N=2) who did not provide sufficient information to estimate the CO<sub>2</sub> of their firms. We also exclude 126 managers who reported highly implausible cost ratios.<sup>7</sup> The final sample size for our main analysis thus contains 1,017 managers.

In [Appendix G](#), we examine potential selection bias in survey participation. We find statistically significant selection with respect to firm size, but not with respect to firms’ energy intensity. Selection by firm size is common in establishment surveys and easily corrected by our weighting procedure; selection by energy intensity would have been more problematic as it might have led to bias, as firms that are more affected by carbon pricing might express views that are not representative of the overall firm population.

[Table 1](#) describes the sample along different characteristics. The typical firm in our sample has a median of 19 full-time employees and CHF 4.8 million in turnover, though the distribution is highly right-skewed (mean of 78 FTE and CHF 44.9 million). Firms report modest energy cost shares (median 2% of total costs). CO<sub>2</sub> emissions are nonetheless important (median 54 tons). Interestingly, a large majority perceive their own emissions as lower than those of other firms in their sector (96%) and their sector’s emissions as lower than those of other sectors (86%). Respondents are predominantly male (88%), senior decision-makers (77% CEO or CFO), with a median age of 52 and an average of 17 years at their firms; when asked about their ideal carbon price, the median response is CHF 100 per ton of CO<sub>2</sub>, rising to CHF 120 if they were to be exempted.

**Aggregation of survey responses.** As part of our analysis of the survey responses, we construct post-stratification weights and aggregate individual responses to the sectoral and economy-wide level. To ensure representativeness and correct for sample biases due to, e.g., the aforementioned oversampling of large firms, we apply an aggregation scheme that combines employment-based scaling with cell-size adjustment. [Section H.1](#) discusses

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<sup>7</sup>We drop responses with total costs per employee above CHF 1 million or below CHF 5,000, or with a total-costs-to-turnover ratio above 5.

the aggregation scheme in detail.

We also use this aggregation scheme when reporting strategic interaction parameters and managers’ belief errors regarding other firms’ outcomes. After obtaining individual-specific values, we use the post-stratification weights defined by the aggregation procedure to compute weighted means and other summary statistics.

As a robustness check, we also report unweighted averages of firms’ responses, interaction parameters, and belief errors. In addition, we employ two alternative aggregation schemes, which are discussed in [Section H.2](#).

## 4 Results

### 4.1 Survey responses

[Table 2](#) reports the mean outcomes for all the different impacts we elicited for a national carbon tax reform. We distinguish the means across tax levels that different managers were assigned to (100 CHF/tCO<sub>2</sub> or 300 CHF/tCO<sub>2</sub>). Note that this variation is not required for our main identification of strategic interaction parameters or prior beliefs. We use this between-subject variation to show that managers are sensitive to the level of the carbon tax under different scenarios. The first row of [Table 2](#) corresponds to the increase in energy costs resulting from the carbon tax. These are the estimates we computed using the cost and energy share information collected from managers. Note also that all values are reweighted, using the approach described in [Section H.1](#), such that the averages are representative of the Swiss economy. The 100 CHF/tCO<sub>2</sub> tax translates into a 8.06% increase in energy cost on average, and the 300 CHF/tCO<sub>2</sub> tax has an impact that is three times larger, as expected. Turning to firm responses, we first observe that managers expect a reduction in their firm’s output of about -1.44% for the lower tax, and a decrease of -2.61% for the higher tax—a difference that is statistically significant (p-value=0.02). Managers also expect to pass these costs to consumers. They report that they will increase the price of their main final product by 2.51% to 3.02%, for the low and high tax levels, respectively. For the other outcomes, i.e., energy consumption, employment (measured with full-time employee), and profits, we also observe the expected sign and sensitivity with respect to the level of tax. On average, managers report that they will lower their energy consumption by -1.16% or -1.87%, depending on the tax level. The impacts on labor and profits are negative. For profits, they are larger and statistically different at the higher tax level compared to the lower tax level.

It is difficult to directly compare our results with observational studies that report elasticities with respect to energy prices. These studies typically focus on a specific sector, notably the industrial sector, and use natural experiments, such as short-term price shocks arising

from energy crises, that may not readily be comparable to an anticipated carbon pricing reform. Nonetheless, the implied elasticities for the different outcomes we measure are in line with the range of elasticities found in observational studies. For instance, several studies focusing on the industrial sector find that the short-run price elasticities of energy demand are typically small, even in response to large price shocks (Alpino et al., 2024). A meta-analysis by Labandeira et al. (2017), which primarily focuses on residential energy demand, reports average short-run price elasticities of -0.17 and -0.22 for the industrial and commercial sectors. In Table 2, if we consider the ratio of the change in real output on the change in energy costs, we have an output elasticity that ranges from -0.10 to -0.21 depending on the level of the carbon tax.

With respect to other outcomes, some observational studies also report employment elasticities with respect to energy prices. Marin and Vona (2021), for instance, found output elasticities with respect to energy prices that range from -0.2 to -0.3, and about -0.25 for employment, which is slightly above our estimates. Our average output and employment elasticities are -0.13 and -0.09, respectively.<sup>8</sup>

Studies examining the impact of the EU-ETS are also relevant. One important caveat is that these studies mostly focus on the manufacturing sector and on the initial phases of the policy when permit prices were low. These ex-post analyses typically find no to minimal impacts on employment and output (Martin et al., 2014; Dechezleprêtre et al., 2023; Colmer et al., 2025). Our estimates, on the other hand, suggest managers expect a negative impact for a substantially higher carbon price.

Finally, note that our responses do increase with the tax level but do not scale linearly. Our elasticities thus tend to be larger at the lowest tax level. The highest tax level we present corresponds to a carbon price that firm managers in Switzerland, or anywhere else, have never experienced. It is thus possible that managers may not foresee the full extent of such a large change. At the same time, it is possible that responses reflect truly non-linear effects of tax changes of different size.

Next, we investigate how managers responded to the scope of the carbon tax reform across the three scenarios we presented to each of them. As shown in Table 3 (Panel A), managers report larger, i.e., more negative, impacts from a carbon pricing scheme that targets only their firm, versus their sector, or the whole economy. For a national scheme, managers expect a reduction in output of -1.18%, on average, which decreases to -1.54% for a sectoral scheme, and to -2.68% for firm-specific tax. Note that for all those estimates, we have normalized the impact for a 100 CHF/tCO<sub>2</sub> tax.<sup>9</sup> We also weight all responses using post-stratification

<sup>8</sup>These elasticities are readily available from Table 2. The output elasticity for the whole sample is given by the ratio: -1.97/15.46, whereas the employment elasticity is given by: -1.44/15.46.

<sup>9</sup>To do so, we have divided the expected impacts of managers that received the 300 CHF/tCO<sub>2</sub> tax by three.

weights constructed as described in [Section H.1](#). [Table 3](#) thus reports aggregate responses that are representative of the Swiss economy.

Finally, we also investigate variation in managers’ beliefs about the impact of carbon tax reform on their own output versus other firms’ output. Remember that, for a national carbon tax, we elicit, in addition to the change in their own production, what they believe the change in production across the whole economy and in their sector will be. In [Table 3](#) (Panel B), we show that managers more pessimistic higher-order beliefs, on average, especially about the national impacts. The average firm’s reduction in output is less pronounced than the (average) expected impact on the whole economy. Managers thus systematically believe that the economy, and to a lesser extent their sector, would be hit harder than what they collectively report about the impact on their own firm.

[Table 3](#) also reports the share of managers that expect a negative, nil, or positive production impact for a given carbon tax scenario. Whereas nearly 50% of managers expect a negative impact on their own or others’ production, a significant share expect no impact. Finally, about 15% of all expected output changes, across scenarios and outcomes, are positive. Examining these responses more closely, we find that most managers who reported at least one positive output change in fact reported positive changes across all scenarios. This suggests they may have been prone to a sign error due to inattention. As we discuss below, we present various robustness tests for our main estimates, showing that excluding or recoding those observations does not qualitatively change our results.

## 4.2 Policy elasticity and strategic interaction parameters

We now report the estimates for the policy elasticity and strategic interaction parameters. We use the system of equations [\(23\)](#) to recover the vector of parameters  $\{E_{in}(\sigma_{in}), \alpha_{in}^a, \alpha_{in}^s\}$  for each manager who responded to the survey.<sup>10</sup> [Figure 2](#) presents the cumulative distribution of each of the three parameters. For the policy elasticity parameter, most of the distribution lies in the positive domain, as we would expect. Panel A of [Table 4](#) reports a mean value of 2.68, meaning that managers expect a carbon tax of CHF 100 per ton of CO<sub>2</sub> to reduce their firm’s output by 2.68% on average, absent any strategic response. This is an economically meaningful effect, suggesting that managers view the direct cost impact of the tax as substantial even before accounting for how competitors might react.

The strategic interaction parameters are similarly important, but they are far more heterogeneous, as shown in Panel B of [Figure 2](#). Both parameters span negative and positive values, but their means are negative overall (Panel A of [Table 4](#)), which indicates that output

<sup>10</sup>We solve the system iteratively. When solving for  $\alpha_{in}^a$  and  $\alpha_{in}^s$ , denominators can be zero, which yields undetermined parameter values. As a result, the distributions and statistics we report for each of the three parameters have different sample sizes.

responses are strategic substitutes on average. In other words, when managers expect competitors to reduce output, they tend to expect their own firm to increase output somewhat in response, and vice versa. The strategic response at the sectoral level is more pronounced than at the national level, with a parameter that is roughly twice the magnitude of the national level parameter.

These averages, however, mask substantial variation when the two parameters are considered jointly. [Figure 3](#) plots the discrete joint density of  $\alpha_{in}^a$  and  $\alpha_{in}^s$ , which offers two useful takeaways. First, about 25% of managers are not strategic in either dimension at all. Second, although we find some probability mass in every quadrant, a larger share of managers fall into the region corresponding to strategic substitutability. Strategic substitutes therefore tend to dominate in this setting.

We next examine heterogeneity by correlating the estimated parameters with observable firm and manager characteristics. We focus on three variables: the perceived level of competition within the sector, the firm’s estimated CO<sub>2</sub> emissions, and the ideal nationwide carbon price that managers would set for the Swiss economy. The first variable captures exposure to sectoral competition and, we expect, should be correlated with the strategic parameter at the sectoral level. The firm’s estimated CO<sub>2</sub> emissions capture overall exposure to carbon pricing, and thus should be correlated primarily with the policy elasticity. The third variable is a proxy for overall political support for a carbon pricing scheme. [Figure 4](#) presents a binned scatter plot for each combination of parameter and variable. As expected, the sectoral strategic interaction parameter correlates strongly with perceived competition intensity: managers who perceive more intense competition from other firms in their sector are more likely to respond strategically, factoring in other firms’ output responses. This pattern holds only at the sectoral level, however, since the corresponding correlation at the national level is much weaker. The policy elasticity is also strongly and positively correlated with competition intensity, suggesting that even absent strategic motives, managers in more competitive sectors perceive themselves as facing higher abatement costs. Turning to CO<sub>2</sub> emissions, the strategic interaction parameters display little correlation with this variable, unlike the policy elasticity, which is strongly and positively correlated with it. Managers thus expect higher marginal abatement costs when their firm emits more, a pattern that is again quite intuitive. Finally, with respect to support for carbon pricing, proxied by the managers’ ideal carbon price, all three parameters display some correlation with this variable, although the pattern appears to be driven in part by possible outliers, where higher support for carbon pricing is associated with a stronger strategic motive and with lower marginal abatement costs.

We test the robustness of these correlations using regressions that include a large set of

observables.<sup>11</sup> We run separate regressions for each parameter, using  $\sigma_{in}$ ,  $\alpha_{in}^s$ , and  $\alpha_{in}^a$  in turn as the outcome variable. We also run linear probability models in which we consider a dummy for whether  $\sigma_{in}$  is positive as the outcome, and, similarly, whether  $\alpha_{in}^s$  (or  $\alpha_{in}^a$ ) is negative. Results are presented in [Table 5](#). In Column I,  $\sigma_{in}$  is the outcome variable, and the dummy for  $\sigma_{in} > 0$  is the outcome for Column II. In Column III, the dependent variable is the parameter  $\alpha_{in}^s$ , whereas in Column IV, the dependent variable is a dummy variable that takes a value of 1 if the parameter  $\alpha_{in}^s$  is negative. Similarly, Columns V and VI use the parameter  $\alpha_{in}^a$  and a dummy variable indicating whether  $\alpha_{in}^a$  is negative, respectively. We find that the intensity of competition remains correlated with the policy elasticity and with the strategic parameters at the sectoral level. However, for the sectoral strategic parameter, the result is sensitive to whether we consider the parameter itself or the corresponding dummy variable as the outcome. We also include in the regression other variables related to competition, such as the export share and whether competitors are only Swiss. Interestingly, firms whose competitors are only Swiss tend to have a larger strategic parameter at the national level, meaning that they might be more prone to behave as a strategic complement with respect to the national output response. Firms with a higher export share display the same pattern, that is, they tend to be less of a strategic substitute at the national level.

Note that higher exposure to a carbon tax, captured by the log of CO<sub>2</sub> emissions, remains correlated with the policy elasticity and with the strategic parameters at the sectoral level. We also find evidence that firm size, captured by the log of the number of full-time employees, is correlated with all three parameters. Finally, the correlation with the proxy for carbon support remains strongest for the strategic parameter at the sectoral level, where higher support translates into a higher likelihood that the firm is prone to strategic substitutability at the sectoral level.

We perform several robustness checks to assess the sensitivity of the parameters to various assumptions. First, in [Appendix I.1](#) we reproduce all the main tables using unweighted firm-level means instead of post-stratification-weighted estimates. The results are qualitatively similar. Both the magnitude and sign of the output responses and estimated parameters are

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<sup>11</sup>The set of observables includes the firm’s export share and whether the firm’s competitors are only Swiss. It also includes the share of energy costs in total costs and total turnover (in logs). We further include indicators for whether the firm operates in the manufacturing sector, whether it uses electricity, and whether it uses fossil fuels, together with indicators for whether the firm’s own CO<sub>2</sub> emissions are lower than those of other firms in its sector and whether its sector’s emissions are lower than those of other sectors. We also include several manager characteristics: an indicator for being a CEO, CFO, or director, the manager’s years in the current role, gender, and highest level of educational attainment. Finally, all specifications include an indicator for assignment to the CHF 300 tax treatment and the survey duration in minutes.

unaffected by the weighting. Second, we present an exhaustive set of robustness checks for the estimation of the policy elasticity and strategic interaction parameters. In [Appendix I.3](#), we show that excluding or recoding managers who appear to have made sign errors raises the policy elasticity but leaves the strategic interaction parameters largely unchanged. Widening the truncation of extreme responses has little effect. We also exclude managers who responded to the survey quickly (the lower decile) and obtain the same results. The strategic parameters are more sensitive to the treatment of zero denominators: imputing a small negative value makes them considerably more negative, while setting missing values to zero yields a conservative lower bound. Overall, across the various robustness checks, we find that our qualitative conclusions are unchanged: both strategic interaction parameters are negative on average, but more so at the sectoral level, and the policy elasticity remains positive.

### 4.3 Noisy beliefs

How accurate are managers' beliefs about the effects of carbon pricing on other firms? To answer this question, we compute aggregate higher-order errors at both national and sectoral levels. We first construct the relevant aggregate output change by aggregating managers' estimates of their own firms' output changes. We then define a manager's error as the difference between the manager's expectations of the impact on other firms and the corresponding aggregate impact. The corresponding national error for manager  $i$  is

$$\varepsilon_{in}^a = E_{in}(q) - q. \quad (24)$$

and the corresponding sectoral error, for the sector  $n$ , for the same manager  $i$  is

$$\varepsilon_{in}^s(n) = E_{in}(q_n) - q_n. \quad (25)$$

Note that both the quantities  $q$  and  $q_n$  are aggregated with the post-stratification weights attached to firm  $i$ , where [Section H.1](#) describes the construction of these weights.

We average national and sectoral errors across managers and test whether the mean errors differ from zero. Panel B of [Table 4](#) shows that managers systematically overestimate the impact of carbon pricing on other firms, relative to the aggregate of their estimates of their own firms' output changes. This pattern is particularly pronounced for the national effects of a national carbon tax increase. Put differently, on average, managers expect the rest of the economy to be more severely affected than their own firm.

As for the policy elasticity and strategic interaction parameters, there is much heterogeneity across managers in the magnitude and sign of these errors. Using the same observables as before, we investigate how the errors correlated with different variables. In [Table 6](#), we

report the estimates for the intensity of competition, export share, whether the firm has only Swiss competitors, CO<sub>2</sub> emissions, reported ideal carbon price, and firm size. Again, we find that the intensity of competition is significantly correlated with the beliefs, but the strongest correlation is with the level of CO<sub>2</sub> emissions. Managers in firms with higher emissions are more likely to hold pessimistic beliefs. Interestingly, we also find some evidence that support for a higher carbon price comes with less pessimistic beliefs, but this correlation is only statistically significant for the sectoral error.

Why do managers hold pessimistic beliefs, and how would their expected response to carbon pricing change if we were to provide better information? We answer these questions in the next section, where we investigate the source of such overly pessimistic beliefs about the impact of a carbon pricing reform and what would happen if managers had access to better information.

## 5 The role of public versus private information

Survey instruments reveal how firm managers anticipate the effects of a proposed policy reform. These anticipations have real consequences. For instance, firms might invest in lobbying efforts and even threaten to relocate to influence the early design of a policy. When those anticipations are noisy or prone to errors, as revealed by our empirical analysis, this creates a rationale for interventions. Policy makers could correct how managers perceive, and, thus, eventually support a given policy by providing better information. But exactly how should such information interventions be designed, and what are their potential impacts? We next show how our survey instrument, coupled with our model, can be used to explore these questions.

To do so, we first need to be more specific about the information structure, particularly the role of public versus private signals that managers have access to when forming anticipations. We then show that, with a parsimonious information structure, we can estimate the model using moments in the data. Finally, having estimated key parameters of the information structure, we can perform information counterfactuals. Specifically, we can investigate the role of errors in public priors or precision in private signals about abatement costs. In this setting, public priors refer to the general sentiment about carbon pricing in the economy. Whereas, private signals are the results of firms' internal capacity to forecast the cost of carbon pricing. As we further discuss below, both could be targeted by specific information interventions.

## 5.1 Public priors and higher-order errors

We now characterize a more specific information structure that is a special case of Assumption 1 and that distinguishes between public and private information:

**Assumption 2** (Public and private information). *We assume that the elasticity decomposition (21) holds, and that the prior distributions of  $\sigma^a$ ,  $\sigma_n^s$  and  $\sigma_{in}^f$  are Gaussian and mutually independent and follow  $\sigma^a \sim \mathcal{N}(x^a, \pi_a^{-1})$ ,  $\sigma_n^s \sim \mathcal{N}(0, \kappa_s^{-1})$ ,  $\sigma_{in}^f \sim \mathcal{N}(0, \kappa_f^{-1})$ .*

*We additionally assume that the firm observes two private signals: a sector-level signal on the sectoral elasticity  $\sigma_n$ ,*

$$y_n^s = \sigma_n + e_n^s = \sigma^a + \sigma_n^s + e_n^s$$

*and an individual signal on the firm-level elasticity  $\sigma_{in}$ ,*

$$z_{in}^f = \sigma_{in} + e_{in}^f = \sigma^a + \sigma_n^s + \sigma_{in}^f + e_{in}^f$$

*with  $e_n^s \sim N(0, (\pi^s)^{-1})$  and  $e_{in}^f \sim N(0, (\pi^f)^{-1})$  two mutually independent noise shocks.*<sup>12</sup>

$x^a$  is the common prior about the economy-wide component  $\sigma^a$ , and  $\pi_a$  is the precision of this prior. It constitutes the public information about  $\sigma^a$ . The other prior assumptions simply state that the sectoral average of  $\sigma_{in}^f$  and the economy-wide average of  $\sigma_n^s$  are equal to zero. This public prior can be decomposed into the actual fundamental and some “surprise”, or noise:

$$x^a = \sigma^a + e^a$$

where  $e^a$  follows the distribution  $e^a \sim N(0, (\pi_a)^{-1})$ .

It is natural to assume that firms within a sector observe the same information about the sectoral elasticity. It is also natural to assume that they observe their own elasticity  $\sigma_{in}$  imperfectly, but that there is no systematic aggregate or sectoral noise in that signal. Conditional on this private signal, some firms may overestimate or underestimate their elasticity, but on average, they observe it correctly.

While the prior  $x^a$  is common knowledge,  $y_n^s$  and  $z_{in}^f$  are private signals.  $z_{in}^f$  is only observed by the firm, and  $y_n^s$  is only observed by firms within sector  $n$ .

Using this more specific information structure, we derive the following proposition, which shows that aggregate first-order and higher-order errors appear (see proof in [Appendix A.2](#)):

<sup>12</sup>This is a special case of Assumption 1, with

$$\Sigma_{in}^0 = \Sigma^0 = \begin{pmatrix} x^a \\ 0 \\ 0 \end{pmatrix}, P^0 = \begin{pmatrix} (\kappa^a)^{-1} & 0 & 0 \\ 0 & (\kappa^s)^{-1} & 0 \\ 0 & 0 & (\kappa^f)^{-1} \end{pmatrix}, \quad h = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}, R = \begin{pmatrix} (\pi^s)^{-1} & 0 \\ 0 & (\pi^f)^{-1} \end{pmatrix}$$

**Proposition 2** (First- and higher-order errors). *Under Assumptions 2, 3, 4, 5 and 6 (see Appendix A.1 for the latter four), the average error on the firms' own elasticity, and on the aggregate and sectoral output responses to an economy-wide tax  $\tau$ , are proportional to  $e^a$ , the noise in the public prior:*

$$\bar{E}(\sigma_{in}) - \sigma^a = \delta^x e^a \quad (26)$$

$$\bar{E}(q) - q = -(\gamma^y + \gamma^z) \delta_a^x e^a \tau \quad (27)$$

$$\bar{E}(q_n) - q = -\gamma^z \tilde{\delta}^x e^a \tau \quad (28)$$

where  $\delta^x$ ,  $\delta_a^x$  and  $\tilde{\delta}^x$  are the weights put by the firm on the public prior  $x^a$  when forecasting, respectively, its own elasticity, the aggregate elasticity, and the sectoral elasticity, while  $(\gamma^y, \gamma^z)$  are the responses of individual output to the private signals  $(y_n^s, z_{in}^f)$  for each unit of aggregate tax  $\tau$ .

Additionally, we show the following:

- (i) **Signs**  $\delta^x \geq 0$ ,  $\delta_a^x \geq 0$  and  $\tilde{\delta}^x \geq 0$ . Additionally, if  $|\alpha^a|^{\frac{1}{2}} + |\alpha^s|^{\frac{1}{2}} < 1$ , then  $\gamma^y + \gamma^z \geq 0$  and  $\gamma^z \geq 0$ .
- (ii) **Perfect private information** If either  $z_{in}^f$  or  $y_n^s$  are perfect signals ( $\pi^f = +\infty$  or  $\pi^s = +\infty$ ), then  $\delta^x = 0$  and there is no aggregate error on  $\sigma_{in}$ .
- (iii) **No private information** If, on the opposite, neither  $z_{in}^f$  nor  $y_n^s$  is informative ( $\pi^f = \pi^s = 0$ ), then  $\gamma^y + \gamma^z = 0$  and there is no aggregate error on  $q$ .
- (iv) **No firm-level private information** If  $z_{in}^f$  is not informative ( $\pi^f = 0$ ), then  $\gamma^z = 0$  and there is no aggregate error on  $q_n$ .

First-order errors on the elasticity appear on average when there is aggregate noise:  $e^a \neq 0$ . Firms combine the public prior  $x^a$  with the private signals  $y_n^s$  and  $z_{in}^f$  to infer their elasticity. As a result, the noise in the public prior  $e^a$  drives the average error. When the prior about the aggregate elasticity is too high ( $e^a > 0$ ), firms overestimate their own elasticity on average. This is true as long as the private signals are not perfect; otherwise, the firms would not rely on public information.

Aggregate noise also leads firms to make higher-order errors on quantities, on average. This is due to the presence of private information. Since individual quantities respond to the firms' private signals, the aggregate and sectoral quantities will reflect information that is not common knowledge. As a result, to form expectations on  $q$  and  $q_n$ , firms must infer the aggregate and sectoral elasticities, based on their imperfect information, and they make use of the public signal  $x^a$ , which will generate some average error driven by  $e^a$ .<sup>13</sup>

<sup>13</sup>For firms to be unable to perfectly forecast aggregate output, some private information is required either

The higher-order errors on quantities thus inform us of the first-order errors firms make on their own elasticity. Conditional on the assumption that strategic interaction parameters are not unrealistically large ( $|\alpha^a|^{\frac{1}{2}} + |\alpha^s|^{\frac{1}{2}} < 1$ ), the errors on quantities and the error on the firms' own elasticity are of opposite signs. If firms underestimate aggregate output ( $\bar{E}(q) - q < 0$ ), as we have documented, it indicates that they get an overly pessimistic public signal on the aggregate elasticity ( $e^a > 0$ ), which implies that they are also over-pessimistic about their own elasticity on average ( $\bar{E}(\sigma_{in}) - \sigma^a > 0$ ).

Combined with Equation (20), these results imply that noise in the public signal  $e^a$  has aggregate consequences. First, a positive average error on  $\sigma_{in}$ , driven by a positive  $e^a$ , leads firms to respond excessively negatively to a tax  $\tau$ . Moreover, it would imply that the errors on  $q$  and  $q_n$  would be negative on average. The way these errors contribute to aggregate output depends on the strategic parameters  $\alpha^a$  and  $\alpha^s$ . Since these parameters are negative on average, then the effect of the noise  $e^a$  on output will be dampened, or even reversed.

However, to study the aggregate consequences of  $e^a$ , it is necessary to estimate  $e^a$ ,  $\sigma^a$ , and the whole set of information parameters jointly. In what follows, we match these parameters to some key moments in the data.

## 5.2 Fitted moments and information parameter estimates

The estimation targets a set of moments capturing average responses, average errors, and selected second moments. Second moments focus on cross-sectoral variances and covariances of the few variables ( $q^{sc.1}$  and  $q_{in}^{sc.3}$ ) that have homogeneous coefficients, so that these moments are not driven by the heterogeneity of the coefficients or by measurement error at the firm level. The cross-individual variance of the responses cannot be used as targets, because they are subject to measurement errors. However, we want to make sure that our estimation picks a realistic level of individual variance, so we impose the constraint that the theoretical variance of  $q_{in}^{sc.1}$  is lower than the empirical one, the difference being explained by measurement error and parameter heterogeneity. All empirical moments are weighted using the weights defined in Section H.1, so that they correspond to the economy-wide response. For instance, the weighted average of individual responses  $q_{in}$  in a given scenario approximates the aggregate response  $q$ .

Table 7 reports targeted and untargeted moments in the data and in the model. We minimize the sum of the squared distances between the theoretical and empirical moments. All moments are weighted with an equal weight of one. The model hits the targeted moments reasonably well. It closely matches the average of firms' own outcomes in Scenario 1 and

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at the firm or sector level, through either  $y_n^s$  or  $z_{in}^f$ . However, the sectoral signal  $y_n^s$  is common knowledge within a sector. Therefore, for firms to be unable to perfectly forecast sectoral output, some private information at the firm level is necessary, as captured by  $z_{in}^f$ .

Scenario 3, as well as the average errors on the aggregate and sectoral outcomes in Scenario 1. It also captures the variance of expectations about aggregate output in Scenario 1, the variance of firms' own outcomes in Scenario 3, and the covariance between the two.

The model also delivers a reasonable fit for untargeted moments. Although not explicitly targeted in the estimation, the average of firms' own outcomes in Scenario 2 is broadly in line with its empirical counterpart. The average error on the sectoral outcomes in Scenario 2 is overestimated, but the sign and relative magnitude are correct.

**Table 8** reports the estimated structural parameters. First, the public prior  $\sigma^a + e^a = 4.62$  is large, but about half of it is due to noise  $e^a$ . The public prior is therefore excessively pessimistic. Another striking feature of the estimation is the prominent role played by private information: the sectoral and individual signals are more precise than the public prior ( $\pi^s$  is about four times larger than  $\pi^a$ , and  $\pi^f$  about 50% larger). Besides, the volatility of the individual elasticity is driven mainly by the individual component  $\sigma_{in}^f$  rather than the sectoral component  $\sigma_{in}^s$  ( $\kappa^f$ , the inverse of the variance of the individual component, is about three times smaller than  $\kappa^s$ , the inverse of the variance of the sectoral component).

The model is non-linear, and the different parameters combine in a complex way to account for the moments. However, some moments play a crucial role for some key parameters. The noise  $e^a$  is identified as being positive and relatively large. That comes from the large errors, especially  $(\bar{E}(q) - q)^{sc.1}$ , as suggested by Proposition 2.<sup>14</sup> The “true” elasticity  $\sigma^a$  that is estimated is the one that is both compatible with the estimated noise  $e^a$  and the large expected output drops in Scenarios 1 and 3.

Another key feature of the data is that the error on sectoral output  $(\bar{E}(q_n) - q)^{sc.1}$  (-0.12%) is low relative to the error on aggregate output  $(\bar{E}(q) - q)^{sc.1}$  (-0.64%). This is key for our estimation of a precise sectoral signal  $y_n^s$ . Indeed, as Proposition 2 shows, the error on sectoral output is proportional to the response of individual output to the individual signal,  $\gamma^f$ , while the error on aggregate output is proportional to the response of individual output to both the individual and sectoral signals,  $\gamma^s + \gamma^f$ . This is because, within a sector, the sectoral signal is common knowledge, so, when firms within a sector respond to sectoral information, this will not generate an error on sectoral output. By contrast, when firms in the economy respond to their sectoral signal, which is not observed by the firms in other sectors of the economy, this will generate an additional error on aggregate output. A significantly larger error on aggregate output thus means that firms respond strongly to their sectoral signal, which is compatible with a high level of precision in that signal ( $\pi^s$  is high).

The second moments hint at how firms use their signals and, indirectly, at how they

<sup>14</sup>Note that these large errors would also be compatible with a larger precision of the public prior  $\pi^a$ —i.e., with a higher weight put on the public prior—and a smaller noise  $e^a$ . The identification of precision versus noise relies on the fact that the precision affects the second moments, while noise does not.

perceive the variance of noise and fundamentals. Notably, the cross-sectoral variance of the firm's own outcomes  $\text{Var}_n(q_{in}^{sc.3})$  (2.07) is three times larger than the cross-sectoral variance of the firms' expectations about the aggregate outcome  $\text{Var}_n(E_{in}(q)^{sc.1})$  (0.66). This indicates that firms put a larger weight on their individual signal  $z_{in}^f$  when forming expectations about their own elasticity (which is driving their own response) than when forming expectations about the aggregate elasticity (which is driving their expectation on the aggregate output). This can be explained by a high volatility in the individual component of the elasticity  $\sigma_{in}^f$  (a low  $\kappa^f$ ). In that case, for a given precision  $\pi^f$ , the individual signal is a poor signal of  $\sigma^a$ , because  $\sigma_{in}^f$  is noise, but is a good signal of  $\sigma^a + \sigma_n^s + \sigma_{in}^f$  because  $\sigma_{in}$  is fundamental. Consistently, firms put a higher weight on that signal when inferring their own response than when inferring aggregate output, and the sectoral component in that signal then drives a higher sectoral volatility in the firm's own response. Importantly for the next section, this implies that firms are able to precisely assess their own elasticity while still making significant errors on the aggregate elasticity.

### 5.3 The role of private information

Using our estimated model, we can conduct a decomposition exercise to show the role of private information. We progressively remove each component of the private information signals, viz., firm-specific and sectoral signals and show how it impacts anticipations and aggregate errors. The first column of [Table 9](#) reports the results for our baseline estimation with all the signals. The second column considers the case where firms do not have any private individual information, but can rely on the public prior and on the private sectoral signal. The third column considers the case where firms only rely on the public prior. Panel A of the table reports the aggregate real output effect. Panel B presents the effects of the different components in Equation (20): the higher-order error on aggregate output, the higher-order error on sectoral output, the first-order error on the elasticity, and the fundamental elasticity. Panel C shows the contribution of each component to the aggregate output effect. While the fundamental elasticity (2.45) and its contribution (-1.37pp) are identical across the three experiments, the first- and higher-order errors differ across information structures.

Consider the baseline case in which firms use all the information. In that case, the expected fall in output is 1.24%. As [Proposition 2](#) suggested, consistent with the fact that the higher-order errors  $\bar{E}(q) - q$  and  $\bar{E}(q_n) - q$  are negative (firms are excessively pessimistic on the aggregate and sectoral output), the first-order errors on  $\sigma_{in}$  are positive on average (firms over-estimate their own elasticity), as they are both driven by the public prior. As a result, these first-order errors contribute to making firms more pessimistic on their outlook. However, because our estimation found that the private information is precise, these first-order errors are small on average (0.06). As a result, their contribution is small (-0.03pp).

The excessively pessimistic higher-order errors (-0.69% and -0.20% respectively), on the other hand, contribute to making firms more optimistic, because of the strategic substitutabilities. The resulting positive contributions (respectively, 0.11pp and 0.06pp) more than offset the negative contribution of the first-order beliefs. All in all, the errors have a positive impact on firms' outlook.

Now consider the case with sectoral and public information only. Firms become more pessimistic, as the average outlook moves from -1.24% to -1.44%. This comes from two distinct effects of removing individual private information. First, it makes the private information on  $\sigma_{in}$  less precise. Firms then rely more heavily on the public prior when assessing their own elasticity. Then over-estimate it by 0.28, which reduces firms' outlooks by -0.16pp. Second, firms now have common knowledge within their sector, so they correctly assess sectoral output (they share the same information as every other firm in their sector). The sectoral error contributed positively to their output in the baseline, but it no longer does so here.

Finally, consider the case with only public information. All information is common knowledge, so there are no higher-order errors anymore. Besides, with no private information whatsoever, the negative public prior generates an even larger error in  $\sigma_{in}$  (2.17 on average), which further lowers the firms' outlook. As a result, the average outlook moves to -2.59%, more than twice the baseline estimate.

In a nutshell, this analysis shows that public information on the costs of carbon taxation is highly pessimistic, reflecting a very negative public sentiment about the economic impact of carbon policies. However, the impact of that noise on the firms' expectations is mitigated because (i) they have access to other sources of information (the sectoral and private signals), (ii) higher-order errors mitigate (and even reverse) the impact because of strategic substitutabilities.

## 5.4 Information interventions

We now turn to explore how policymakers could influence managers' anticipations of carbon pricing with information interventions. We examine two alternative policies: one targeting firm-specific private signals, and the other targeting public priors. The first policy consists of providing firms with a technology such that they can perfectly observe  $\sigma_{in}$ . Uncertainty on the first-order beliefs about  $\sigma_{in}$ , thus, disappears. We refer to this policy as "capacity building" as the firm has all the resources to perfectly know its own abatement costs. Note that in this scenario, firms would still need to infer the aggregate and sectoral components  $\sigma_n^s$  and  $\sigma^a$  using their sectoral and public information to form higher-order beliefs.

The second policy consists of communicating the estimated  $\sigma^a$  to the firms. We assume that in that case, firms would take the value of  $\sigma^a$  as the true realization, so they would not have any uncertainty on  $\sigma^a$  anymore. However, they would still need to infer their sectoral

and individual elasticity components  $\sigma_n^s$  and  $\sigma_{in}^f$  using their sectoral and individual signals. We refer to this policy as “public communication”. We next show how these policies affect expectations about the effects of carbon pricing and whether those effects are economically relevant.

The results are shown in [Table 10](#). We report the resulting mean and variance of  $q_{in}$ , and the proportion of firms with a large negative impact, along with the decomposition of the average output into the contributions of the fundamental, first-order, and higher-order errors for the baseline, the two policy scenarios, and their combination.

Under capacity building, the small first-order error on  $\sigma_{in}$  disappears, as a result, the average output improves, but only slightly. The variance of  $q_{in}$  increases slightly because it now better reflects the true individual fundamentals. Overall, the effects are quite small, capturing the fact that firms are already well informed on  $\sigma_{in}$ .

Under a public communication intervention, firms’ assessments of their own elasticity (their first-order beliefs) are not impacted, but it does alter their assessment of the sectoral and aggregate impacts (their higher-order beliefs). This intervention essentially corrects errors in higher-order beliefs, which, because of strategic substitutability, results in a more negative outlook on average. As a result, this shifts the anticipation about aggregate output change from -1.24% to -1.37%.<sup>15</sup> This may seem paradoxical. The planner, by communicating good news about carbon pricing: namely, that it is less impactful than what was reflected by public priors, induce anticipations that are less favorable to carbon pricing. This could notably contribute to reducing political support for carbon pricing.

Overall, the policy scenarios suggest that interventions targeting errors in public priors could have a large impact, unlike interventions targeting firms’ private signals. But surprisingly, a public communication campaign that shared good news about carbon pricing—i.e., the possibility of lower-than-expected economic costs—might have an unexpected effect: it could lower support for the policy. This result shows the importance of accounting for the interaction between strategic motives and anticipation errors when studying beliefs about carbon pricing. Ultimately, both must be accounted for to determine how managers would view such a policy reform.

## 6 Conclusion

In this paper, we propose a new survey-based methodology to recover firm-level primitives governing strategic interactions and noisy beliefs in response to a policy reform. We apply this method to the context of carbon pricing reform in Switzerland. By presenting

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<sup>15</sup>Note that the variance of  $q_{in}$  increases. This is because, thanks to better information on  $\sigma^a$ , private signals are better indicators of  $\sigma_n^s$  and  $\sigma_{in}^f$ , leading firms to rely more on their private information.

firm managers with scenarios that vary the scope of a carbon, we can jointly identify cost semi-elasticities, strategic interaction parameters, and higher-order beliefs without imposing strong structural assumptions. The approach is portable to any policy shock with a natural scope dimension. Three main findings emerge. First, managers expect moderate but meaningful output costs due to carbon pricing: a CHF 100/ton nationwide carbon tax increase is associated with an expected 1.18% decline in aggregate output over three years, consistent with existing estimates from observational studies. Second, we provide the first direct evidence that quantity setting features strategic substitutabilities, especially at the sectoral level. Managers who expect competitors to contract less reduce their own planned response, consistent with competitive interactions within sectors. Third, firms hold overly pessimistic beliefs about the economy-wide impact of carbon pricing—roughly half of the public prior on the aggregate elasticity reflects noise rather than fundamentals. Correcting this excess pessimism through public communication does not, however, translate into a more optimistic outlook: because output responses are strategic substitutes, telling firms that the aggregate impact is milder than they feared makes them more optimistic about the economy but more pessimistic about their own output, deepening the expected output decline from 1.24% to 1.37%.

Our approach and results have important policy implications, especially for the design of carbon tax reforms that aim to minimize opposition by firms. They caution against a natural presumption: that correcting firms’ overly pessimistic beliefs about the costs of carbon pricing would soften their resistance to it. Because quantity responses are strategic substitutes, a public communication campaign conveying that economy-wide costs are smaller than firms fear can make firms more pessimistic about their own prospects, and hence potentially more, rather than less, opposed. Whether belief correction raises or lowers firms’ support thus hinges on the sign and strength of strategic interactions, which our framework is designed to measure. Future work could extend the framework to study optimal differentiated carbon taxes that balance belief correction against misallocation costs, introduce information treatments to causally test the role of aggregate beliefs, or apply the methodology to other policy contexts where strategic interactions determine firms’ expectations.

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## 7 Tables

Table 1: Summary Statistics – Analysis Sample

|                                                           | N     | Mean | SD    | p10  | p50  | p90  |
|-----------------------------------------------------------|-------|------|-------|------|------|------|
| <i>Panel A: Firm characteristics</i>                      |       |      |       |      |      |      |
| Employees (FTE)                                           | 1,017 | 78   | 245   | 2    | 19   | 166  |
| Turnover (million CHF)                                    | 1,017 | 44.9 | 352.5 | 0.4  | 4.8  | 50.0 |
| Operating expenses (million CHF)                          | 1,017 | 27.1 | 258.6 | 0.3  | 3.9  | 37.5 |
| Export share (%)                                          | 1,017 | 12.2 | 25.7  | 0.0  | 0.0  | 55.0 |
| Competition intensity (1–7)                               | 1,016 | 5.16 | 1.45  | 3.00 | 5.00 | 7.00 |
| Energy share in total costs (%)                           | 1,017 | 4.5  | 7.0   | 0.5  | 2.0  | 10.0 |
| Estimated CO <sub>2</sub> emissions (tons)                | 1,017 | 1139 | 12818 | 3    | 54   | 935  |
| Ideal CO <sub>2</sub> price for all (CHF/ton)             | 996   | 156  | 343   | 0    | 100  | 250  |
| Ideal CO <sub>2</sub> price if exempted (CHF/ton)         | 984   | 164  | 404   | 0    | 120  | 300  |
| <i>Panel B: Binary firm characteristics</i>               |       |      |       |      |      |      |
| Only/mostly CH competition (0/1)                          | 1,017 | 0.43 |       |      |      |      |
| Uses electricity (0/1)                                    | 1,017 | 0.95 |       |      |      |      |
| Uses fossil fuel (0/1)                                    | 1,017 | 0.71 |       |      |      |      |
| Uses other energy sources (0/1)                           | 1,017 | 0.13 |       |      |      |      |
| Own CO <sub>2</sub> less than other firms in sector (0/1) | 1,017 | 0.96 |       |      |      |      |
| Own sector CO <sub>2</sub> less than other sectors (0/1)  | 1,017 | 0.86 |       |      |      |      |
| Company subject to CO <sub>2</sub> tax (0/1)              | 1,017 | 0.85 |       |      |      |      |
| Company part of CH ETS (0/1)                              | 1,017 | 0.02 |       |      |      |      |
| Company has made formal commitment (0/1)                  | 1,017 | 0.09 |       |      |      |      |
| <i>Panel C: Manager characteristics</i>                   |       |      |       |      |      |      |
| CEO, CFO (0/1)                                            | 1,017 | 0.77 |       |      |      |      |
| Male (0/1)                                                | 1,017 | 0.88 |       |      |      |      |
| Bachelor’s degree or higher (0/1)                         | 1,017 | 0.32 |       |      |      |      |
| Years at company                                          | 1,003 | 16.9 | 11.5  | 3.0  | 15.0 | 34.0 |
| Survey duration (minutes)                                 | 1,014 | 39.5 | 196.8 | 9.3  | 16.2 | 40.5 |

*Notes:* Entries are means, standard deviations, and percentiles for the analysis sample ( $N = 1017$ ). For binary (0/1) variables we report only the share.

Table 2: Firm Responses to a National Carbon Tax Increase

|                                 | All   | Tax increase (CHF/Ton CO <sub>2</sub> ) |       |                    | S.E. |
|---------------------------------|-------|-----------------------------------------|-------|--------------------|------|
|                                 |       | 100                                     | 300   | $\Delta_{100-300}$ |      |
| $\Delta$ Energy costs (%)       | 15.46 | 8.06                                    | 24.17 | -16.11             | 0.84 |
| $\Delta$ Real output (%)        | -1.97 | -1.44                                   | -2.61 | 1.18               | 0.51 |
| $\Delta$ Sales prices (%)       | 2.74  | 2.51                                    | 3.02  | -0.51              | 0.42 |
| $\Delta$ Energy consumption (%) | -1.48 | -1.16                                   | -1.87 | 0.72               | 0.44 |
| $\Delta$ Employment (%)         | -1.44 | -0.98                                   | -2.00 | 1.03               | 0.41 |
| $\Delta$ Profits (%)            | -3.04 | -2.51                                   | -3.67 | 1.16               | 0.56 |
| Observations                    | 1,017 | 508                                     | 509   | 1,017              | .    |

*Notes:* The table reports managers' expectations of the effects of an increase in the national carbon tax on outcomes for their own firms. Entries are weighted mean expected percentage changes across responding managers, using post-stratification weights constructed as described in Appendix H. Results are shown for the full sample and separately for managers randomly assigned to carbon tax increases of CHF 100 and CHF 300 per ton of CO<sub>2</sub>.  $\Delta_{100-300}$  reports the difference between the two treatment groups, with corresponding standard errors.

Table 3: Firms' Own Output Responses and Beliefs About Others' Responses

| <b>Panel A: Own output across tax scenarios</b>                    |                    |                    |                    |                             |                           |
|--------------------------------------------------------------------|--------------------|--------------------|--------------------|-----------------------------|---------------------------|
|                                                                    | National<br>tax    | Sectoral<br>tax    | Firm<br>tax        | Difference:<br>Nat – Sector | Difference:<br>Nat – Firm |
| $\Delta$ Real output (%)                                           | -1.18<br>(0.19)    | -1.54<br>(0.20)    | -2.68<br>(0.25)    | 0.37<br>(0.27)              | 1.50<br>(0.31)            |
| Share < 0                                                          | 0.49               | 0.49               | 0.52               | .                           | .                         |
| Share = 0                                                          | 0.37               | 0.39               | 0.37               | .                           | .                         |
| <b>Panel B: Own output vs. beliefs about others (national tax)</b> |                    |                    |                    |                             |                           |
|                                                                    | Own firm<br>impact | Sectoral<br>impact | National<br>impact | Difference:<br>Sector – Own | Difference:<br>Nat – Own  |
| $\Delta$ Real output (%)                                           | -1.18<br>(0.19)    | -1.29<br>(0.19)    | -1.82<br>(0.17)    | 0.12<br>(0.26)              | 0.64<br>(0.25)            |
| Share < 0                                                          | 0.49               | 0.50               | 0.62               | .                           | .                         |
| Share = 0                                                          | 0.37               | 0.35               | 0.26               | .                           | .                         |

*Notes:* Panel A reports managers' expected effects of three carbon tax scenarios—a national tax increase, a sector-specific tax increase, and a firm-specific tax increase—on real output of their own firm. Tax increases are normalized to CHF 100 per ton of CO<sub>2</sub>. Panel B compares managers' expected effects of the national carbon tax increase on real output of their own firm, their own sector, and the national economy. Entries in rows labeled  $\Delta$  Real output (%) are weighted mean expected percentage changes across respondents, using post-stratification weights constructed as described in Appendix H. Standard errors are shown in parentheses below each estimate. Rows labeled “Share < 0” and “Share = 0” report the fractions of respondents expecting negative and zero output effects, respectively. All values are for the main sample (N=1017).

Table 4: Policy Elasticity, Strategic Interactions, and Higher-order Errors

|                                                                        | Mean  | S.E. | Share < 0 | Share > 0 | # Obs |
|------------------------------------------------------------------------|-------|------|-----------|-----------|-------|
| <b>Panel A: Policy elasticity and strategic interactions</b>           |       |      |           |           |       |
| Policy elasticity $E_{in}(\sigma_{in})$                                | 2.68  | 0.25 | 0.11      | 0.52      | 1,017 |
| Sectoral-level strategic $\alpha_{in}^s$                               | -0.46 | 0.10 | 0.45      | 0.20      | 645   |
| National-level strategic $\alpha_{in}^a$                               | -0.25 | 0.13 | 0.42      | 0.20      | 609   |
| <b>Panel B: Errors in beliefs about national and sectoral outcomes</b> |       |      |           |           |       |
| National error with national tax                                       | -0.64 | 0.17 | 0.44      | 0.56      | 1,017 |
| Sectoral error with national tax                                       | -0.12 | 0.18 | 0.39      | 0.61      | 1,017 |
| Sectoral error with sectoral tax                                       | -0.03 | 0.23 | 0.34      | 0.66      | 1,017 |

*Notes:* Panel A reports estimates of the policy elasticity and strategic interaction parameters for respondents with non-missing values for each parameter. Panel B reports national and sectoral higher-order errors, defined as the difference between managers' expectations of economy-wide or sectoral output effects and the corresponding effects implied by aggregating respondents' expectations for their own firms. Entries report weighted means computed using post-stratification weights constructed described in Appendix H, along with standard errors and shares of negative and positive values.

Table 5: Heterogeneity: Policy Elasticity and Strategic Interaction Parameters

| Dependent Variable                 | $\sigma_{in}$<br>(I) | $\mathbb{1}\{\sigma_{in} > 0\}$<br>(II) | $\alpha_{in}^s$<br>(III) | $\mathbb{1}\{\alpha_{in}^s < 0\}$<br>(IV) | $\alpha_{in}^a$<br>(V) | $\mathbb{1}\{\alpha_{in}^a < 0\}$<br>(VI) |
|------------------------------------|----------------------|-----------------------------------------|--------------------------|-------------------------------------------|------------------------|-------------------------------------------|
| Intense compet. (1:low-7:high)     | 0.534***<br>(0.140)  | 0.034**<br>(0.013)                      | -0.075<br>(0.064)        | 0.030*<br>(0.018)                         | -0.071<br>(0.094)      | 0.015<br>(0.022)                          |
| Only Swiss compet. (bin.)          | -1.267**<br>(0.606)  | -0.103**<br>(0.044)                     | 0.124<br>(0.269)         | -0.056<br>(0.059)                         | 0.802**<br>(0.324)     | -0.028<br>(0.063)                         |
| Export share (%)                   | -0.629<br>(1.044)    | -0.054<br>(0.086)                       | -0.047<br>(0.657)        | -0.133<br>(0.114)                         | 0.760**<br>(0.372)     | -0.211**<br>(0.107)                       |
| CO2 emissions (log) (Mt)           | 0.655***<br>(0.213)  | 0.068***<br>(0.017)                     | -0.100<br>(0.068)        | 0.059***<br>(0.019)                       | 0.071<br>(0.085)       | 0.005<br>(0.025)                          |
| Ideal carbon price (,000 CHF/tCO2) | -0.585<br>(0.402)    | -0.056<br>(0.070)                       | -0.427<br>(0.353)        | 0.152***<br>(0.045)                       | 0.019<br>(0.334)       | 0.085*<br>(0.050)                         |
| FT employees (log) (,000)          | -0.174<br>(0.292)    | -0.046*<br>(0.024)                      | 0.088<br>(0.083)         | -0.071***<br>(0.026)                      | -0.004<br>(0.108)      | -0.053*<br>(0.031)                        |
| Observations                       | 970                  | 970                                     | 621                      | 621                                       | 588                    | 588                                       |
| Adj. R-squared                     | 0.138                | 0.131                                   | 0.020                    | 0.066                                     | 0.032                  | 0.023                                     |

*Notes:* The table reports coefficient estimates from linear regressions estimated with observations weighted by post-stratification weights constructed as described in Appendix H. The intensity of competition indicator variable is measured with a 1 to 7 scale, where 1 corresponds to the lowest level of intra-sector competition perceived by the manager. The regression has also a dummy variable taking a value of one if the firm has only Swiss competitors and zero otherwise, and export share. The variable ideal carbon price is the value of a nationwide carbon price a manager would recommend for the Swiss economy. Firm size is captured by the number of full-time employees (in log). All specifications also include a constant and additional controls for assignment to the CHF 300 treatment group, turnover, manufacturing sector, use of fossil fuels and electricity, survey duration, and manager characteristics, including indicators for being a CEO, CFO, or director, years in the current role, gender, and highest educational attainment. All estimated coefficients for these additional controls can be found in Section I.2. Robust standard errors are reported in parentheses.

\* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

Table 6: Heterogeneity: National and Sectoral Errors

| Dependent Variable                 | National Error<br>(I) | $\mathbb{1}\{\text{Nat. Error} < 0\}$<br>(II) | Sectoral Error<br>(III) | $\mathbb{1}\{\text{Sect. Error} < 0\}$<br>(IV) |
|------------------------------------|-----------------------|-----------------------------------------------|-------------------------|------------------------------------------------|
| Intense compet. (1:low-7:high)     | -0.220**<br>(0.110)   | 0.041***<br>(0.013)                           | -0.233**<br>(0.110)     | 0.027*<br>(0.014)                              |
| Only Swiss compet. (bin.)          | 0.032<br>(0.445)      | 0.007<br>(0.047)                              | 0.320<br>(0.489)        | -0.071<br>(0.047)                              |
| Export share (%)                   | 0.190<br>(0.709)      | 0.012<br>(0.088)                              | 0.706<br>(0.773)        | -0.058<br>(0.083)                              |
| CO2 emissions (log) (Mt)           | -0.417***<br>(0.138)  | 0.057***<br>(0.018)                           | -0.198<br>(0.136)       | 0.033*<br>(0.017)                              |
| Ideal carbon price (,000 CHF/tCO2) | 0.640<br>(0.475)      | -0.078<br>(0.089)                             | 0.949***<br>(0.364)     | -0.116***<br>(0.038)                           |
| FT employees (log) (,000)          | 0.352<br>(0.217)      | -0.050**<br>(0.024)                           | 0.273<br>(0.226)        | -0.037<br>(0.024)                              |
| Observations                       | 970                   | 970                                           | 970                     | 970                                            |
| Adj. R-squared                     | 0.036                 | 0.065                                         | 0.042                   | 0.076                                          |

*Notes:* The table reports coefficient estimates from linear regressions estimated with observations weighted by post-stratification weights constructed as described in Appendix H. The dependent variables are managers' higher-order errors regarding national and sectoral outcomes under the national tax scenario and indicators equal to one if the corresponding higher-order errors are negative. The intensity of competition indicator variable is measured with a 1 to 7 scale, where 1 corresponds to the lowest level of intra-sector competition perceived by the manager. The regression has also a dummy variable taking a value of one if the firm has only Swiss competitors and zero otherwise, and export share. The variable ideal carbon price is the value of a nationwide carbon price a manager would recommend for the Swiss economy. Firm size is captured by the number of full-time employees (in log). All specifications include a constant and additional controls for assignment to the CHF 300 treatment group, manufacturing sector, use of fossil fuels and electricity, survey duration, and manager characteristics, including indicators for being a CEO, CFO, or director, years in the current role, gender, and highest educational attainment. All estimated coefficients for these additional controls can be found in Section I.2. Robust standard errors are reported in parentheses. \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

Table 7: Fitted Moments: Data and Model Comparison

|                                                 | Model | Data  |
|-------------------------------------------------|-------|-------|
| <b>Panel A: Targeted moments</b>                |       |       |
| $q^{sc.1}$                                      | -1.24 | -1.18 |
| $(\bar{E}(q) - q)^{sc.1}$                       | -0.69 | -0.64 |
| $(\bar{E}(q_n) - q)^{sc.1}$                     | -0.20 | -0.12 |
| $q^{sc.3}$                                      | -2.51 | -2.68 |
| $\text{Var}_n(E_{in}(q)^{sc.1})$                | 0.66  | 0.66  |
| $\text{Cov}_n(E_{in}(q)^{sc.1}, q_{in}^{sc.3})$ | 0.88  | 0.88  |
| $\text{Var}_n(q_{in}^{sc.3})$                   | 2.07  | 2.07  |
| <b>Panel B: Untargeted moments</b>              |       |       |
| $q^{sc.2}$                                      | -1.81 | -1.54 |
| $(\bar{E}(q_n) - q)^{sc.2}$                     | -0.21 | -0.03 |

*Notes:* The table reports empirical and model-implied moments. Model moments are computed by simulating the model at the estimated parameter values. Parameters are estimated by minimizing the weighted sum of squared deviations between model-implied and empirical moments. All moments receive unit weight except  $q^{sc.1}$ , which receives a weight of five. Untargeted moments are not included in the estimation objective and are reported to assess model fit.

Table 8: Parameter Estimates

| $\sigma^a$ | $e^a$ | $\pi^a$ | $\pi^s$ | $\pi^f$ | $\kappa^s$ | $\kappa^f$ |
|------------|-------|---------|---------|---------|------------|------------|
| 2.45       | 2.17  | 0.13    | 0.50    | 0.19    | 0.17       | 0.05       |

*Notes:* The table reports estimates of the structural parameters of the model. Parameters are estimated by simulated method of moments, minimizing the weighted sum of squared deviations between model-implied and empirical moments.

Table 9: Public vs. Private Information

|                                                         | Baseline | Public & sectoral info only | Public info only |
|---------------------------------------------------------|----------|-----------------------------|------------------|
| <b>Panel A: Aggregate real output effect (in %)</b>     |          |                             |                  |
| $q$                                                     | -1.24    | -1.44                       | -2.59            |
| <b>Panel B: Components of <math>q</math></b>            |          |                             |                  |
| $\bar{E}(q) - q$                                        | -0.69    | -0.59                       | -0.00            |
| $\bar{E}(q_n) - q$                                      | -0.20    | -0.00                       | -0.00            |
| $\bar{E}(\sigma_{in}) - \sigma^a$                       | 0.06     | 0.28                        | 2.17             |
| $\sigma^a$                                              | 2.45     | 2.45                        | 2.45             |
| <b>Panel C: Contributions to <math>q</math> (in pp)</b> |          |                             |                  |
| $\bar{E}(q) - q$                                        | 0.11     | 0.09                        | 0.00             |
| $\bar{E}(q_n) - q$                                      | 0.06     | 0.00                        | 0.00             |
| $\bar{E}(\sigma_{in}) - \sigma^a$                       | -0.03    | -0.16                       | -1.22            |
| $\sigma^a$                                              | -1.37    | -1.37                       | -1.37            |

*Notes:* The table reports a decomposition of the average real output response to the national tax scenario under alternative information structures. The baseline corresponds to the estimated model. The second column removes firm-specific signals, while the third column removes both firm-specific and sector-specific signals. Panel A reports the aggregate real output response  $q$  to the national tax scenario (in percent). Panel B reports the components of  $q$  in Equation (20). Panel C reports contributions of each component to  $q$  in percentage points (pp).

Table 10: Policy Counterfactuals

|                                                         | Baseline | Capacity building | Public info | Both  |
|---------------------------------------------------------|----------|-------------------|-------------|-------|
| <b>Panel A: Aggregate outcomes</b>                      |          |                   |             |       |
| $q$                                                     | -1.24    | -1.19             | -1.37       | -1.37 |
| $\text{Var}(q_{in})$                                    | 17.23    | 20.61             | 18.05       | 21.80 |
| Share $q_{in} < -3$                                     | 0.34     | 0.35              | 0.35        | 0.36  |
| <b>Panel B: Components of <math>q</math></b>            |          |                   |             |       |
| $\bar{E}(q) - q$                                        | -0.69    | -0.71             | 0.00        | -0.00 |
| $\bar{E}(q_n) - q$                                      | -0.20    | -0.25             | 0.00        | -0.00 |
| $\bar{E}(\sigma_{in}) - \sigma^a$                       | 0.06     | 0.00              | 0.00        | 0.00  |
| $\sigma^a$                                              | 2.45     | 2.45              | 2.45        | 2.45  |
| <b>Panel C: Contributions to <math>q</math> (in pp)</b> |          |                   |             |       |
| $\bar{E}(q) - q$                                        | 0.11     | 0.11              | -0.00       | 0.00  |
| $\bar{E}(q_n) - q$                                      | 0.06     | 0.07              | -0.00       | 0.00  |
| $\bar{E}(\sigma_{in}) - \sigma^a$                       | -0.03    | -0.00             | -0.00       | -0.00 |
| $\sigma^a$                                              | -1.37    | -1.37             | -1.37       | -1.37 |

*Notes:* The table reports policy counterfactuals for the real output response to the national tax scenario based on the estimated model. The baseline corresponds to the estimated economy. “Capacity building” eliminates errors in managers’ assessments of their own elasticity (first-order beliefs), while “Public info” eliminates errors in their assessments of sectoral and aggregate outcomes (higher-order beliefs). “Both” combines both interventions. Panel A reports aggregate outcomes. Panel B reports the components of the aggregate output response  $q$  in Equation (20). Panel C reports contributions of each component to  $q$  in percentage points (pp).

## 8 Figures

Figure 1: Carbon Tax Cost Estimate

Given how much you reported spending on different energy sources, we estimate that your CO2 emissions associated with the following energy sources, for the year 2024, were:

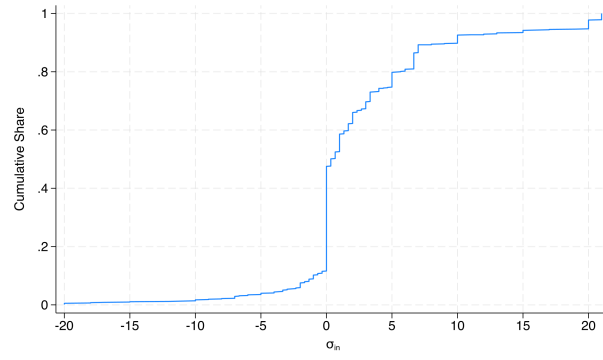
|                      |           |             |
|----------------------|-----------|-------------|
| Electricity          | 133.333   | tons of CO2 |
| Fossile fuel         | 560.000   | tons of CO2 |
| Other energy sources | 632.000   | tons of CO2 |
| Total                | 1'325.333 | tons of CO2 |

**We estimate that an additional CO2 tax of 100 CHF per ton of CO2 would then increase your annual after-tax energy costs by 132'500 CHF.**

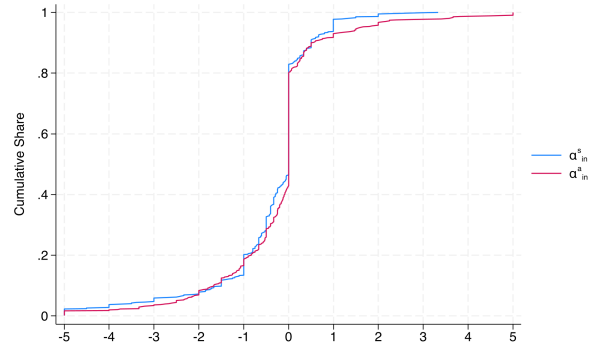
Of course, if you change your energy consumption or energy mix, the cost impact would be different.

*Notes:* The values shown are illustrative and correspond to reported operating expenditures of CHF 10,000,000, an energy expenditure share of 10%, and energy source shares of 40% electricity, 40% fossil fuels, and 20% other sources.

Figure 2: Parameter Distributions



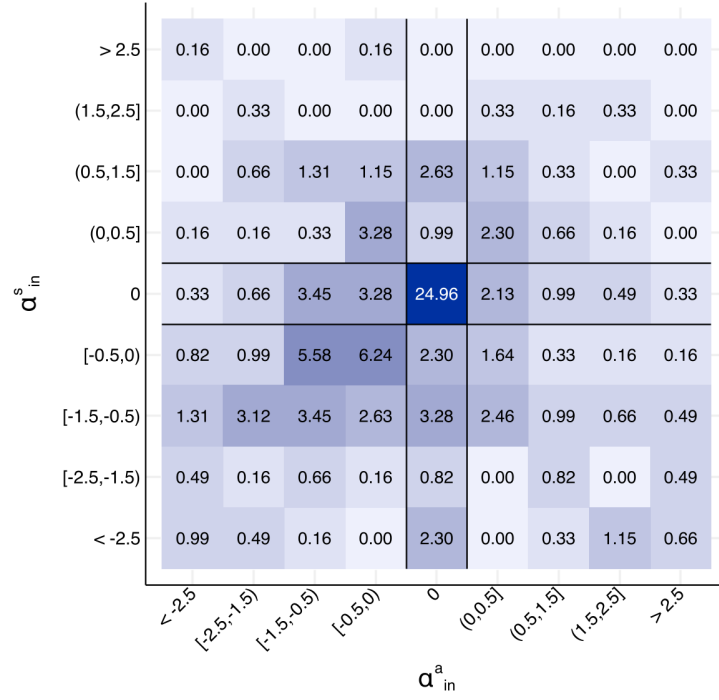
(a) Policy elasticity  $\sigma_{in}$



(b) Sector-level strategic interaction  $\alpha_{in}^s$  versus national-level strategic interaction  $\alpha_{in}^a$

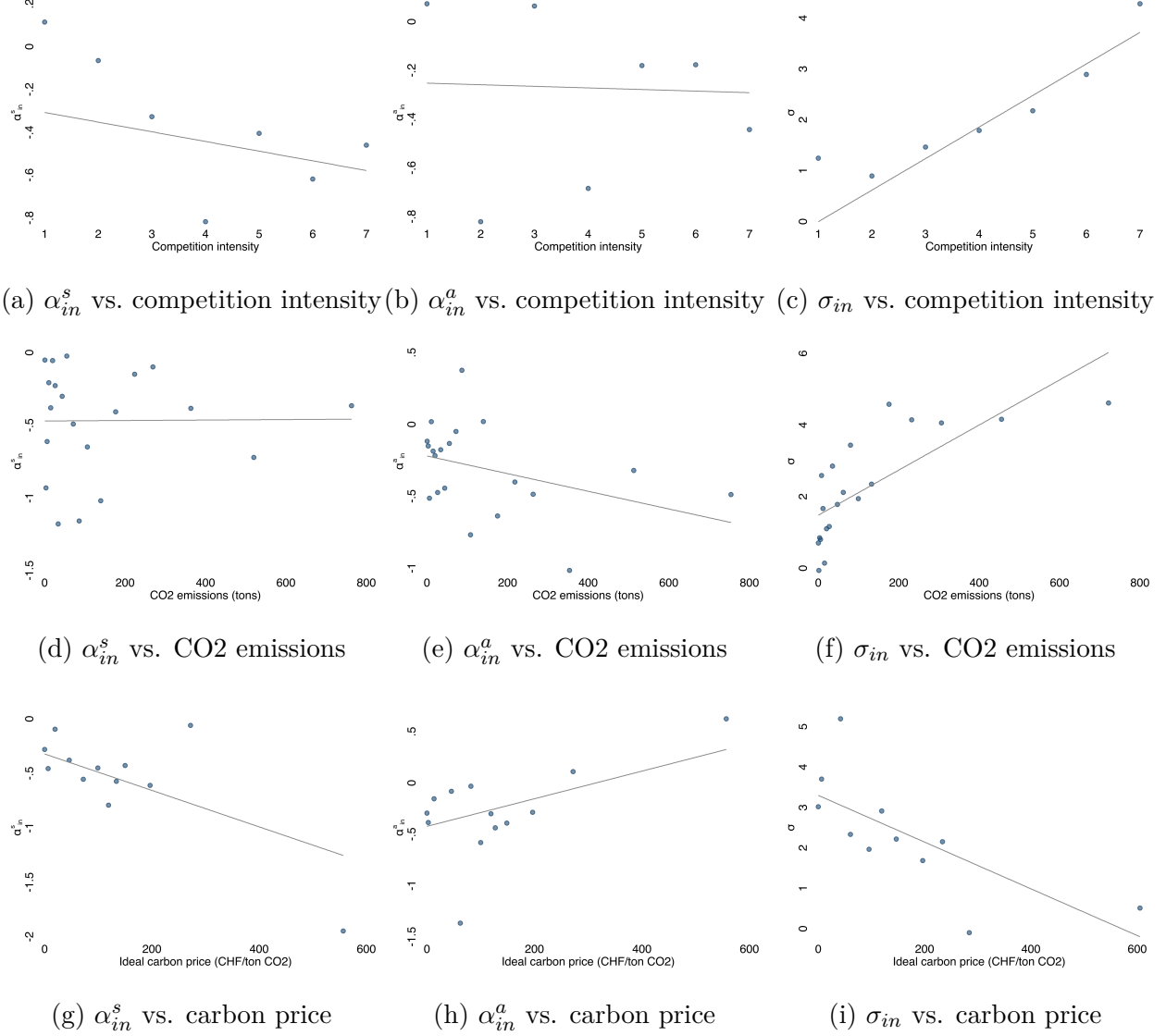
*Notes:* Cumulative density estimates of structural parameters across firms.

Figure 3: Relationship Between Sector- and National-Level Strategic Interaction



*Notes:* Discrete joint density plot of sector-level strategic interaction parameters ( $\alpha_{in}^s$ ) against national-level interaction parameters ( $\alpha_{in}^a$ ). Each square corresponds to a density point.

Figure 4: Correlation with competition intensity, CO<sub>2</sub> emissions, ideal carbon price



*Notes:* Binscatter plots of sector-level strategic interaction parameters ( $\alpha_{in}^s$ ), national-level strategic interaction parameters ( $\alpha_{in}^a$ ), and policy elasticity parameter ( $\sigma_{in}$ ) against measures of market and input-cost conditions. The top row relates these parameters to perceived competition intensity in the firm's sector (1–7 scale; 1 = very low competition, 7 = very intense competition). The second row relates to the level of CO<sub>2</sub> emissions (tons of CO<sub>2</sub>) estimated for each firm. The bottom row relates to the ideal carbon price each manager reported. Lines indicate OLS fits.

Strategic Managers with Noisy Beliefs:  
Evidence from Carbon Pricing

– Online Appendix –

K. Benhima, A. Fuster, S. Houde, and H. Mikosch

# A Proofs

## A.1 Proof of Proposition 1

To prove Proposition 1, we first characterize expectation formation by firms under the general information structure described in Assumption 1, then turn to the proof itself.

### A.1.1 Expectation formation under general information structure

Consider that Assumption 1 holds. We define the information vector

$$S_{in} = \begin{pmatrix} \Sigma_{in} \\ s_{in} \end{pmatrix}$$

as the vector of prior and signals.  $S_{in}$  has  $3 + m$  elements. Define

$$u_{in} = \Sigma_{in} - \Sigma_{in}^0$$

as the prior error. Define the vectors of aggregate, sector-specific, and firm-specific errors as

$$U_{in} = \begin{pmatrix} u \\ u_n - u \\ u_{in} - u_n \end{pmatrix}, \quad V_{in} = \begin{pmatrix} v \\ v_n - v \\ v_{in} - v_n \end{pmatrix}$$

where  $(u, v)$  and  $(u_n, v_n)$  are the economy-wide and sectoral averages of  $(u_{in}, v_{in})$ .  $U_{in}$  and  $V_{in}$  have respectively 9 and  $3m$  elements. Both  $u_{in}$  and  $v_{in}$  can be decomposed into their aggregate, sectoral, and idiosyncratic components:  $u_{in} = u + (u_n - u) + (u_{in} - u_n)$  and  $v_{in} = v + (v_n - v) + (v_{in} - v_n)$ . We can write this decomposition as follows:

$$u_{in} = \begin{pmatrix} Id_3 & Id_3 & Id_3 \end{pmatrix} U_{in} \quad v_{in} = \begin{pmatrix} Id_m & Id_m & Id_m \end{pmatrix} V_{in} \quad (29)$$

Define  $P_0^U$  and  $P_0^V$  as  $9 \times 9$  and  $3m \times 3m$  matrices such that  $U_{in} \sim \mathcal{N}(0, P_0^U)$  and  $V_{in} \sim \mathcal{N}(0, P_0^V)$ .  $P_0^U$  and  $P_0^V$  may have zero elements in the diagonal if the aggregate, sectoral or idiosyncratic elements of  $u_{in}$  and  $v_{in}$  are irrelevant. We then define

$$\xi_{in} = \begin{pmatrix} \Sigma_{in} \\ U_{in} \\ V_{in} \end{pmatrix}$$

$\xi_{in}$  has  $12 + 3m$  elements.

The following lemma summarizes the firms' expectation formation:

**Lemma 1** (Expectations). *Under Assumption 1, the rational expectation of  $\xi_{in}$  by firm  $i$  of sector  $n$  conditional on  $S_{in} = (\Sigma_{in}^0, s_{in}')'$  is given by*

$$E_{in}(\xi_{in}) = KS_{in}$$

where  $K$  is a  $(12 + 3m) \times (3 + m)$  matrix:

$$K = \begin{pmatrix} Id_3 - kh & k \\ k_U kh & -k_U k \\ -k_V h(Id_3 - kh) & k_V (Id_m - hk) \end{pmatrix} \quad (30)$$

with  $k = P_0 h' (h P_0 h' + R)^{-1}$ ,  $k_U = P_0^U \begin{pmatrix} Id_3 \\ Id_3 \\ Id_3 \end{pmatrix} P_0^{-1}$  and  $k_V = P_0^V \begin{pmatrix} Id_m \\ Id_m \\ Id_m \end{pmatrix} R^{-1}$ , where  $Id_3$  is the  $3 \times 3$  identity matrix and  $Id_m$  is the  $m \times m$  identity matrix.

*Proof.* We are looking for the firm's posterior expectation about the components of the elasticity  $\Sigma_{in} = (\sigma^a, \sigma_n^s, \sigma_{in}^f)'$ , given the prior mean  $\Sigma_{in}^0$ , prior variance  $P_0$  and conditional on the signals  $s_{in}$  under Assumption 1.

The measurement equation is

$$s_{in} = h \Sigma_{in} + v_{in}$$

where  $h$  is a  $m \times 3$  matrix and  $v_{in}$  is a vector of  $m$  noise shocks:  $v_{in} \sim \mathcal{N}(0, R)$ , with  $R$  a  $m^2$  positive diagonal matrix.

We use the Kalman gain formula:

$$\begin{aligned} E_{in}(\Sigma_{in}) &= E(\Sigma_{in} | \Sigma_{in}^0, s_{in}) = \Sigma_{in}^0 + k (s_{in} - E(s_{in} | \Sigma_{in}^0)) \\ &= \Sigma_{in}^0 + k (s_{in} - h \Sigma_{in}^0) = (Id_3 - kh) \Sigma_{in}^0 + k s_{in} \\ &= \begin{pmatrix} Id_3 - kh & k \end{pmatrix} \begin{pmatrix} \Sigma_{in}^0 \\ s_{in} \end{pmatrix} = \begin{pmatrix} Id_3 - kh & k \end{pmatrix} S_{in} \end{aligned} \quad (31)$$

where  $k$  is defined in the lemma and is of dimension  $3 \times m$ .

Then, note that

$$\begin{aligned} E_{in}(v_{in}) &= E_{in}(s_{in} - h \Sigma_{in}) = s_{in} - h E_{in}(\Sigma_{in}) \\ &= \begin{pmatrix} -h(Id_3 - kh) & Id_m - hk \end{pmatrix} S_{in} \end{aligned} \quad (32)$$

and

$$\begin{aligned} E_{in}(u_{in}) &= E_{in}(\Sigma_{in}^0 - \Sigma_{in}) = \Sigma_{in}^0 - E_{in}(\Sigma_{in}) \\ &= \begin{pmatrix} kh & -k \end{pmatrix} S_{in} \end{aligned} \quad (33)$$

The measurement equations for  $u_{in}$  and  $v_{in}$  are (29) with  $U_{in} \sim \mathcal{N}(0, P_0^U)$  and  $V_{in} \sim \mathcal{N}(0, P_0^V)$  the prior distributions. Then, conditional on  $u_{in}$  and  $v_{in}$ , the Kalman gain formulas are

$$\begin{aligned} E(U_{in}|u_{in}) &= k_U u_{in} \\ E(V_{in}|v_{in}) &= k_V v_{in} \end{aligned}$$

with

$$\begin{aligned} k_U &= P_0^U \begin{pmatrix} Id_3 \\ Id_3 \\ Id_3 \end{pmatrix} \left( \begin{pmatrix} Id_3 & Id_3 & Id_3 \end{pmatrix} P_0^U \begin{pmatrix} Id_3 \\ Id_3 \\ Id_3 \end{pmatrix} \right)^{-1} \\ k_V &= P_0^V \begin{pmatrix} Id_m \\ Id_m \\ Id_m \end{pmatrix} \left( \begin{pmatrix} Id_m & Id_m & Id_m \end{pmatrix} P_0^V \begin{pmatrix} Id_m \\ Id_m \\ Id_m \end{pmatrix} \right)^{-1} \end{aligned}$$

$k_U$  is a  $9 \times 3$  matrix and  $k_V$  is a  $3m \times m$  matrix.

Now, note that, according to (29),

$$\begin{aligned} P_0 &= \begin{pmatrix} Id_3 & Id_3 & Id_3 \end{pmatrix} P_0^U \begin{pmatrix} Id_3 \\ Id_3 \\ Id_3 \end{pmatrix} \\ R &= \begin{pmatrix} Id_m & Id_m & Id_m \end{pmatrix} P_0^V \begin{pmatrix} Id_m \\ Id_m \\ Id_m \end{pmatrix} \end{aligned}$$

hence the expressions for  $k_U$  and  $k_V$ .

Conditional on the firm's information, this yields

$$E_{in}(U_{in}) = k_U E_{in}(u_{in}) \tag{34}$$

$$E_{in}(V_{in}) = k_V E_{in}(v_{in}) \tag{35}$$

Combining (31)-(35), we can then derive the full solution for  $K$  in Lemma 1:

$$E_{in}(\xi_{in}) = E_{in} \begin{pmatrix} \Sigma_{in} \\ U_{in} \\ V_{in} \end{pmatrix} = \begin{pmatrix} Id_3 - kh & k \\ k_U kh & -k_U k \\ -k_V h(Id_3 - kh) & k_V (Id_m - hk) \end{pmatrix} S_{in}$$

□

### A.1.2 Proof

We guess and verify that the solution is of the form

$$q_{in} = (A_{in}^{(1)} S_{in})\tau + (A_{in}^{(2)} S_{in})(\tau_n - \tau) + (A_{in}^{(3)} S_{in})(\tau_{in} - \tau_n) \quad (36)$$

with  $A_{in}^{(j)} = A^{(j)}(\Theta_{in})$ , for  $j = 1, 2, 3$ .

Define  $A_n^{(1)}$ ,  $A_n^{(2)}$  and  $A_n^{(3)}$  as the sector averages of  $A_{in}^{(1)}$ ,  $A_{in}^{(2)}$  and  $A_{in}^{(3)}$ , and  $A_n^{(1)}$ ,  $A_n^{(2)}$  and  $A^{(3)}$  as the economy-wide averages.

We have defined  $S_{in} = (\Sigma_{in}' s_{in}')'$  as the vector of priors and signals and  $\xi_{in} = (\Sigma_{in}', U_{in}', V_{in}')'$  as the vector of underlying shocks. Note that the vector  $S_{in}$  is a function of the shocks:

$$S_{in} = H\xi_{in} \quad (37)$$

with

$$H = \begin{pmatrix} h & 0_{m \times 3} & 0_{m \times 3} & 0_{m \times 3} & Id_m & Id_m & Id_m \\ Id_3 & Id_3 & Id_3 & Id_3 & 0_{3 \times m} & 0_{3 \times m} & 0_{3 \times m} \end{pmatrix}$$

where  $0_{n \times n'}$  is a  $n \times n'$  matrix with zeros.  $H$  is a  $(m+3) \times (12+3m)$  matrix.

Using Equation (37), our guess can be rewritten as a function of shocks:

$$q_{in} = (A_{in}^{(1)} H\xi_{in})\tau + (A_{in}^{(2)} H\xi_{in})(\tau_n - \tau) + (A_{in}^{(3)} H\xi_{in})(\tau_{in} - \tau_n) \quad (38)$$

Define  $Id$  as the  $(12+3m) \times (12+3m)$  identity matrix, and  $Id^s$  and  $Id^a$  as the identity matrices that select only sectoral and aggregate shocks:

$$Id^s = \begin{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} & 0_{3 \times 3} & 0_{3 \times 3} & 0_{3 \times 3} & 0_{3 \times m} & 0_{3 \times m} & 0_{3 \times m} \\ 0_{3 \times 3} & Id_3 & 0_{3 \times 3} & 0_{3 \times 3} & 0_{3 \times m} & 0_{3 \times m} & 0_{3 \times m} \\ 0_{3 \times 3} & 0_{3 \times 3} & Id_3 & 0_{3 \times 3} & 0_{3 \times m} & 0_{3 \times m} & 0_{3 \times m} \\ 0_{3 \times 3} & 0_{3 \times 3} & 0_{3 \times 3} & 0_{3 \times 3} & 0_{3 \times m} & 0_{3 \times m} & 0_{3 \times m} \\ 0_{m \times 3} & 0_{m \times 3} & 0_{m \times 3} & 0_{m \times 3} & Id_m & 0_{m \times m} & 0_{m \times m} \\ 0_{m \times 3} & 0_{m \times 3} & 0_{m \times 3} & 0_{m \times 3} & 0_{m \times m} & Id_m & 0_{m \times m} \\ 0_{m \times 3} & 0_{m \times 3} & 0_{m \times 3} & 0_{m \times 3} & 0_{m \times m} & 0_{m \times m} & 0_{m \times m} \end{pmatrix}$$

$$Id^a = \begin{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} & 0_{3 \times 3} & 0_{3 \times 3} & 0_{3 \times 3} & 0_{3 \times m} & 0_{3 \times m} & 0_{3 \times m} \\ 0_{3 \times 3} & Id_3 & 0_{3 \times 3} & 0_{3 \times 3} & 0_{3 \times m} & 0_{3 \times m} & 0_{3 \times m} \\ 0_{3 \times 3} & 0_{3 \times 3} & 0_{3 \times 3} & 0_{3 \times 3} & 0_{3 \times m} & 0_{3 \times m} & 0_{3 \times m} \\ 0_{3 \times 3} & 0_{3 \times 3} & 0_{3 \times 3} & 0_{3 \times 3} & 0_{3 \times m} & 0_{3 \times m} & 0_{3 \times m} \\ 0_{m \times 3} & 0_{m \times 3} & 0_{m \times 3} & 0_{m \times 3} & Id_m & 0_{m \times m} & 0_{m \times m} \\ 0_{m \times 3} & 0_{m \times 3} & 0_{m \times 3} & 0_{m \times 3} & 0_{m \times m} & 0_{m \times m} & 0_{m \times m} \\ 0_{m \times 3} & 0_{m \times 3} & 0_{m \times 3} & 0_{m \times 3} & 0_{m \times m} & 0_{m \times m} & 0_{m \times m} \end{pmatrix}$$

Denote  $\xi_n = Id^s \xi_{in}$  and  $\xi = Id^a \xi_{in}$  the vectors of shocks that select only the sectoral and aggregate shocks.

We first assume that there is a large number of firms and sectors:

**Assumption 3** (Large numbers of firms and sectors). *Each sector contains a large number of firms and the economy contains a large number of sectors:  $N^n \rightarrow \infty$  for every sector  $n$  and  $N \rightarrow \infty$ .*

We also assume the following independence assumption:

**Assumption 4** (Independence). *The firm-specific shocks and parameters are independent: for  $j = 1, 2, 3$ , we have*

$$\begin{aligned} \frac{1}{N^n} \sum_i (A_{in}^{(j)} - A_n^{(j)}) H(\xi_{in} - \xi_n) (\tau_{in} - \tau_n) &= 0 \\ \frac{1}{N^n} \sum_i (A_{in}^{(j)} - A_n^{(j)}) H(\xi_{in} - \xi_n) &= 0 \\ \frac{1}{N^n} \sum_i (A_{in}^{(j)} - A_n^{(j)}) (\tau_{in} - \tau_n) &= 0 \\ \frac{1}{N^n} \sum_i (\xi_{in} - \xi_n) (\tau_{in} - \tau_n) &= 0 \end{aligned}$$

*The sector-specific shocks and parameters are independent: for  $j = 1, 2$ , we have*

$$\begin{aligned} \frac{1}{N} \sum_n (A_n^{(j)} - A^{(j)}) H(\xi_n - \xi) (\tau_n - \tau) &= 0 \\ \frac{1}{N} \sum_n (A_n^{(j)} - A^{(j)}) H(\xi_n - \xi) &= 0 \\ \frac{1}{N} \sum_n (A_n^{(j)} - A^{(j)}) (\tau_n - \tau) &= 0 \\ \frac{1}{N} \sum_n (\xi_n - \xi) (\tau_n - \tau) &= 0 \end{aligned}$$

We first establish the following lemma:

**Lemma 2** (Sectoral and Economy-wide averages). *Under Assumption 4,*

$$\begin{aligned} \frac{1}{N^n} \sum_i (A_{in}^{(3)} H \xi_{in}) (\tau_{in} - \tau_n) &= 0 \\ \frac{1}{N^n} \sum_i (A_{in}^{(2)} H \xi_{in}) (\tau_n - \tau) &= (A_n^{(2)} H \xi_n) (\tau_n - \tau) \\ \frac{1}{N^n} \sum_i (A_{in}^{(1)} H \xi_{in}) \tau &= (A_n^{(1)} H \xi_n) \tau \\ \frac{1}{N} \sum_n (A_n^{(2)} H \xi_n) (\tau_n - \tau) &= 0 \\ \frac{1}{N} \sum_n (A_n^{(1)} H \xi_n) \tau &= (A^{(1)} H \xi) \tau \end{aligned}$$

*Proof.* We use the decomposition  $A_{in}^l = (A_{in}^l - A_n^l) + A_n^l$ ,  $A_n^l = (A_n^l - A^l) + A^l$ , for  $l = a, s, f$ ,  $\xi_{in} = (\xi_{in} - \xi_n) + \xi_n$  and  $\xi_n = (\xi_n - \xi) + \xi$ , replace in the above averages, and use Assumptions 3 and 4, as well as the definitions of  $\tau$ ,  $(\tau_n - \tau)$ ,  $(\tau_{in} - \tau_n)$ ,  $A_n^l$  and  $A^l$ , for  $l = a, s, f$ . Assumption 3 ensures that a law of large numbers applies, so that the within-sector averages of idiosyncratic terms and the across-sector averages of sector-specific terms converge to zero.  $\square$

We additionally assume that the following assumptions are satisfied:

**Assumption 5** (Heterogeneity). Define  $\alpha_{in} = (\alpha_{in}^a, \alpha_{in}^s)'$ . Let  $\alpha_n$  and  $\alpha$  be the sectoral and economy-wide average of  $\alpha_{in}$ . We assume that  $|(\alpha_{in} - \alpha_n)(\alpha_{in} - \alpha_n)'|$  is small compared to  $|\alpha_n|$ , and  $|(\alpha_n - \alpha)(\alpha_n - \alpha)'|$  is small compared to  $|\alpha|$ .

**Assumption 6** (Invertibility).  $Id - \alpha^s Id^s K H - \alpha^a Id^a K H$  and  $Id - \alpha_n^s Id^s K H$  are invertible.

**Guess and verify** Now we verify that our guess is correct.

Suppose that Equation (36) is true, then, under Assumptions 3 and 4,

$$\begin{aligned} q_n &= (A_n^{(1)} H \xi_n) \tau + (A_n^{(2)} H \xi_n) (\tau_n - \tau) \\ q &= (A^{(1)} H \xi) \tau \end{aligned}$$

Where we used Lemma 2. We then replace  $\xi_n = Id^s \xi_{in}$  and  $\xi = Id^a \xi_{in}$ :

$$\begin{aligned} q_n &= (A_n^{(1)} H Id^s \xi_{in}) \tau + (A_n^{(2)} H Id^s \xi_{in}) (\tau_n - \tau) \\ q &= (A^{(1)} H Id^a \xi_{in}) \tau \end{aligned}$$

We now take expectation and use the linearity of expectations:

$$\begin{aligned} E_{in}(q_n) &= (A_n^{(1)} H Id^s E_{in}(\xi_{in})) \tau + (A_n^{(2)} H Id^s E_{in}(\xi_{in})) (\tau_n - \tau) \\ E_{in}(q) &= (A^{(1)} H Id^a E_{in}(\xi_{in})) \tau \end{aligned}$$

Using Lemma 2, we obtain

$$E_{in}(q_n) = (A_n^{(1)} H Id^s K S_{in}) \tau + (A_n^{(2)} H Id^s K S_{in}) (\tau_n - \tau) \quad (39)$$

$$E_{in}(q) = (A^{(1)} H Id^a K S_{in}) \tau \quad (40)$$

Note also that  $\sigma_{in} = X \xi_{in}$  with  $X = (1, 0, 1, 0, 1, 0)$ . Therefore, using similar steps, we can show

$$E_{in}(\sigma_{in}) = X K S_{in} \quad (41)$$

Now, we replace  $E_{in}(q_n)$ ,  $E_{in}(q)$  and  $E_{in}(\sigma_{in})$  in Equation (18) using Equations (39), (40) and (41) and obtain

$$\begin{aligned}
q_{in} &= \alpha_{in}^a (A^{(1)} HId^a K S_{in}) \tau + \alpha_{in}^s \left[ (A_n^{(1)} HId^s K S_{in}) \tau + (A_n^{(2)} HId^s K S_{in}) (\tau_n - \tau) \right] - X K S_{in} (\tau_{in} - \tau_n) \\
&= \left[ \alpha_{in}^a (A^{(1)} HId^a K) + \alpha_{in}^s (A_n^{(1)} HId^s K) - X K \right] S_{in} \tau \\
&\quad + \left[ \alpha_{in}^s (A_n^{(2)} HId^s K) - X K H \right] S_{in} (\tau_n - \tau) \\
&\quad - X K S_{in} (\tau_{in} - \tau_n)
\end{aligned} \tag{42}$$

Identifying each element in our guess (36) and in Equation (42), we obtain

$$A_{in}^{(1)} = \alpha_{in}^a (A^{(1)} HId^a K) + \alpha_{in}^s (A_n^{(1)} HId^s K) - X K \tag{43}$$

$$A_{in}^{(2)} = \alpha_{in}^s (A_n^{(2)} HId^s K) - X K \tag{44}$$

$$A_{in}^{(3)} = -X K \tag{45}$$

$A_{in}^{(3)}$  is explicitly defined. To find the solution for  $A_{in}^{(2)}$ , we solve first for  $A_n^{(2)}$  by taking the sectoral average of Equation (44):

$$\begin{aligned}
A_n^{(2)} &= \alpha_n^s (A_n^{(2)} HId^s K) - X K \\
\Leftrightarrow A_n^{(2)} (Id - \alpha_n^s HId^s K) &= -X K \\
\Leftrightarrow A_n^{(2)} &= -X K (Id - \alpha_n^s HId^s K)^{-1}
\end{aligned} \tag{46}$$

where we used Assumption 6. Equation (46) gives us a solution for  $A_n^{(2)}$ . Then  $A_{in}^{(2)}$  is simply obtained by replacing  $A_n^{(2)}$  in Equation (44).

To find the solution for  $A_{in}^{(1)}$ , we solve first for  $A^{(1)}$  by taking the economy-wide average of Equation (43):

$$\begin{aligned}
A^{(1)} &= \alpha^a (A^{(1)} HId^a K) + \alpha^s (A^{(1)} HId^s K) - X K \\
\Leftrightarrow A^{(1)} (Id - \alpha^s HId^s K - \alpha^a HId^a K) &= -X K \\
\Leftrightarrow A^{(1)} &= -X K (Id - \alpha^s HId^s K - \alpha^a HId^a K)^{-1}
\end{aligned} \tag{47}$$

where we used Assumptions 5 and 6. We use Assumption 5 to approximate  $\frac{1}{N} \sum_{n=1}^N \alpha_n^s A_n^{(1)} \simeq \alpha^s A^{(1)}$ . Indeed,

$$\begin{aligned}
A_{in}^{(1)} &= A^{(1)}(\Theta_{in}) = A^{(1)}(\theta_{in}^a, \theta_{in}^s) \\
&\simeq A^{(1)} + (\alpha_{in}^a - \alpha_n^a) \frac{\partial A^{(1)}}{\partial (\alpha_{in}^a - \alpha_n^a)} + (\alpha_{in}^s - \alpha_n^s) \frac{\partial A^{(1)}}{\partial (\alpha_{in}^s - \alpha_n^s)} \\
&\quad + (\alpha^a - \alpha_n^a) \frac{\partial A^{(1)}}{\partial (\alpha^a - \alpha_n^a)} + (\alpha^s - \alpha_n^s) \frac{\partial A^{(1)}}{\partial (\alpha^s - \alpha_n^s)}
\end{aligned}$$

where  $A^{(1)}$  is the value of  $A^{(1)}(\Theta_{in})$  when  $\alpha_{in} = \alpha_n = \alpha$ , with  $\alpha$  the economy-wide average of  $\alpha_{in}$ . Therefore,  $A_n^{(1)}$ , the sectoral average of  $A_{in}^{(1)}$  can be approximated as

$$A_n^{(1)} \simeq A^{(1)} + (\alpha^a - \alpha_n^a) \frac{\partial A^{(1)}}{\partial (\alpha^a - \alpha_n^a)} + (\alpha^s - \alpha_n^s) \frac{\partial A^{(1)}}{\partial (\alpha^s - \alpha_n^s)}$$

As a result,

$$\begin{aligned} \frac{1}{N} \sum_{n=1}^N \alpha_n^s A_n^{(1)} &= \frac{1}{N} \sum_{n=1}^N \alpha^s A_n^{(1)} + \frac{1}{N} \sum_{n=1}^N (\alpha_n^s - \alpha^s) A_n^{(1)} \\ &= \alpha^s A^{(1)} + \underbrace{\frac{1}{N} \sum_{n=1}^N (\alpha_n^s - \alpha^s) A^{(1)}}_{=0} + \frac{1}{N} \sum_{n=1}^N (\alpha_n^s - \alpha^s) (A_n^{(1)} - A^{(1)}) \\ &\simeq \alpha^s A^{(1)} + \frac{1}{N} \sum_{n=1}^N (\alpha_n^s - \alpha^s) \left( (\alpha^a - \alpha_n^a) \frac{\partial A^{(1)}}{\partial (\alpha^a - \alpha_n^a)} + (\alpha^s - \alpha_n^s) \frac{\partial A^{(1)}}{\partial (\alpha^s - \alpha_n^s)} \right) \\ &\simeq \alpha^s A^{(1)} \end{aligned}$$

where the last line is a direct consequence of Assumption 5. This enables us to aggregate equation (43) into the simple linear expression (47).

Equation (47) then gives us a solution for  $A^{(1)}$ . To find  $A_n^{(1)}$ , we use the sectoral average of Equation (43):

$$\begin{aligned} A_n^{(1)} &= \alpha_n^a (A^{(1)} H I d^a K) + \alpha_n^s (A_n^{(1)} H I d^s K) - X K \\ \Leftrightarrow A_n^{(1)} (I d - \alpha_n^s H I d^s K) &= \alpha_n^a (A^{(1)} H I d^a K) - X K \\ \Leftrightarrow A_n^{(1)} &= [\alpha_n^a (A^{(1)} H I d^a K) - X K] (I d - \alpha_n^s H I d^s K)^{-1} \end{aligned} \quad (48)$$

where we used Assumption 6. Equation (48) gives us a solution for  $A_n^{(1)}$ . Then  $A_{in}^{(1)}$  is simply obtained by replacing  $A^{(1)}$  and  $A_n^{(1)}$  in Equation (43).

$$\begin{aligned} \text{Combining Equations (43) to (48), we can show that the solution for } (A_{in}^{(1)}, A_{in}^{(2)}, A_{in}^{(3)}) \text{ is} \\ A_{in}^{(1)} &= -X K [I d + \alpha_n^a (I d - \alpha^s H I d^s K - \alpha_n^a H I d^a K)^{-1} I d^a K] [I d + \alpha_{in}^s (I d - \alpha_n^s H I d^s K)^{-1} I d^s K] \\ &\quad - X K (\alpha_{in}^a - \alpha_n^a) (I d - \alpha^s H I d^s K - \alpha_n^a H I d^a K)^{-1} I d^a K \\ A_{in}^{(2)} &= -X K [I d + \alpha_{in}^s (I d - \alpha_n^s H I d^s K)^{-1} I d^s K] \\ A_{in}^{(3)} &= -X K \end{aligned} \quad (49)$$

This satisfies our guess and provides an explicit solution for Equation (36).

Then, the solution for  $q_{in}$  (line 1 of Equation (22)) can be derived, with  $\mathcal{Q}^{(j)}(\Theta_{in}, S_{in}) = A^{(j)}(\Theta_{in}) S_{in}$ ,  $j = 1, 2, 3$ , with  $A^{(j)}(\Theta_{in}) = A_{in}^{(j)}$ , as defined in (49).

The solutions for  $E_{in}(q_n)$  and  $E_{in}(q)$  (lines 2 and 3 of Equation (22)) can be obtained by aggregating Equation (36) at the sectoral and aggregate levels, using Assumptions 3 and 4:

$$\begin{aligned} q_n &= (A_n^{(1)} H \xi_n) \tau + (A_n^{(2)} \xi_n) (\tau_n - \tau) \\ q &= (A^{(1)} H \xi) \tau \end{aligned}$$

We take expectations:

$$\begin{aligned} E_{in}(q_n) &= A_n^{(1)} H E_{in}(\xi_n) \tau + A_n^{(2)} H E_{in}(\xi_n) (\tau_n - \tau) \\ E_{in}(q) &= A^{(1)} H E_{in}(\xi) \tau \end{aligned}$$

We then use Lemma 1 to replace the expectations in the right-hand side:

$$\begin{aligned} E_{in}(q_n) &= A_n^{(1)} Id^s H K S_{in} \tau + A_n^{(2)} Id^s H K S_{in} (\tau_n - \tau) \\ E_{in}(q) &= A^{(1)} Id^a H K S_{in} \tau \end{aligned}$$

We thus have

$$\begin{aligned} E_{in}(q_n) &= A_n^{s(1)} S_{in} \tau + A_n^{s(2)} S_{in} (\tau_n - \tau) \\ E_{in}(q) &= A^{a(1)} S_{in} \tau \end{aligned}$$

with

$$\begin{aligned} A_n^{s(1)} &= A_n^{(1)} Id^s H K \\ A_n^{s(2)} &= A_n^{(2)} Id^s H K \\ A^{a(1)} &= A^{(1)} Id^a H K \end{aligned} \tag{50}$$

Therefore, Equation (22) is satisfied, with  $\mathcal{Q}^{k(j)}(\Theta_{in}, S_{in}) = A^{k(j)}(\Theta_{in}) S_{in}$ , for  $(j, k) = (1, s), (2, s), (1, a)$ , with  $A^{s(1)}(\Theta_{in}) = A_n^{s(1)}$ ,  $A^{s(2)}(\Theta_{in}) = A_n^{s(2)}$  and  $A^{a(1)}(\Theta_{in}) = A^{a(1)}$  as defined in (50).

## A.2 Proof of Proposition 2

To prove proposition 2, we first characterize expectation formation by firms under the more specific information structure described in Assumption 2, then turn to the proof itself.

### A.2.1 Expectation formation with private and public information

The following lemma summarizes the signal extraction by firms:

**Lemma 3** (Expectations). *Under Assumption 2, the rational expectation of  $\xi_{in} = (\sigma^a, e^a, \sigma_n^s, e_n^s, \sigma_{in}^f, e_{in}^f)'$  by firm  $i$  of sector  $n$  conditional on  $S_{in} = (x^a, y_n^s, z_{in}^f)'$  is given by*

$$E_{in}(\xi_{in}) = K S_{in}$$

where  $K$  is a  $6 \times 3$  matrix:

$$K = \begin{pmatrix} (1 - \delta_a^y - \delta_a^z) & \delta_a^y & \delta_a^z \\ \delta_a^y + \delta_a^z & -\delta_a^y & -\delta_a^z \\ -\delta_s^y - \delta_s^z & \delta_s^y & \delta_s^z \\ -(1 - \delta_a^y - \delta_a^z - \delta_s^y - \delta_s^z) & 1 - \delta_a^y - \delta_s^y & -\delta_a^z - \delta_s^z \\ -(\delta_f^z - \delta_f^y) & -\delta_f^y & \delta_f^z \\ -(1 - \delta_a^y - \delta_a^z - \delta_s^y - \delta_s^z - (\delta_f^z - \delta_f^y)) & 1 - \delta_a^y - \delta_s^y + \delta_f^y & -\delta_a^z - \delta_s^z - \delta_f^z \end{pmatrix} \tag{51}$$

with

$$\begin{aligned}
\delta_a^y &= \frac{\kappa^s \pi^s (\kappa^f + \pi^f)}{\Omega} \\
\delta_a^z &= \frac{\kappa^f \kappa^s \pi^f}{\Omega} \\
\delta_s^y &= \frac{\pi^a \pi^s (\kappa^f + \pi^f)}{\Omega} \\
\delta_s^z &= \frac{\kappa^f \pi^a \pi^f}{\Omega} \\
\delta_f^y &= \frac{\pi^f \pi^s (\pi^a + \kappa^s)}{\Omega} \\
\delta_f^z &= \frac{\pi^f \pi^s (\pi^a + \kappa^s) + \kappa^s \pi^a \pi^f}{\Omega}
\end{aligned} \tag{52}$$

and

$$\Omega = \kappa^f \kappa^s \pi^a + \kappa^f \kappa^s \pi^f + \kappa^f \kappa^s \pi^s + \kappa^f \pi^a \pi^f + \kappa^f \pi^a \pi^s + \kappa^s \pi^a \pi^f + \kappa^s \pi^f \pi^s + \pi^a \pi^f \pi^s$$

*Proof.* We are looking for the firm's posterior expectation about the components of the elasticity  $\Sigma_{in} = (\sigma^a, \sigma_n^s, \sigma_{in}^f)'$ , under Assumption 2. This information structure assumes a prior mean  $(x^a, 0, 0)'$ , a prior variance

$$P_0 = \begin{pmatrix} (\pi^a)^{-1} & 0 & 0 \\ 0 & (\kappa^s)^{-1} & 0 \\ 0 & 0 & (\kappa^f)^{-1} \end{pmatrix}$$

and the signals  $s_{in} = (y_n^s, z_{in}^f)'$ .

The measurement equation is

$$s_{in} = h \Sigma_{in} + v_{in}$$

with

$$h = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}$$

and

$$v_{in} = \begin{pmatrix} e_n^s \\ e_{in}^f \end{pmatrix} \sim N \left( \begin{pmatrix} 0 \\ 0 \end{pmatrix}, R \right), \quad R = \begin{pmatrix} (\pi^s)^{-1} & 0 \\ 0 & (\pi^f)^{-1} \end{pmatrix}$$

We use the Kalman gain formula:

$$E_{in}(\Sigma_{in}) = E(\Sigma_{in} | x^a, s_{in}) = \begin{pmatrix} x^a \\ 0 \\ 0 \end{pmatrix} + k \left( s_{in} - \begin{pmatrix} x^a \\ x^a \end{pmatrix} \right)$$

with

$$k = P_0 h' (h P_0 h' + R)^{-1}$$

Some computation gives us the solution for  $k$ :

$$k = \begin{pmatrix} \delta_a^y & \delta_a^z \\ \delta_s^y & \delta_s^z \\ -\delta_f^y & \delta_f^z \end{pmatrix}$$

where the  $\delta$ s are given in (52). This gives us the expectations on  $(\sigma^a, \sigma_n^s, \sigma_{in}^f)$ . Noting that

$$\begin{aligned} E_{in}(e^a) &= x^a - E_{in}(\sigma^a) \\ E_{in}(e_n^s) &= y_n^s - E_{in}(\sigma^a) - E_{in}(\sigma_n^s) \\ E_{in}(e_{in}^f) &= z_{in}^f - E_{in}(\sigma^a) - E_{in}(\sigma_n^s) - E_{in}(\sigma_{in}^f) \end{aligned}$$

we can then derive the full solution for  $K$  in Lemma 1. □

It is useful to derive the elasticities' expectations in the following corollary:

**Corollary 1** (Expectations). *The expectations of the individual, sectoral, and aggregate elasticities are*

$$\begin{pmatrix} E_{in}(\sigma_{in}) \\ E_{in}(\sigma_n) \\ E_{in}(\sigma) \end{pmatrix} = \begin{pmatrix} E_{in}(\sigma^a + \sigma_n^s + \sigma_{in}^f) \\ E_{in}(\sigma^a + \sigma_n^s) \\ E_{in}(\sigma^a) \end{pmatrix} = \begin{pmatrix} \delta^x & \delta^y & \delta^z \\ \tilde{\delta}^x & \tilde{\delta}^y & \tilde{\delta}^z \\ \delta_a^x & \delta_a^y & \delta_a^z \end{pmatrix} \begin{pmatrix} x^a \\ y_n^s \\ z_{in}^f \end{pmatrix} \quad (53)$$

with

$$\begin{aligned} \delta^y &= \delta_a^y + \delta_s^y + \delta_f^y \\ \delta^z &= \delta_a^z + \delta_s^z - \delta_f^z \\ \delta^x &= 1 - \delta^y - \delta^z \\ \tilde{\delta}^y &= \delta_a^y + \delta_s^y \\ \tilde{\delta}^z &= \delta_a^z + \delta_s^z \\ \tilde{\delta}^x &= 1 - \tilde{\delta}^y - \tilde{\delta}^z \\ \delta_a^x &= 1 - \delta_a^y - \delta_a^z \end{aligned} \quad (54)$$

where  $(\delta_a^y, \delta_s^y, \delta_f^y, \delta_a^z, \delta_s^z, \delta_f^z)$  are given in (52).

### A.2.2 Proof

We prove Proposition 2 in four steps.

**Step 1: Error on elasticity** According to Corollary 1, the firm expectation on its elasticity  $\sigma_{in}$  is

$$\begin{aligned} E_{in}(\sigma_{in}) &= E_{in}(\sigma^a) + E_{in}(\sigma_n^s) + E_{in}(\sigma_{in}^f) \\ &= \delta^x x^a + \delta^y y_n^s + \delta^z z_{in}^f \end{aligned} \quad (55)$$

Aggregating across all firms:

$$E_{in}(\sigma_{in}) = \sigma^a + \delta^x e^a$$

which yields Equation (26).

**Step 2: Errors on quantities** Proposition 1 implies that the solution for  $q_{in}$ , in the case of an economy-wide tax  $\tau$ , is of the form

$$q_{in} = -[\gamma_{in}^x x^a + \gamma_{in}^y y_n^s + \gamma_{in}^z z_{in}^f] \tau \quad (56)$$

Simply put, the output response is a linear combination of the firm's signals and prior.<sup>16</sup> Among these,  $x^a$  is common knowledge in the whole economy,  $y_n^s$  is private to the sector but common knowledge within the sector and  $z_{in}^f$  is private to the firm.

Assuming that Assumptions 3 and 4 hold, and aggregating at the sector and aggregate level implies that the following expressions hold for output at the sector and aggregate level.

$$q_n = -[\gamma_n^x x^a + \gamma_n^y y_n^s + \gamma_n^z (\sigma^a + \sigma_n^s)] \tau \quad (57)$$

$$q = -[\gamma^x x^a + (\gamma^y + \gamma^z) \sigma^a] \tau \quad (58)$$

When summing these individual quantities across firms, the private signals sum to the fundamental value of the aggregate elasticity  $\sigma^a$  at the aggregate level and to the sectoral elasticity  $\sigma^a + \sigma_n^s$  at the sectoral level.

When forming an expectation of the aggregate output  $q$ , (a higher-order belief), firms must not only forecast the part that they know,  $x^a$  (because it is public information), but also the part that they don't know,  $\sigma^a$  (because it arises from the others' private information). As a consequence, the error on  $q$  is driven by the error on  $\sigma^a$ , but only to the extent that  $q$  depends on the firms' private individual and sectoral information ( $\gamma^z + \gamma^y \neq 0$ ):

$$E_{in}(q) - q = -(\gamma^z + \gamma^y)[E_{in}(\sigma^a) - \sigma^a] \tau$$

The aggregate error is then

$$\bar{E}(q) - q = -(\gamma^z + \gamma^y)[\bar{E}(\sigma^a) - \sigma^a] \tau = -(\gamma^z + \gamma^y) \delta_a^x e^a \tau \quad (59)$$

---

<sup>16</sup>This corresponds to the solution (22) with  $\tau_n - \tau = 0$ ,  $\tau_{in} - \tau_n = 0$ , and  $\mathcal{Q}^{(1)} = A^{(1)} S_{in}$  with  $A^{(1)} = (-\gamma_{in}^x, -\gamma_{in}^y, -\gamma_{in}^z)'$ .

where we used Corollary 1.

Similarly, the expectation error about sectoral output can be expressed as a function of the errors on both the aggregate and sectoral components:

$$E_{in}(q_n) - q_n = -\gamma_n^z [E_{in}(\sigma^a) - \sigma^a + E_{in}(\sigma_n^s) - \sigma_n^s] \tau$$

The aggregate error on sectoral output is then

$$\bar{E}(q_n) - q = -\gamma^z \tilde{\delta}^x e^a \tau \quad (60)$$

where we used Corollary 1.

Equations (59) and (60) correspond to Equations (27) and (28) of Proposition 2.

**Step 3: Sign of  $\gamma^z$  and  $\gamma^y$**  We now derive a closed-form expressions for  $\gamma^z$  and  $\gamma^y$ .

Taking expectations, and using the solution for expectations from Corollary 1, we obtain

$$\begin{aligned} E_{in}(q_n) &= -(\gamma_n^x x^a + \gamma_n^y y_n^s + \gamma_n^z (E_{in}(\sigma^a) + E_{in}(\sigma_n^s))) \tau \\ &= -(\gamma_n^x x^a + \gamma_n^y y_n^s + \gamma_n^z \tilde{\delta}^x x^a + \tilde{\delta}^y y_n^s + \tilde{\delta}^z z_{in}^f) \tau \\ &= -((\gamma_n^x + \gamma_n^z \tilde{\delta}^x) x^a + (\gamma_n^y + \gamma_n^z \tilde{\delta}^y) y_n^s + \gamma_n^z \tilde{\delta}^z z_{in}^f) \tau \\ E_{in}(q) &= -(\gamma^x x^a + (\gamma^y + \gamma^z) E_{in}(\sigma^a)) \tau \\ &= -(\gamma^x x^a + (\gamma^y + \gamma^z) [\delta_a^x x^a + \delta_a^y y_n^s + \delta_a^z z_{in}^f]) \tau \\ &= -((\gamma^x + \gamma^z \delta_a^x) x^a + (\gamma^y + \gamma^z) \delta_a^y y_n^s + (\gamma^y + \gamma^z) \delta_a^z z_{in}^f) \tau \end{aligned}$$

Using these expressions, along with (55), we can substitute the terms in Equation (18) and obtain:

$$\begin{aligned} q_{in} &= -\alpha_{in}^a ((\gamma^x + \gamma^z \delta_a^x) x^a + (\gamma^y + \gamma^z) \delta_a^y y_n^s + (\gamma^y + \gamma^z) \delta_a^z z_{in}^f) \tau \\ &\quad - \alpha_{in}^s ((\gamma_n^x + \gamma_n^z \tilde{\delta}^x) x^a + (\gamma_n^y + \gamma_n^z \tilde{\delta}^y) y_n^s + \gamma_n^z \tilde{\delta}^z z_{in}^f) \tau \\ &\quad - (\delta^x x^a + \delta^y y_n^s + \delta^z z_{in}^f) \tau \\ &= -(\alpha_{in}^a (\gamma^x + \gamma^z \delta_a^x) + \alpha_{in}^s (\gamma_n^x + \gamma_n^z \tilde{\delta}^x) + \delta^x) x^a \tau \\ &\quad - (\alpha_{in}^a (\gamma^y + \gamma^z) \delta_a^y + \alpha_{in}^s (\gamma_n^y + \gamma_n^z \tilde{\delta}^y) + \delta^y) y_n^s \tau \\ &\quad - (\alpha_{in}^a (\gamma^y + \gamma^z) \delta_a^z + \alpha_{in}^s \gamma_n^z \tilde{\delta}^z + \delta^z) z_{in}^f \tau \end{aligned}$$

Identifying the coefficients of  $y_n^s$  and  $z_{in}^f$  across the above expression and (56), we obtain:

$$\begin{aligned} \gamma_{in}^y &= \alpha_{in}^a (\gamma^y + \gamma^z) \delta_a^y + \alpha_{in}^s (\gamma_n^y + \gamma_n^z \tilde{\delta}^y) + \delta^y \\ \gamma_{in}^z &= \alpha_{in}^a (\gamma^y + \gamma^z) \delta_a^z + \alpha_{in}^s \gamma_n^z \tilde{\delta}^z + \delta^z \end{aligned} \quad (61)$$

These equations can be aggregated at the sector level:

$$\begin{aligned}\gamma_n^y &= \alpha_n^a(\gamma^y + \gamma^z)\delta_a^y + \alpha_n^s(\gamma_n^y + \gamma_n^z\tilde{\delta}^y) + \delta^y \\ \gamma_n^z &= \alpha_n^a(\gamma^y + \gamma^z)\delta_a^z + \alpha_n^s\gamma_n^z\tilde{\delta}^z + \delta^z\end{aligned}\tag{62}$$

Under Assumption 5, we can approximate:

$$\frac{1}{N} \sum_{n=1}^N \alpha_n^s \gamma_n^y \simeq \alpha^s \gamma^y, \quad \frac{1}{N} \sum_{n=1}^N \alpha_n^s \gamma_n^z \simeq \alpha^s \gamma^z$$

To show this, we follow the same steps as in the proof of Proposition 1. As a result, the sectoral equations can be aggregated as follows:

$$\begin{aligned}\gamma^y &= \alpha^a(\gamma^y + \gamma^z)\delta_a^y + \alpha^s(\gamma^y + \gamma^z\tilde{\delta}^y) + \delta^y \\ \gamma^z &= \alpha^a(\gamma^y + \gamma^z)\delta_a^z + \alpha^s\gamma^z\tilde{\delta}^z + \delta^z\end{aligned}\tag{63}$$

After some further manipulation of these equations, we can show that

$$\begin{aligned}\gamma^y + \gamma^z &= \frac{(1 - \alpha^s\tilde{\delta}^z)\delta^y + (1 - \alpha^s(1 - \tilde{\delta}^y))\delta^z}{(1 - \alpha^s\tilde{\delta}^z)(1 - \alpha^a(\delta_a^y + \delta_a^z) - \alpha^s) + \alpha^a\delta_a^z\alpha^s(1 - \tilde{\delta}^y - \tilde{\delta}^z)} \\ \gamma^z &= \frac{(1 - \alpha^a\delta_a^y - \alpha^s)\delta^z + \alpha^a\delta^y\delta_a^z}{(1 - \alpha^s\tilde{\delta}^z)(1 - \alpha^a(\delta_a^y + \delta_a^z) - \alpha^s) + \alpha^a\delta_a^z\alpha^s(1 - \tilde{\delta}^y - \tilde{\delta}^z)}\end{aligned}\tag{64}$$

These two coefficients share the same denominator. We can show that this denominator is positive when  $|\alpha^a|^{\frac{1}{2}} + |\alpha^s|^{\frac{1}{2}} < 1$ :

$$\begin{aligned}(1 - \alpha^s\tilde{\delta}^z)(1 - \alpha^a(\delta_a^y + \delta_a^z) - \alpha^s) + \alpha^a\delta_a^z\alpha^s(1 - \tilde{\delta}^y - \tilde{\delta}^z) &\geq (1 - |\alpha^s|)(1 - |\alpha^a| - |\alpha^s|) - |\alpha^a||\alpha^s| \\ &\geq 1 - |\alpha^a| - 2|\alpha^s| + |\alpha^s|^2 \\ &\geq (1 - |\alpha^s|)^2 - |\alpha^a|\end{aligned}$$

We used the fact that  $\tilde{\delta}^z$ ,  $\delta_a^y + \delta_a^z$ ,  $\delta_a^z$  and  $1 - \tilde{\delta}^y - \tilde{\delta}^z$  are positive and lower than one, and the fact that  $|\alpha^a|^{\frac{1}{2}} + |\alpha^s|^{\frac{1}{2}} < 1$  implies that  $|\alpha^a| < 1$ ,  $|\alpha^s| < 1$  and  $|\alpha^a| + |\alpha^s| < 1$ .

Now, notice that  $|\alpha^s|^{\frac{1}{2}} > |\alpha^s|$ . As a result,  $|\alpha^a|^{\frac{1}{2}} + |\alpha^s| < |\alpha^a|^{\frac{1}{2}} + |\alpha^s|^{\frac{1}{2}} < 1$ , yielding  $1 - |\alpha^s| > |\alpha^a|^{\frac{1}{2}}$ , and hence  $(1 - |\alpha^s|)^2 > |\alpha^a|$ . This implies that the denominator is strictly positive.

Using similar arguments, we can show that the numerator of  $\gamma^y + \gamma^z$  is strictly positive:

$$(1 - \alpha^s\tilde{\delta}^z)\delta^y + (1 - \alpha^s(1 - \tilde{\delta}^y))\delta^z \geq (1 - |\alpha^s|)\delta^y + (1 - |\alpha^s|)\delta^z \geq 0$$

and that the numerator of  $\gamma^z$  is strictly positive:

$$\begin{aligned}(1 - \alpha^a\delta_a^y - \alpha^s)\delta^z + \alpha^a\delta^y\delta_a^z &= \left(1 - \alpha^a\left(\delta_a^y - \delta^y\frac{\delta_a^z}{\delta^z}\right) - \alpha^s\right)\delta^z \\ &\geq (1 - |\alpha^a| - |\alpha^s|)\delta^z \geq 0\end{aligned}$$

where we used the fact that  $\delta_a^z/\delta^z$  is between zero and one.

Consequently, both  $\gamma^y + \gamma^z$  and  $\gamma^z$  are positive.

**Step 4: The role of private information** Remember that, according to Corollary 1,  $\delta^y + \delta^z = \delta_a^y + \delta_s^y - \delta_f^y + \delta_a^z + \delta_s^z + \delta_f^z$  and  $\delta^x = 1 - \delta^y - \delta^z$ . Using Lemma 1, this implies that

$$\delta^x = \frac{\kappa^f \kappa^s \pi^a}{\Omega}$$

As a consequence,  $\delta^x$  goes to zero when either  $\pi^f$  goes to infinity or  $\pi^s$  goes to infinity.

If  $\pi^f = 0$ , then, using Lemma 1 and Corollary 1, we can show that  $\delta^z = \delta_a^z = 0$ , and therefore, according to (64),  $\gamma^z = 0$ . Similarly, if  $\pi^f = \pi^s = 0$ , then  $\delta^y = \delta^z = 0$  and  $\gamma^y + \gamma^z = 0$ .

## B Identification

According to Proposition 1, we can write  $P_{in}$  as the product of two matrices:

$$P_{in} = \underbrace{\begin{pmatrix} \tau_1 & \tau_{n,1} - \tau_1 & \tau_{in,1} - \tau_{n,1} \\ \tau_2 & \tau_{n,2} - \tau_2 & \tau_{in,2} - \tau_{n,2} \\ \tau_3 & \tau_{n,3} - \tau_3 & \tau_{in,3} - \tau_{n,3} \end{pmatrix}}_{T_{in}} \underbrace{\begin{pmatrix} \mathcal{Q}^{a(1)} & \mathcal{Q}^{s(1)} & 1 \\ 0 & \mathcal{Q}^{s(2)} & 1 \\ 0 & 0 & 1 \end{pmatrix}}_{D_{in}}$$

$T_{in}$  is the set of alternative scenarios on the different tax components (aggregate, sector-specific and firm-specific), and  $D_{in}$  is the set of elasticities of expected aggregate and sectoral quantities, and of the individual tax to these tax components.  $P_{in}$  is invertible if and only if both  $T_{in}$  and  $D_{in}$  are invertible.  $D_{in}$  is an upper-triangular matrix invertible as long as  $\mathcal{Q}^{a(1)}$  and  $\mathcal{Q}^{s(2)}$  are different from zero. This is a mild assumption. Intuitively, this requires that firms believe that an economy-wide tax will have some impact on the economy in general and that a sectoral tax will have an impact on their sector.

$T_{in}$  describes the set of tax scenarios that are provided to firms. It is invertible if we set for instance:

$$T_{in} = \begin{pmatrix} \epsilon & 0 & 0 \\ 0 & \epsilon & 0 \\ 0 & 0 & \epsilon \end{pmatrix}$$

$\tau_{in}$  is defined as the deviation of the tax from its baseline level. This amounts to presenting firms with the following three scenarios: (1) an increase  $\epsilon$  in the economy-wide tax; (2) an increase  $\epsilon$  in the tax in the sector where the firm operates (but not in the rest of the economy); (3) an increase  $\epsilon$  in the tax that applies to the firm (but not to the rest of the economy, including the sector where the firm operates). In each of these scenarios, we ask firms about their own production, the production in their sector, and aggregate production.

Note that with these scenarios, we ensure that  $P_{in}$  is upper-triangular, with non-zero diagonal elements, as expressed by Equation (23), which ensures invertibility.

## C Robustness of Structural Parameter Estimates

It is noteworthy that our identification strategy renders the firm-level strategic interaction parameters  $\alpha_{in}^a$  and  $\alpha_{in}^s$  robust to individual-specific response biases.

To fix ideas, suppose firm manager  $i$  exhibits a multiplicative bias  $\lambda_i$  affecting all assessments:

$$\hat{q}_{in,1} = \lambda_i q_{in,1}, \quad \hat{q}_{in,2} = \lambda_i q_{in,2}, \quad \hat{q}_{in,3} = \lambda_i q_{in,3},$$

$$\widehat{E_{in}(q_1)} = \lambda_i E_{in}(q_1), \quad \widehat{E_{in}(q_{n,1})} = \lambda_i E_{in}(q_{n,1}), \quad \widehat{E_{in}(q_{n,2})} = \lambda_i E_{in}(q_{n,2}).$$

Equation (23) becomes

$$\begin{pmatrix} \lambda_i q_{in,1} \\ \lambda_i q_{in,2} \\ \lambda_i q_{in,3} \end{pmatrix} = \begin{pmatrix} \lambda_i E_{in}(q_1) & \lambda_i E_{in}(q_{n,1}) & \epsilon \\ 0 & \lambda_i E_{in}(q_{n,2}) & \epsilon \\ 0 & 0 & \epsilon \end{pmatrix} \cdot \begin{pmatrix} \alpha_{in}^a \\ \alpha_{in}^s \\ E_{in}(\sigma_{in}) \end{pmatrix}.$$

In this case, the measurement of the expected elasticity does not correspond to the actually expected elasticity, but  $\alpha_{in}^a$  and  $\alpha_{in}^s$  are still accurately measured. To see this, note that the system can be solved recursively starting from the third equation:

$$\lambda_i q_{in,3} = E_{in}(\sigma_{in}),$$

so that

$$\widehat{E_{in}(\sigma_{in})} = \lambda_i E_{in}(\sigma_{in}).$$

Substituting into the second equation yields

$$\lambda_i q_{in,2} = \alpha_{in}^s \lambda_i E_{in}(q_{n,2}) + \lambda_i E_{in}(\sigma_{in}),$$

which simplifies to

$$q_{in,2} = \alpha_{in}^s E_{in}(q_{n,2}) + E_{in}(\sigma_{in}).$$

Substituting the same expression for  $E_{in}(\sigma_{in})$  into the first equation gives

$$\lambda_i q_{in,1} = \alpha_{in}^a \lambda_i E_{in}(q_1) + \alpha_{in}^s \lambda_i E_{in}(q_{n,1}) + \lambda_i E_{in}(\sigma_{in}),$$

which reduces to

$$q_{in,1} = \alpha_{in}^a E_{in}(q_1) + \alpha_{in}^s E_{in}(q_{n,1}) + E_{in}(\sigma_{in}).$$

Thus, the bias  $\lambda_i$  cancels out from both equations, implying that the mapping from observables to  $(\alpha_{in}^a, \alpha_{in}^s)$  is invariant to constant individual-specific scaling biases in the managers' assessments.

# D Questionnaire

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Bitte wählen Sie Ihre Sprache.  
Veuillez sélectionner votre langue.  
Please select your language.

☐ Deutsch

☐ Français

☒ English

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Our research team at ETH Zürich, EPFL and University of Lausanne would like to find out **how Swiss firms react to public policies that affect their energy costs**.

Completing the survey will take about **15 minutes**.

**No right or wrong answers**  
We are interested in your assessments regarding the impacts of specific policies. Your answers cannot be right or wrong.

**Why participate?**

- You will make an important contribution to the understanding of the Swiss economy.
- To thank you for your participation, an exclusive report with the results will be shared with you.
- All data collected will be used for research only. The data will be anonymized, remain strictly confidential and are subject to Swiss data protection law.

Participation is completely voluntary, and you may exit the survey at any time without giving reasons or incurring any disadvantages. The survey does not pose any risks or inconvenience.

If you have any questions, please contact us at [firmsurvey@demoscope.ch](mailto:firmsurvey@demoscope.ch).

By clicking on «Continue», you acknowledge that you voluntarily choose to participate in this study.

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To start with, we have a few questions about your company's key figures.

In which sector is your company active, or which products or services generate the majority of your company's turnover?

Please enter keywords for the industry name in the field below and select the category that best matches your industry.

Your selection:

BACK

CONTINUE

Please provide us with the 4-digit postal code of the production site of your company or business unit.  
If your company or business unit has multiple production sites, please provide the main site.

Production can include the manufacture of goods and/or services.

Postal code

BACK

CONTINUE

What was the number of **employees** at your company in Switzerland (converted into full-time equivalents (FTE)) at the end of 2024? Approximate information is sufficient.

FTE

BACK

CONTINUE

What was the **turnover** (excluding VAT) of your company in Switzerland in 2024, including goods/services delivered abroad? Approximate information is sufficient.

CHF

BACK

CONTINUE

What proportion of your total sales was generated **abroad** or **by customers residing abroad in the year 2024**?

0%

100%

BACK

CONTINUE

On the scale below, how would you rate the **intensity of competition** in the sector in which your company operates? Please make a subjective assessment.

very low competition

1

2

3

4

5

6

7

very intense competition

Are your main competitors based in Switzerland or abroad?

☐ All in Switzerland

☐ Mostly in Switzerland, some abroad

☐ Roughly half and half

☐ Mostly abroad, some in Switzerland

☐ All abroad

BACK

CONTINUE

What were your company's **operating expenditures/expenses** (including employee costs, cost of materials, other operating expenses, depreciation; excluding investments) in Switzerland in the year 2024? Approximate indication is sufficient.

CHF

BACK

CONTINUE

In 2024, what proportion of your total operating expenditures was directly spent on **energy products** such as electricity, oil or gas fuels, motor fuels, industrial, waste, coal, or wood? If you don't know the exact percentage, you can give an approximation.

Please enter a percentage value between 0 and 100%.

%

BACK

CONTINUE

Which sources of energy does your company directly use?  
Please select all options that apply.

☐ Electricity

☐ Fossil fuels (natural gas, heating oil, gasoline and diesel, including for motor vehicles)

☐ Other energy sources: waste, coal, or wood

BACK

CONTINUE

Which sources of energy does your company directly use?  
Please select all options that apply.

☒ Electricity

☒ Fossil fuels (natural gas, heating oil, gasoline and diesel, including for motor vehicles)

☒ Other energy sources: waste, coal, or wood

BACK

CONTINUE

In 2024, how were your expenditures on energy **split between different types of energy sources**: electricity, fossil fuels (natural gas, heating oil, gasoline and diesel, including for motor vehicles), and other energy sources (waste, coal, or wood)?

Please report the **percentage share of electricity expenditures in your total energy expenditures**. For instance, if one-third of your energy expenditures were associated with electricity, one-third with fossil fuels, and one-third with other energy sources, report a share of 33% for each energy source. It is ok to approximate these percentages if you don't know the exact figure.

Please enter a percentage value between 0 and 100%.

Share of **electricity** expenditures in your total energy expenditures

0% 100%

☐ Fix value

Share of **fossil fuel** expenditures in your total energy expenditures

0% 100%

☐ Fix value

Share of **other energy sources** expenditures in your total energy expenditures

0% 100%

☐ Fix value

BACK

CONTINUE

Which sources of energy does your company directly use?  
Please select all options that apply.

☒ Electricity

☒ Fossil fuels (natural gas, heating oil, gasoline and diesel, including for motor vehicles)

☐ Other energy sources: waste, coal, or wood

BACK

CONTINUE

In 2024, how were your expenditures on energy expenditure **split between electricity and energy sources from fossil fuels** (natural gas, heating oil, gasoline and diesel, including for motor vehicles)?

Please report the **percentage share of electricity expenditures in your total energy expenditures**. For instance, if 50% of your energy expenditures were associated with electricity, report a share of 50%. It is ok to approximate this percentage if you don't know the exact figure.

Please enter a percentage value between 0 and 100%.

Share of **electricity** expenditures in your total energy expenditures

0%

100%

☐ Fix value

Share of **fossil fuel** expenditures in your total energy expenditures

0%

100%

☐ Fix value

BACK

CONTINUE

Which sources of energy does your company directly use?  
Please select all options that apply.

☒ Electricity

☐ Fossil fuels (natural gas, heating oil, gasoline and diesel, including for motor vehicles)

☒ Other energy sources: waste, coal, or wood

BACK

CONTINUE

In 2024, how were your expenditures on energy expenditure **split between electricity and other energy sources** (waste, coal, or wood)?

Please report the **percentage share of electricity expenditures in your total energy expenditures**. For instance, if 50% of your energy expenditures were associated with electricity, report a share of 50%. It is ok to approximate this percentage if you don't know the exact figure.

Please enter a percentage value between 0 and 100%.

Share of **electricity** expenditures in your total energy expenditures

0%

100%

☐ Fix value

Share of **other energy sources** expenditures in your total energy expenditures

0%

100%

☐ Fix value

BACK

CONTINUE

Which sources of energy does your company directly use?  
Please select all options that apply.

☐ Electricity

☒ Fossil fuels (natural gas, heating oil, gasoline and diesel, including for motor vehicles)

☒ Other energy sources: waste, coal, or wood

BACK

CONTINUE

In 2024, how were your expenditures on energy expenditure **split between energy sources from fossil fuels** (natural gas, heating oil, gasoline and diesel, including for motor vehicles) and **other energy sources** (waste, coal, or wood)?

Please report the **percentage share of electricity expenditures in your total energy expenditures**. For instance, if 50% of your energy expenditures were associated with electricity, report a share of 50%. It is ok to approximate this percentage if you don't know the exact figure.

Please enter a percentage value between 0 and 100%.

Share of **fossil fuel** expenditures in your total energy expenditures

☐ Fix value

Share of **other energy sources** expenditures in your total energy expenditures

☐ Fix value

BACK

CONTINUE

In the next section of the survey, we would like to know how an **additional CO2 tax** would impact your decisions and your expectations. Beforehand, we would like to help you think about how an additional CO2 tax of 100 CHF per ton of CO2 would impact your energy costs.

(In all scenarios that follow, assume that your company will not be exempt from the additional tax, even if it is currently exempt from the tax.)

BACK

CONTINUE

Given how much you reported spending on different energy sources, we estimate that your CO2 emissions associated with the following energy sources, for the year 2024, were:

|                      |           |             |
|----------------------|-----------|-------------|
| Electricity          | 133.333   | tons of CO2 |
| Fossile fuel         | 560.000   | tons of CO2 |
| Other energy sources | 632.000   | tons of CO2 |
| Total                | 1'325.333 | tons of CO2 |

We estimate that an additional CO2 tax of 100 CHF per ton of CO2 would then increase your annual after-tax energy costs by **132'500 CHF**.

Of course, if you change your energy consumption or energy mix, the cost impact would be different.

BACK

CONTINUE

Note: The values shown on the preceding and following pages are illustrative and correspond to reported operating expenditures of CHF 10,000,000, an energy expenditure share of 10%, and energy source shares of 40% electricity, 40% fossil fuels, and 20% other sources. The reproduced pages correspond to the treatment group facing a CO2 tax increase of CHF 100; the alternative treatment group faced a CO2 tax increase of CHF 300.

We would now like you to consider the impacts of an **additional CO2 tax of 100 CHF per ton of CO2** for the following three scenarios. As previously stated, we estimate that an additional CO2 tax of 100 CHF per ton of CO2 would then increase your annual after-tax energy costs by **132'500 CHF**. For all scenarios, consider that the tax would be introduced in 2025.

**In Scenario 1**, the CO2 tax is levied on **all firms in Switzerland, including your firm**.

**In Scenario 2**, the CO2 tax is targeted **only at Swiss firms in your sector, including your firm**. All firms outside your sector are not subject to the tax.

**In Scenario 3**, the CO2 tax is targeted **only on your firm**. All other firms in Switzerland are not subject to the tax. (This third scenario is more hypothetical in nature.)

First, let's consider the impact on **production of services and goods**: In your opinion, for each of the three scenarios described above, how would an additional tax of 100 CHF (per ton of CO2) impact:

- (1.) the overall economic production in Switzerland,
- (2.) your Swiss competitors' production (in case there are firms located in Switzerland among your competitors),
- (3.) your company's production?

BACK

CONTINUE

We would now like you to consider the impacts of an **additional CO2 tax of 100 CHF per ton of CO2** for the following three scenarios. As previously stated, we estimate that an additional CO2 tax of 100 CHF per ton of CO2 would then increase your annual after-tax energy costs by **132'500 CHF**. For all scenarios, consider that the tax would be introduced in 2025.

**In Scenario 1**, the CO2 tax is levied on **all firms in Switzerland, including your firm**.

**In Scenario 2**, the CO2 tax is targeted **only at Swiss firms in your sector, including your firm**. All firms outside your sector are not subject to the tax.

**In Scenario 3**, the CO2 tax is targeted **only on your firm**. All other firms in Switzerland are not subject to the tax. (This third scenario is more hypothetical in nature.)

First, let's consider the impact on **production of services and goods**: In your opinion, for each of the three scenarios described above, how would an additional tax of 100 CHF (per ton of CO2) impact:

- (1.) the overall economic production in Switzerland,
- (2.) your Swiss competitors' production (in case there are firms located in Switzerland among your competitors),
- (3.) your company's production?

Please assess: By how many percent will annual production volume be lower or higher in 2028 (i.e., in the medium run) due to the tax, compared to a situation in which there is no additional tax?

Please enter a **negative number** if you believe the tax will reduce production. To give you a point of reference, the worst 3-year decline in Swiss economic production (i.e., in Swiss real GDP) over the last 75 years was -7.3%. Enter a **positive number** if you believe that the tax will increase production.

Impact on Switzerland's volume of produced services and goods (measured at current sales prices):

Scenario 1: The 100 CHF CO2 tax is levied on **all firms in Switzerland**.



Impact on your competitors' volume of produced services and/or goods (measured at current sales prices):

Scenario 1: The 100 CHF CO2 tax is levied on **all firms in Switzerland**.



Scenario 2: The 100 CHF CO2 tax is levied on **only on firms in your sector**.



Impact on your firm's volume of produced services and/or goods (measured at current sales prices):

Scenario 1: The 100 CHF CO2 tax is levied on **all firms in Switzerland**.

-20% 20%

☐ less than -20% ☐ more than +20%

Scenario 2: The 100 CHF CO2 tax is levied on **only on firms in your sector**.

-20% 20%

☐ less than -20% ☐ more than +20%

Scenario 3: The 100 CHF CO2 tax is levied on **only on your firm**.

-20% 20%

☐ less than -20% ☐ more than +20%

BACK

CONTINUE

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As previously stated, we estimate that an additional CO2 tax of 100 CHF per ton of CO2 would then increase your after-tax energy costs by **132'500 CHF**. Now, let's consider the impact on **prices**:

Please assess: By how many percent will **prices be lower or higher in 2028** (i.e., in the medium run) due to the **tax**, compared to a situation in which there is no additional tax?

Please enter a **positive number** if you believe the tax will increase prices. Enter a **negative number** if you believe that the tax will lead to a decrease in prices.

Impact on Switzerland's prices of produced services and goods:

Scenario 1: The 100 CHF CO2 tax is levied on **all firms in Switzerland**.

-20% 20%

☐ less than -20% ☐ more than +20%

Impact on your competitors' prices of produced services and/or goods:

Scenario 1: The 100 CHF CO2 tax is levied on **all firms in Switzerland**.

-20% 20%

☐ less than -20% ☐ more than +20%

Scenario 2: The 100 CHF CO2 tax is levied on **only on firms in your sector**.

-20% 20%

☐ less than -20% ☐ more than +20%

Impact on your firm's prices of produced services and/or goods:

Scenario 1: The 100 CHF CO2 tax is levied on **all firms in Switzerland**.

-20% 20%

☐ less than -20% ☐ more than +20%

Scenario 2: The 100 CHF CO2 tax is levied on **only on firms in your sector**.

-20% 20%

☐ less than -20% ☐ more than +20%

Scenario 3: The 100 CHF CO2 tax is levied on **only on your firm**.

-20% 20%

☐ less than -20% ☐ more than +20%

BACK

CONTINUE



Relative to other firms in your sector in Switzerland, do you think **your firm** produces fewer or more CO2 emissions per unit volume of services and/or goods produced than other firms?

☐ Less than the other firms in my sector (lower CO2 intensity)

☐ Same as the other firms

☐ More than the other firms in my sector (higher CO2 intensity)

☐ I don't know

BACK

CONTINUE

Relative to all other sectors in Switzerland, do you think that **the firms in your sector** produce more or less CO2 emissions per unit volume of services and/or goods produced than firms in the other sectors?

☐ Less than the other sectors (lower CO2 intensity)

☐ Same as the other sectors

☐ More than the other sectors (higher CO2 intensity)

☐ I don't know

BACK

CONTINUE

We would like to know if your firm is currently subject to a CO2 regulation or exemption.

Is your company exempt from the Swiss CO2 tax?

☐ I don't know

Does your company take part in the Swiss Emission Trading System (CH EHS/SEQE)?

☐ I don't know

Has your company made a formal commitment to the Confederation to reduce greenhouse gas emissions?

☐ I don't know

BACK

CONTINUE

Finally, a few questions about you personally and your assessment of this questionnaire.

How many years have you been working at your current company?

BACK

CONTINUE

What position do you hold in the company?

- ☐ CEO
- ☐ CFO
- ☐ Other C-Levels
- ☐ Senior Management
- ☐ Junior Management
- ☐ Other:

BACK

CONTINUE

How old are you?

- ☐ Under 25
- ☐ 25-29
- ☐ 30-34
- ☐ 35-39
- ☐ 40-44
- ☐ 45-49
- ☐ 50-54
- ☐ 55-59
- ☐ 60-64
- ☐ 65-69
- ☐ 70 or older
- ☐ No answer

BACK

CONTINUE

What is your gender?

- ☐ Male
- ☐ Female
- ☐ Other
- ☐ No answer

BACK

CONTINUE

What is the highest level of education you have completed?

☐ Compulsory education attended but not completed

☐ Compulsory education

☐ 1-year education and training course / bridging course

☐ Upper-secondary specialised school

☐ Apprenticeship in enterprise (VET) / basic vocational training / vocational school / commercial school

☐ Academic baccalaureate / teacher training certificate

☐ Vocational baccalaureate / specialised baccalaureate

☐ Federal PET examination with federal PET diploma / higher vocational examination with federal diploma/Master certificate

☐ College of higher vocational education and training

☐ Bachelor' degree: university, ETH/EPF, university of applied sciences, university of teacher education (incl. diploma from university of applied sciences/university of teacher education)

☐ Master's degree: university, ETH/EPF, university of applied sciences, university of teacher education (incl. licentiate, university/ETH/EPF diploma)

☐ Doctorate, post-doctoral qualification (habilitation)

☐ No answer

With your participation in the survey you have been supporting our research on the Swiss economy. We would like to conduct further company surveys in the future. To this end, we are planning to set up a panel, for which we are dependent on your support.

May we invite you again for future surveys? If you would like to take part in future surveys (approximately twice a year), you can enter your name and e-mail address below. Once we have this information, we can also send you an exclusive report on the results of the survey. Only survey participants will have access to this report.

If you accept our invitation, you can of course still have your name removed from our e-mail list at any time in the future and decide individually for each survey whether you would like to participate or not.

☐ Yes, please invite me for future surveys.

☐ No, do not invite me for future surveys.

BACK

CONTINUE

With your participation in the survey you have been supporting our research on the Swiss economy. We would like to conduct further company surveys in the future. To this end, we are planning to set up a panel, for which we are dependent on your support.

May we invite you again for future surveys? If you would like to take part in future surveys (approximately twice a year), you can enter your name and e-mail address below. Once we have this information, we can also send you an exclusive report on the results of the survey. Only survey participants will have access to this report.

If you accept our invitation, you can of course still have your name removed from our e-mail list at any time in the future and decide individually for each survey whether you would like to participate or not.

☒ Yes, please invite me for future surveys.

☐ No, do not invite me for future surveys.

After completing the survey, you will receive an e-mail from our service partner DemoSCOPE to confirm your e-mail address.

First name

Surname

To simplify the invitation process, we would be grateful if you could provide us with an e-mail address to which we can send future invitations to these surveys.



E-mail address

Confirm e-mail address

BACK

CONTINUE

Do you want to receive the report on the results of the survey?

☐ Yes

☐ No

BACK

CONTINUE

Do you want to receive the report on the results of the survey?

☒ Yes

☐ No

First name

Surname

To simplify the invitation process, we would be grateful if you could provide us with an e-mail address to which we can send future invitations to these surveys.



E-mail address

Confirm e-mail address

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CONTINUE

How interesting or uninteresting did you find the questions in the survey?

Very uninteresting

Rather uninteresting

Neither interesting nor uninteresting

Rather interesting

Very interesting

BACK

CONTINUE

How easy or difficult was it for you to answer the questions in the survey?

Very difficult

Rather difficult

Neither difficult nor easy

Rather light

Very light

BACK

CONTINUE

Thank you for completing the questionnaire!

Do you have any feedback on the questions or the survey process?

BACK

CONTINUE

## E Carbon Tax Cost Calculation

This appendix describes how we estimate each firm’s annual CO<sub>2</sub> emissions from the information elicited in the survey, and how these estimates translate into the firm-specific carbon tax cost presented to managers ([Figure 1](#)). The objective of this procedure is to provide each manager with a private, firm-specific signal about the cost impact of the carbon tax increase, as discussed in [Section 3.1](#).

**Elicited inputs.** We rely on three survey items, which precede the carbon tax scenarios in the questionnaire (see [Appendix D](#)): (i) the firm’s total operating expenses, denoted  $X_{in}$  for firm  $i$  in sector  $n$ ; (ii) the share of these expenses devoted to energy, denoted  $s_{in}$ ; and (iii) the composition of energy expenditures across three energy source categories: electricity, fossil fuels (natural gas, heating oil, gasoline and diesel, including for motor vehicles), and other energy sources (waste, coal, or wood). For the third item, managers first indicated which energy source categories their firm uses. Managers were then presented with sliders to allocate expenditure shares only across the categories they selected, with shares constrained to sum to one.<sup>17</sup> We denote by  $\omega_{in}^k$  the resulting share of energy expenditures allocated to source  $k \in \{e, f, o\}$  (electricity, fossil fuels, other).

**From expenditures to emissions.** For each energy source category  $k$ , we convert expenditures into physical energy consumption using an average price  $p^k$  (in CHF/kWh), and then into emissions using an average emission coefficient  $\varepsilon^k$  (in g CO<sub>2</sub>/kWh). The firm’s estimated annual emissions, in tons of CO<sub>2</sub>, are

$$\text{CO}_{2,in} = \frac{X_{in} \cdot s_{in}}{10^6} \sum_{k \in \{e, f, o\}} \omega_{in}^k \frac{\varepsilon^k}{p^k}, \quad (65)$$

where the factor  $10^{-6}$  converts grams into tons. Panel A of [Table E.1](#) reports the values of the price and emission parameters, together with their sources. For electricity, we use the official price for medium-sized commercial consumers (consumption profile C3) from the Swiss Federal Electricity Commission (ElCom) price observatory ([Swiss Federal Electricity Commission ElCom, 2024](#)), and an emission coefficient consistent with official estimates of the carbon content of the Swiss consumer electricity mix of approximately 100 g CO<sub>2</sub>/kWh ([Swiss Federal Council, 2022](#); [Horocarbon, 2024](#)). Because the survey does not distinguish between individual fuels within the fossil and other energy source categories, we use average prices and emission coefficients for these two categories. The averages are computed from

<sup>17</sup>Pre-testing revealed that a single allocation question covering all three categories was too complex for respondents. Since most firms use only two energy source categories, the screening question reduces the task, for most respondents, to reporting a single percentage.

fuel-level prices reported by ElCom ([Swiss Federal Electricity Commission ElCom, 2024](#)) and Avenenergy Suisse ([Avenenergy Suisse, 2024](#)), fuel-level emission factors published by the U.S. Energy Information Administration ([U.S. Energy Information Administration, 2023](#)), and the energy statistics of the Swiss Federal Office of Energy ([Swiss Federal Office of Energy, 2024](#)).

**Carbon tax cost estimate.** The firm-specific annual cost of the carbon tax increase communicated to managers is obtained by multiplying estimated emissions by the randomly assigned tax increase  $\tau \in \{100, 300\}$  CHF per ton of CO<sub>2</sub>:

$$\Delta \text{Cost}_{in} = \tau \cdot \text{CO}_{2,in}. \quad (66)$$

Managers were informed that this estimate holds their energy consumption and energy mix fixed, and that the actual cost impact would differ if the firm adjusted either margin. Panel B of [Table E.1](#) illustrates the calculation for a hypothetical firm. Note that Equation (65) captures emissions associated with the firm’s direct energy use, including the (indirect) emissions embodied in purchased electricity via the average carbon content of the Swiss consumer electricity mix; it does not capture non-energy process emissions or emissions embodied in other inputs.

Table E.1: Estimation of Firm-Level CO<sub>2</sub> Emissions: Parameters and Example

| <b>Panel A: Energy prices and emission coefficients</b>       |                 |                                  |                                                                                                              |
|---------------------------------------------------------------|-----------------|----------------------------------|--------------------------------------------------------------------------------------------------------------|
| Parameter                                                     | Notation        | Value                            | Source                                                                                                       |
| Electricity price                                             | $p^e$           | 0.30 CHF/kWh                     | <a href="#">Swiss Federal Electricity Commission ElCom (2024)</a>                                            |
| Fossil fuel price                                             | $p^f$           | 0.16 CHF/kWh                     | <a href="#">Swiss Federal Electricity Commission ElCom (2024)</a> ; <a href="#">Avenenergy Suisse (2024)</a> |
| Other sources price                                           | $p^o$           | 0.10 CHF/kWh                     | <a href="#">Swiss Federal Office of Energy (2024)</a>                                                        |
| Electricity coefficient                                       | $\varepsilon^e$ | 100 g CO <sub>2</sub> /kWh       | <a href="#">Swiss Federal Council (2022)</a>                                                                 |
| Fossil fuel coefficient                                       | $\varepsilon^f$ | 224 g CO <sub>2</sub> /kWh       | <a href="#">U.S. Energy Information Administration (2023)</a>                                                |
| Other sources coefficient                                     | $\varepsilon^o$ | 316 g CO <sub>2</sub> /kWh       | <a href="#">U.S. Energy Information Administration (2023)</a>                                                |
| <b>Panel B: Example calculation for a hypothetical firm</b>   |                 |                                  |                                                                                                              |
| Input / step                                                  | Value           |                                  |                                                                                                              |
| Total operating expenses ( $X_{in}$ )                         | CHF 10,000,000  | Elicited                         |                                                                                                              |
| Energy share of expenses ( $s_{in}$ )                         | 3%              | Elicited                         |                                                                                                              |
| Energy expenditures ( $X_{in} \cdot s_{in}$ )                 | CHF 300,000     |                                  |                                                                                                              |
| Share electricity / fossil ( $\omega_{in}^e, \omega_{in}^f$ ) | 60% / 40%       | Elicited                         |                                                                                                              |
| CO <sub>2</sub> from electricity                              | 60 tons         | $180,000 \times (100/0.30)/10^6$ |                                                                                                              |
| CO <sub>2</sub> from fossil fuels                             | 168 tons        | $120,000 \times (224/0.16)/10^6$ |                                                                                                              |
| Total CO <sub>2</sub> emissions                               | 228 tons        | Equation (65)                    |                                                                                                              |
| Annual cost, tax of CHF 100/ton                               | CHF 22,800      | Equation (66)                    |                                                                                                              |
| Annual cost, tax of CHF 300/ton                               | CHF 68,400      | Equation (66)                    |                                                                                                              |

*Notes:* Panel A reports the energy prices and CO<sub>2</sub> emission coefficients used to estimate firms' annual CO<sub>2</sub> emissions from the elicited expenditure information, following Equation (65). The electricity price corresponds to the price for the C3 consumption profile (medium-sized commercial consumers) reported by the Swiss Federal Electricity Commission (ElCom) for 2024. The electricity emission coefficient reflects official estimates of the carbon content of the Swiss consumer electricity mix of approximately 100 g CO<sub>2</sub>/kWh ([Swiss Federal Council, 2022](#); [Horocarbon, 2024](#)). Because the survey does not distinguish between individual fuels within the fossil and other energy source categories, the corresponding prices and emission coefficients are averages over the underlying fuels (natural gas, heating oil, gasoline, and diesel for fossil fuels; waste, coal, and wood for other sources), computed from fuel-level prices reported by ElCom and Avenenergy Suisse, fuel-level emission factors published by the U.S. Energy Information Administration, and the energy statistics of the Swiss Federal Office of Energy. Panel B illustrates the calculation for a hypothetical firm that uses only electricity and fossil fuels. The resulting cost estimate was presented to managers as shown in [Figure 1](#), together with the caveat that the impact would differ if the firm reduced its energy consumption or changed its energy mix.

## F Firm Population and Survey Sample

### F.1 Firm population

Our firm survey covers the entire Swiss economy, with the exception of (i) the Italian- and Romansh-speaking regions, and (ii) sectors outside the market economy—specifically, Sectors O (Public Administration and Defence; Compulsory Social Security), P (Education), T (Activities of Households as Employers; Undifferentiated Goods- and Services-Producing Activities of Households for Own Use), and U (Activities of Extraterritorial Organisations and Bodies).

**Table F.1** presents firm population numbers from the Business and Enterprise Register (BER) maintained by the Swiss Federal Statistical Office, disaggregated by employee size class and economic sector, according to letter codes in the Swiss General Classification of Economic Activities (NOGA). The NOGA classification corresponds to the Statistical Classification of Economic Activities (NACE) used in the European Community.<sup>18</sup>

### F.2 Address sources

We construct our firm contact sample from three sources: the firm panel maintained by the KOF Swiss Economic Institute at ETH Zurich ([KOF Institute, 2025](#)), an address panel obtained from a survey conducted by us in the year 2023 ([Benhima et al., 2024](#)), and the Orbis firm database ([Moody’s Analytics, 2024](#)). All three sources are used exclusively to obtain contact e-mail addresses for firms, stratified by sector and firm size.<sup>19</sup> The three address sources are presented in the following paragraphs.

**KOF address panel.** The KOF Business Tendency Surveys are conducted monthly or quarterly among Swiss firms across a broad range of sectors ([KOF Institute, 2025](#)). For our study, KOF kindly provided access to the address panel underlying the following sectoral surveys: manufacturing, construction, general services, wholesale services, banking, financial, and insurance services, and project engineering services.

**Address panel from 2023 survey.** Between June and August 2023, we conducted a survey on energy prices and firm decision-making among Swiss firms ([Benhima et al., 2024](#)). The survey sample was drawn from the BER using stratified random sampling based on the size-sector distribution of the BER firm population. Firms were randomly selected within

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<sup>18</sup>Firms located in the Italian- and Romansh-speaking regions account for only 6% of all firms in the BER.

<sup>19</sup>Other firm-level information included in these sources is not used in our analysis. To ensure consistency, all relevant firm information is collected directly through the survey.

Table F.1: Firm Population (French- and German-speaking Switzerland)

|                 | A                | B                | C                | D                  | E                | F                 | G                 | H                | I                   |
|-----------------|------------------|------------------|------------------|--------------------|------------------|-------------------|-------------------|------------------|---------------------|
| Micro (1–9)     | 49163<br>(8.38%) | 112<br>(0.02%)   | 29682<br>(5.06%) | 563<br>(0.10%)     | 1163<br>(0.20%)  | 39266<br>(6.70%)  | 59557<br>(10.16%) | 10732<br>(1.83%) | 21109<br>(3.60%)    |
| Small (10–49)   | 1003<br>(0.17%)  | 94<br>(0.02%)    | 6491<br>(1.11%)  | 150<br>(0.03%)     | 315<br>(0.05%)   | 6385<br>(1.09%)   | 7558<br>(1.29%)   | 1520<br>(0.26%)  | 4919<br>(0.84%)     |
| Medium (50–249) | 65<br>(0.01%)    | 15<br>(0.00%)    | 1822<br>(0.31%)  | 69<br>(0.01%)      | 72<br>(0.01%)    | 793<br>(0.14%)    | 1051<br>(0.18%)   | 395<br>(0.07%)   | 498<br>(0.08%)      |
| Large (250+)    | 2<br>(0.00%)     | 2<br>(0.00%)     | 381<br>(0.06%)   | 34<br>(0.01%)      | 5<br>(0.00%)     | 84<br>(0.01%)     | 202<br>(0.03%)    | 119<br>(0.02%)   | 62<br>(0.01%)       |
| Total           | 50233<br>(8.57%) | 223<br>(0.04%)   | 38376<br>(6.54%) | 816<br>(0.14%)     | 1555<br>(0.27%)  | 46528<br>(7.93%)  | 68368<br>(11.66%) | 12766<br>(2.18%) | 26588<br>(4.53%)    |
|                 | J                | K                | L                | M                  | N                | Q                 | R                 | S                | Total               |
| Micro (1–9)     | 23543<br>(4.01%) | 14527<br>(2.48%) | 18873<br>(3.22%) | 95054<br>(16.21%)  | 21455<br>(3.66%) | 65682<br>(11.20%) | 23899<br>(4.08%)  | 50259<br>(8.57%) | 524639<br>(89.46%)  |
| Small (10–49)   | 2179<br>(0.37%)  | 1581<br>(0.27%)  | 976<br>(0.17%)   | 5935<br>(1.01%)    | 2685<br>(0.46%)  | 4919<br>(0.84%)   | 1346<br>(0.23%)   | 2560<br>(0.44%)  | 50616<br>(8.63%)    |
| Medium (50–249) | 450<br>(0.08%)   | 399<br>(0.07%)   | 117<br>(0.02%)   | 730<br>(0.12%)     | 731<br>(0.12%)   | 1657<br>(0.28%)   | 194<br>(0.03%)    | 325<br>(0.06%)   | 9383<br>(1.60%)     |
| Large (250+)    | 73<br>(0.01%)    | 114<br>(0.02%)   | 10<br>(0.00%)    | 109<br>(0.02%)     | 152<br>(0.03%)   | 419<br>(0.07%)    | 32<br>(0.01%)     | 29<br>(0.00%)    | 1829<br>(0.31%)     |
| Total           | 26245<br>(4.48%) | 16621<br>(2.83%) | 19976<br>(3.41%) | 101828<br>(17.36%) | 25023<br>(4.27%) | 72677<br>(12.39%) | 25471<br>(4.34%)  | 53173<br>(9.07%) | 586467<br>(100.00%) |

*Notes:* The table displays the number of active firms and their percentage shares (in parentheses) in the German- and French-speaking regions of Switzerland, disaggregated by NOGA letter sector and employee size class, as recorded in the Business and Enterprise Register of the Swiss Federal Statistical Office. Sectors O, P, T, and U are excluded. Shares are relative to the grand total of 586,467 firms.

each stratum to ensure representative coverage. A total of 353 respondents registered for future survey participation. In line with the criteria outlined in [Appendix F.1](#), we exclude Italian-speaking respondents and respondents from Sectors O, P, T, and U. Further, we exclude certain legal types: cooperatives, clubs, foundations, public-law institutions, public corporation (administration), public corporation (enterprise).

**Orbis address panel.** The Orbis database provides one of the most comprehensive collections of firm-level contact e-mails in Switzerland. We draw our address sample from Orbis using stratified sampling, based on the population benchmarks shown in [Table F.1](#).

Rather than applying purely proportional stratified sampling, we intentionally oversample selected sectors and larger firms. Regards sector oversampling: for each firm size stratum in Sectors C (Manufacturing), D (Energy Supply), E (Water Supply; Sewerage; Waste Management and Remediation), F (Construction), G (Wholesale and Retail Trade; Repair of Motor Vehicles and Motorcycles), and H (Transportation and Storage), the number of contact e-mails is doubled relative to the values in [Table F.1](#). With regard to oversampling of larger firms: in order to achieve the target distribution across firm size categories in the overall contact sample (see [Appendix F.3](#)), and taking into account the availability of firms in the other two address sources, we needed to sample from the Orbis address database 15,897 micro firms (67.8% of all sampled Orbis firms, as compared to micro firms accounting for 89.5% of all firms in the population), 3,648 small firms (15.5% vs. 8.6%), 2,359 medium-sized firms (10.0% vs. 1.6%), and 1,556 large firms (6.6% vs. 0.3%).

The rationale behind the oversampling strategy is as follows: A larger number of observation units is allocated to those strata of the firm population that contribute more heavily to the overall aggregation, either due to their high number of employees or high value added. As a result, these strata, which are important for the aggregate, are more accurately captured, reducing the expected sampling error, i.e., the deviation of the survey aggregates from the unobserved population aggregate, which would have been obtained had the entire population been surveyed. Furthermore, the decision to oversample the Sectors C to H is motivated by the expectation that these sectors are more likely to exhibit pronounced effects in response to the climate shock scenario, along with higher within-stratum variability among firms, relative to other sectors. For other sectors, by contrast, we anticipate the effects to be rather small or even zero, and the cross-firm variance to be relatively low. Since higher variance implies a larger expected sampling error, *ceteris paribus*, it justifies a proportionally larger sample size for these high-variance cells. Notably, during the aggregation of survey results, the over- and undersampling is corrected by reweighting firms according to population statistics.

### F.3 Contact sample

Our contact sample combines the address panels and includes 26,900 firms.<sup>20</sup> Table F.3 reports the composition of the contact sample by firm size and sector. The sample includes 16,950 micro firms (63.0%), 5,192 small firms (19.3%), 3,037 medium-sized firms (11.3%), and 1,721 large firms (6.4%). We set these targets ex ante to oversample firms larger than micro firms (see Appendix F.2).

Table F.2 compares the resulting distribution of firms across size classes with the corresponding population distribution. As intended, medium-sized and large firms account for substantially larger shares of the contact sample than of the population, while micro firms are underrepresented.

Table F.2: Firm Size Distribution in Sample and Population

| Variable                                      | Sample | Population |
|-----------------------------------------------|--------|------------|
| Share of micro firms (1–9 employees)          | 0.63   | 0.90       |
| Share of small firms (10–49 employees)        | 0.19   | 0.09       |
| Share of medium-size firms (50–249 employees) | 0.11   | 0.02       |
| Share of large firms (250+ employees)         | 0.06   | 0.003      |

*Notes:* The table compares the distribution of firms across employment size classes in the survey and population samples.

<sup>20</sup>We capped the number of firms contacted at this level because the KOF survey e-mail system can send 9,000 e-mails per day (i.e., 10,000 minus 1,000 reserved for other communications) and because we sought to complete distribution within three days.

Table F.3: Firm Contact Sample

|                 | A               | B              | C                | D                | E              | F                | G                | H               | I                  |
|-----------------|-----------------|----------------|------------------|------------------|----------------|------------------|------------------|-----------------|--------------------|
| Micro (1–9)     | 1171<br>(4.35%) | 3<br>(0.01%)   | 1560<br>(5.80%)  | 24<br>(0.09%)    | 56<br>(0.21%)  | 2058<br>(7.65%)  | 3009<br>(11.19%) | 540<br>(2.01%)  | 504<br>(1.87%)     |
| Small (10–49)   | 50<br>(0.19%)   | 5<br>(0.02%)   | 1043<br>(3.88%)  | 12<br>(0.04%)    | 31<br>(0.12%)  | 1026<br>(3.81%)  | 927<br>(3.45%)   | 187<br>(0.70%)  | 254<br>(0.94%)     |
| Medium (50–249) | 12<br>(0.04%)   | 3<br>(0.01%)   | 853<br>(3.17%)   | 26<br>(0.10%)    | 26<br>(0.10%)  | 411<br>(1.53%)   | 437<br>(1.62%)   | 156<br>(0.58%)  | 89<br>(0.33%)      |
| Large (250+)    | 1<br>(0.00%)    | 1<br>(0.00%)   | 483<br>(1.80%)   | 44<br>(0.16%)    | 6<br>(0.02%)   | 116<br>(0.43%)   | 260<br>(0.97%)   | 151<br>(0.56%)  | 37<br>(0.14%)      |
| Total           | 1234<br>(4.59%) | 12<br>(0.04%)  | 3939<br>(14.64%) | 106<br>(0.39%)   | 119<br>(0.44%) | 3611<br>(13.42%) | 4633<br>(17.22%) | 1034<br>(3.84%) | 884<br>(3.29%)     |
|                 | J               | K              | L                | M                | N              | Q                | R                | S               | Total              |
| Micro (1–9)     | 656<br>(2.44%)  | 391<br>(1.45%) | 481<br>(1.79%)   | 2489<br>(9.25%)  | 560<br>(2.08%) | 1626<br>(6.04%)  | 602<br>(2.24%)   | 1220<br>(4.54%) | 16950<br>(63.01%)  |
| Small (10–49)   | 195<br>(0.72%)  | 152<br>(0.57%) | 80<br>(0.30%)    | 512<br>(1.90%)   | 185<br>(0.69%) | 307<br>(1.14%)   | 85<br>(0.32%)    | 141<br>(0.52%)  | 5192<br>(19.30%)   |
| Medium (50–249) | 111<br>(0.41%)  | 109<br>(0.41%) | 31<br>(0.12%)    | 165<br>(0.61%)   | 151<br>(0.56%) | 353<br>(1.31%)   | 40<br>(0.15%)    | 64<br>(0.24%)   | 3037<br>(11.29%)   |
| Large (250+)    | 55<br>(0.20%)   | 56<br>(0.21%)  | 8<br>(0.03%)     | 78<br>(0.29%)    | 103<br>(0.38%) | 283<br>(1.05%)   | 19<br>(0.07%)    | 20<br>(0.07%)   | 1721<br>(6.40%)    |
| Total           | 1017<br>(3.78%) | 708<br>(2.63%) | 600<br>(2.23%)   | 3244<br>(12.06%) | 999<br>(3.71%) | 2569<br>(9.55%)  | 746<br>(2.77%)   | 1445<br>(5.37%) | 26900<br>(100.00%) |

*Notes:* The table displays the number of firms in the contact sample and their percentage shares (in parentheses), disaggregated by NOGA letter sector and employee size class. Shares are relative to the grand total of 26,900 firms.

## G Assessment of Sample Selection

Firms that are more exposed to energy costs may be more likely to participate in the survey, as the topic is more salient to their operations. If participation is positively correlated with both policy exposure and expected economic impacts, the response sample may not be representative of the underlying firm population. In particular, energy-intensive firms may report stronger expected effects on output, prices, employment, and profits. As a result, estimated average effects based on the survey data could overstate the magnitude of adjustment effects at the population level.

To assess this potential non-response bias, we test whether survey participation is systematically related to firms’ energy intensity. We first do so in a simple descriptive way, and then turn to regressions where we also control for firm size, another potentially important driver of survey participation.

**Energy-intensity data.** We do not have measures of firm-level energy consumption for the set of firms that were invited to participate but did not do so. We therefore use sectoral energy intensity data from [Swiss Federal Office of Energy \(2026\)](#) and assign it to firms based on their sector affiliation. Our baseline measure (“Energy use, measure 1”) is energy consumption in terajoule per firm at the NOGA 2-digit division level, using firm counts from [Federal Statistical Office \(2026\)](#). Since this construction combines data from distinct sources and may introduce measurement inconsistencies, we consider an alternative measure (“Energy use, measure 2”) based on broader industry aggregates (e.g., 12 manufacturing and seven service sectors), where both energy consumption and firm counts are taken directly from [Swiss Federal Office of Energy \(2026\)](#).

**Descriptive differences and effect size (Cohen’s  $d$ ).** [Table G.1](#) reports mean energy intensity for respondents and non-respondents. Respondents are defined as firms that completed the survey (see [Table 1](#)). We exclude observations that accessed the survey link but did not begin or complete the questionnaire from the difference-in-means tests. While differences are statistically significant, we assess their economic magnitude using Cohen’s  $d$ , defined as  $(\mu_{\text{non-respondents}} - \mu_{\text{respondents}})/s_{\text{pooled}}$ , where  $s_{\text{pooled}}$  denotes the pooled standard deviation. This measure expresses group differences in standard deviation units and follows [Cohen \(1988\)](#). Across both energy-use measures, Cohen’s  $d$  is small, suggesting limited economic relevance of the observed selection differences.

Table G.1: Energy Intensity: Mean Differences by Response Status

| Energy use | Non-respondents |               | Respondents |               | $\Delta$          | Cohen's $d$ |
|------------|-----------------|---------------|-------------|---------------|-------------------|-------------|
|            | $n$             | $\mu$ (s.e.)  | $n$         | $\mu$ (s.e.)  |                   |             |
| Measure 1  | 21593           | 0.705 (0.018) | 1023        | 0.966 (0.091) | -0.261*** (0.083) | 0.101       |
| Measure 2  | 22643           | 0.926 (0.048) | 1057        | 1.399 (0.334) | -0.473** (0.235)  | 0.063       |

*Notes:* Measure 1 (measure 2) is energy use in terajoule per firm at the NOGA 2-digit division level (at the industry level). \*, \*\*, and \*\*\* indicate significance at the 10%, 5%, and 1% levels, respectively.  $\Delta = \mu_{\text{non-respondents}} - \mu_{\text{respondents}}$ . Cohen's  $d = \mu_{\text{non-respondents}} - \mu_{\text{respondents}} / s_{\text{pooled}}$ .

**Regression-based tests of selection.** The previous analysis compares energy intensity between participants and non-participants, but does not control for another variable that may affect selection into the survey, namely firm size. It is common in establishment surveys to find that response rates increase strongly with firm size. A plausible explanation is that the fixed time cost of completing the survey represents a much smaller burden for larger firms than for smaller firms. Even when the respondent is a senior executive in both cases, executives in larger organizations may have greater scope to allocate time to external requests (and may be more used to doing so), whereas executives in smaller firms are more directly involved in day-to-day operations and therefore face higher opportunity costs of participation.<sup>21</sup>

We next estimate probit models of survey response as a function of firm size (as measured by the log of the number of full-time employees) and log firm-level energy intensity, imputed as described above. We do this for three different measures of survey response: (i) whether a respondent at least started the survey (by clicking on the link in the invitation email); (ii) whether they completed the survey (even with some skipped questions); and (iii) whether a response is in our main analysis sample, i.e. it passed the filters described in the main text.

Table G.2 reports the result. We find in each case that participation and completion are strongly related to firm size, as expected. For instance, in the first three columns, a doubling of firm size is associated with a 3 percentage point higher probability of opening the survey (i.e. at least click on the invitation link in the email), relative to a sample mean of 14.5%. The subsequent columns report effects of similar magnitude (when scaled to the mean of the respective depending variables) for completing the survey and ending up in our main sample. More importantly, however, we find that energy intensity, based on either of

<sup>21</sup>It is also possible that, especially for the Orbis sample, the invitation email may not have reached anybody at all, if the firm is no longer in operations. As the exit rate is much higher for small firms than for large firms, this would again lead to higher measured participation by larger firms.

the two measures describe above, does not significantly correlate with survey participation or completion. Furthermore, the economic magnitudes of the point estimates are very small. These estimates thus indicate that potential selection into participation by firms that may be more interested or vested in the topic because of differential energy intensity is likely not a concern.<sup>22</sup>

We finally note that selection by firm size is not problematic for our analysis since we reweight responses to make them representative of the Swiss economy (see [Appendix H](#) for the weighting procedure).

Table G.2: Probit models of selection into survey participation and completion

|                             | Started (0/1)       |                     |                     | Completed (0/1)     |                     |                     | Main sample (0/1)   |                     |                     |
|-----------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                             | (1)                 | (2)                 | (3)                 | (4)                 | (5)                 | (6)                 | (7)                 | (8)                 | (9)                 |
| Log(Employment)             | 0.030***<br>(0.001) | 0.030***<br>(0.001) | 0.030***<br>(0.001) | 0.011***<br>(0.001) | 0.011***<br>(0.001) | 0.011***<br>(0.001) | 0.010***<br>(0.001) | 0.010***<br>(0.001) | 0.010***<br>(0.001) |
| Log(Energy Int., Measure 1) |                     | -0.000<br>(0.002)   |                     |                     | -0.001<br>(0.001)   |                     |                     | -0.000<br>(0.001)   |                     |
| Log(Energy Int., Measure 2) |                     |                     | 0.001<br>(0.002)    |                     |                     | 0.002<br>(0.001)    |                     |                     | 0.002<br>(0.001)    |
| Nr. obs.                    | 26295               | 26295               | 26295               | 26295               | 26295               | 26295               | 26295               | 26295               | 26295               |
| Mean (dep. var.)            | 0.145               | 0.145               | 0.145               | 0.049               | 0.049               | 0.049               | 0.036               | 0.036               | 0.036               |

This table reports average marginal effects from Probit regressions of different binary measures of survey participation on a firm's log employment (in terms of full-time employees) and its log energy intensity (based on the imputation explained in the text). For firms where the energy intensity could not be imputed, we used the sample mean and controlled for the imputation with a dummy variable (not shown). Robust standard errors in parentheses. \*, \*\*, and \*\*\* indicate significance at the 10%, 5%, and 1% levels, respectively.

## H Aggregation

This appendix describes the construction of the post-stratification weights used to aggregate firm-level outcomes elicited in our survey to the sectoral and economy-wide levels.

### H.1 Employment-weighting with cell-size adjustment

We employ an aggregation procedure with employment scaling and cell-size adjustment. Suppose that the economy is a collection of cells, denoted  $c$ . Economy-wide output  $\hat{Y}$  is the

<sup>22</sup>To reconcile these findings with the ones of small but significant differences in [Table G.1](#), note that log employment and log energy intensity are positively correlated ( $\rho = +0.31$  for measure 1 and  $+0.20$  for measure 2).

sum of the population value added  $Y_c^{(p)}$  at the cell level:

$$\hat{Y} = \sum_c Y_c^{(p)}. \quad (67)$$

We can make the sample value added  $Y_c^{(s)}$  appear in the expression:

$$\hat{Y} = \sum_c \frac{Y_c^{(p)}}{Y_c^{(s)}} Y_c^{(s)} = \sum_c \frac{Y_c^{(p)}}{Y_c^{(s)}} \sum_{i \in c} Y_i, \quad (68)$$

where  $Y_i$  are the responses collected in the sample. Because valued added data at the cell level may not be available, the ratio of population to firms' sample value added is approximated using employment:

$$\hat{Y} = \sum_c \frac{E_c^{(p)}}{E_c^{(s)}} \sum_{i \in c} Y_i. \quad (69)$$

where  $E_c^{(p)}$  and  $E_c^{(s)}$  are the population and sample total employments of cell  $c$ . The corresponding growth rate of output can then be approximated as follows (lower-case letters denote growth rates):

$$\hat{y} = \sum_c \frac{E_c^{(p)}}{E_c^{(s)}} \sum_{i \in c} \frac{Y_i}{\hat{Y}} y_i. \quad (70)$$

Again, we approximate value added shares using employment:

$$\hat{y} = \sum_c \sum_{i \in c} \frac{E_c^{(p)}}{E_c^{(s)}} \frac{E_i}{\hat{E}} y_i. \quad (71)$$

where  $\hat{E}$  is the economy's total employment and  $E_i$  is employment reported by firm  $i$ . We now make the assumption that the firm-level growth rate can be decomposed into a cell-specific component  $y_{c(i)}$  and an orthogonal, zero-mean firm-specific component  $\epsilon_i$ :  $y_i = y_{c(i)} + \epsilon_i$ . We replace it in the growth expression:

$$\hat{y} = \sum_c \sum_{i \in c} \frac{E_c^{(p)}}{E_c^{(s)}} \frac{E_i}{\hat{E}} (y_c + \epsilon_i) \quad (72)$$

$$= \sum_c \frac{E_c^{(p)}}{E_c^{(s)}} \underbrace{\sum_{i \in c} \frac{E_i}{\hat{E}} y_c}_{\frac{E_c^{(s)}}{\hat{E}} y_c} + \sum_c \sum_{i \in c} \frac{E_c^{(p)}}{E_c^{(s)}} \frac{E_i}{\hat{E}} \epsilon_i \quad (73)$$

$$= \sum_c \frac{E_c^{(p)}}{\hat{E}} y_c + \sum_c \sum_{i \in c} \frac{E_c^{(p)}}{E_c^{(s)}} \frac{E_i}{\hat{E}} \epsilon_i. \quad (74)$$

Note that  $\sum_c \frac{E_c^{(p)}}{\hat{E}} = 1$  and  $\sum_c \sum_{i \in c} \frac{E_c^{(p)}}{E_c^{(s)}} \frac{E_i}{\hat{E}} = 1$ , so that both terms are weighted averages, respectively of the cell growth  $y_c$  across cells and of the firm-specific terms  $\epsilon_i$  across the sample.

The weighted average of  $\epsilon_i$ ,  $\sum_c \sum_{i \in c} \frac{E_c^{(p)}}{E_c^{(s)}} \frac{E_i}{\bar{E}} \epsilon_i$ , converges to zero in the population. However, because of the small size of the sample within a cell, large firm responses will weigh disproportionately in that sum. To avoid that issue, we replace this term with its population asymptotic value, which is zero.<sup>23</sup> We then obtain:

$$\hat{y}_{\text{total}} \simeq \sum_c \frac{E_c^{(p)}}{\hat{E}} y_c. \quad (75)$$

Now, note that  $y_c$  can be approximated as the simple average of the sample growth rates within cell  $c$ :  $y_c \simeq \sum_{i \in c} y_i / N_c^{(s)}$ , where  $N_c^{(s)}$  is the number of firms in the sample that belong to cell  $c$ . Substituting in the above equation and rearranging, we obtain:

$$\hat{y}_{\text{total}} \simeq \sum_c \sum_{i \in c} \omega_i y_i, \quad (76)$$

where the post-stratification weights are given by

$$\omega_i = \frac{E_c^{(p)}}{\hat{E} N_c^{(s)}}. \quad (77)$$

Equation (77) nicely expresses what the aggregation procedure is all about: Firms are re-weighted according to the employment size of their respective cell in the population through  $E_c^{(p)}$ . In addition, the term  $N_c^{(s)}$  adjusts for over- or under-representation of a given cell in our sample.

For aggregation of firm-level outcomes to the sectoral level, the analogous expression is:

$$\hat{y}_j \simeq \sum_{c \in j} \sum_{i \in c} \omega_i^{(j)} y_i, \quad (78)$$

where the firm-specific post-stratification weights are given by

$$\omega_i^{(j)} = \frac{E_c^{(p)}}{E_j^{(p)} N_c^{(s)}}. \quad (79)$$

The aggregation procedure is simple and transparent, avoiding double weighting or redundancy while yielding stable estimates when employment is strongly correlated with value added within cells. It is also relatively robust to outlier responses from large firms, preventing aggregate estimates from being misleadingly driven by the noisy responses of large firms in our sample.

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<sup>23</sup>Even if the population firm-size distribution is granular ([Gabaix, 2011](#)), so that the population weighted average need not converge to zero, the small number of firms in the sample makes the sample weighted average particularly sensitive to the realization of firm sizes. Setting this term to zero shuts down this source of sample-specific variation and ensures that the results are not driven by the particular concentration of firm sizes in the sample.

## H.2 Alternative aggregation schemes

For robustness, we consider three alternative aggregation schemes.

**Unweighted means.** As a first alternative, we compute unweighted means. Specifically, we aggregate firm-level outcomes to the economy-wide or sectoral level using weights

$$\omega_i = \frac{1}{N^{(s)}}, \quad (80)$$

and

$$\omega_i^{(j)} = \frac{1}{N_j^{(s)}}, \quad (81)$$

where  $N^{(s)}$  denotes the total number of firms in the sample and  $N_j^{(s)}$  the number of sampled firms in sector  $j$ .

**Simple employment-weighting.** As a second alternative, we weight firms according to Equation (71):

$$\omega_i = \frac{E_c^{(p)}}{E_c^{(s)}} \frac{E_i}{\hat{E}}, \quad (82)$$

and

$$\omega_i^{(j)} = \frac{E_c^{(p)}}{E_c^{(s)}} \frac{E_i}{E_j^{(p)}}. \quad (83)$$

This scheme tends to place relatively more weight on large firms within a cell, in contrast to the post-stratification weights in Equations (76)–(77), which equalizes firm weights within cells.

**Two-stage aggregation.** As a third alternative, we employ a two-stage aggregation. First, within each sector, firms are re-weighted according to the employment size of their respective cell in the population. The resulting sector totals are then re-weighted by the sector's share in total value added to produce an economy-wide aggregate. Formally, the procedure can be written as follows:

$$\text{Within-sector: } y_j^{(emp)} = \sum_{c \in j} \sum_{i \in c} \frac{E_c^{(p)}}{E_j^{(p)} N_c^{(s)}} y_i \quad (84)$$

$$\text{Across-sector: } \hat{y}_{\text{total}} = \sum_j \frac{Y_j^{(p)}}{\hat{Y}_{\text{total}}} y_j^{(emp)}, \quad (85)$$

where  $Y_j^{(p)}$  is the population value added of sector  $j$ . The first step applies the one-stage aggregation to sector  $j$ . The second step uses the level relation between total output and

value added at the sector level,  $\hat{Y}_{total} = \sum_j Y_j$ , to derive an approximated relationship between the growth rates.

The two-stage aggregation procedure reflects sectoral economic importance by weighting sectors according to their value-added shares rather than employment alone, making use of the availability of value added information at the sector level, and thereby correcting for unequal sectoral representation in the sample and aligning the resulting aggregate more closely with standard macroeconomic measures.

The idea of weighting survey data in two stages is supported both in official survey manuals and in statistical literature. For instance, [OECD \(2003\)](#) recommends using employment weights within sectors and value-added weights across sectors when aggregating (qualitative) answers in business tendency survey.

## I Additional Results and Robustness

### I.1 Unweighted results

In the main part of the paper, our results are weighted, using the approach described in [Appendix H.1](#), such that the firm-level responses to carbon tax scenarios represent economy-wide aggregate statistics. We now report the same report without weights and show that the results are qualitatively similar. There are two takeaways from this exercise. First, it shows that our aggregation procedure is not the main driver of our results. Second, the composition of our sample, although it diverges with the population distribution in some dimensions (see [Appendix G](#)), does not impact the average responses to carbon tax scenarios we elicit. Put simply, our results are not driven by some firms that have bigger weights in the Swiss economy.

Table I.1: Unweighted Firm Responses to a National Carbon Tax Increase

|                                 | All   | Tax increase (CHF/Ton CO <sub>2</sub> ) |       |                    | S.E. |
|---------------------------------|-------|-----------------------------------------|-------|--------------------|------|
|                                 |       | 100                                     | 300   | $\Delta_{100-300}$ |      |
| $\Delta$ Energy costs (%)       | 15.99 | 8.22                                    | 23.76 | -15.54             | 0.66 |
| $\Delta$ Real output (%)        | -2.11 | -1.76                                   | -2.46 | 0.71               | 0.39 |
| $\Delta$ Sales prices (%)       | 2.79  | 2.55                                    | 3.02  | -0.47              | 0.33 |
| $\Delta$ Energy consumption (%) | -1.44 | -1.34                                   | -1.53 | 0.19               | 0.29 |
| $\Delta$ Employment (%)         | -1.66 | -1.40                                   | -1.93 | 0.53               | 0.29 |
| $\Delta$ Profits (%)            | -3.08 | -2.66                                   | -3.50 | 0.84               | 0.37 |
| Observations                    | 1,017 | 508                                     | 509   | 1,017              | .    |

*Notes:* The table reports managers' expectations of the effects of an increase in the national carbon tax on outcomes for their own firms. Entries are unweighted mean expected percentage changes across responding managers. Results are shown for the full sample and separately for managers randomly assigned to carbon tax increases of CHF 100 and CHF 300 per ton of CO<sub>2</sub>.

Table I.2: Unweighted Firms' Own Output Responses and Beliefs About Others' Responses

| <b>Panel A: Own output across tax scenarios</b>                    |                    |                    |                    |                             |                           |
|--------------------------------------------------------------------|--------------------|--------------------|--------------------|-----------------------------|---------------------------|
|                                                                    | National<br>tax    | Sectoral<br>tax    | Firm<br>tax        | Difference:<br>Nat – Sector | Difference:<br>Nat – Firm |
| $\Delta$ Real output (%)                                           | -1.29<br>(0.14)    | -1.67<br>(0.15)    | -2.57<br>(0.19)    | 0.38<br>(0.21)              | 1.28<br>(0.24)            |
| Share < 0                                                          | 0.49               | 0.50               | 0.53               | .                           | .                         |
| Share = 0                                                          | 0.36               | 0.38               | 0.36               | .                           | .                         |
| <b>Panel B: Own output vs. beliefs about others (national tax)</b> |                    |                    |                    |                             |                           |
|                                                                    | Own firm<br>impact | Sectoral<br>impact | National<br>impact | Difference:<br>Sector – Own | Difference:<br>Nat – Own  |
| $\Delta$ Real output (%)                                           | -1.29<br>(0.14)    | -1.42<br>(0.14)    | -1.95<br>(0.14)    | 0.14<br>(0.20)              | 0.66<br>(0.20)            |
| Share < 0                                                          | 0.49               | 0.52               | 0.62               | .                           | .                         |
| Share = 0                                                          | 0.36               | 0.33               | 0.26               | .                           | .                         |

*Notes:* Panel A reports managers' expected effects of three carbon tax scenarios—a national tax increase, a sector-specific tax increase, and a firm-specific tax increase—on real output of their own firm. Tax increases are normalized to CHF 100 per ton of CO<sub>2</sub>. Panel B compares managers' expected effects of the national carbon tax increase on real output of their own firm, their own sector, and the national economy. Entries in rows labeled  $\Delta$  Output (%) are mean expected percentage changes across respondents. Rows labeled “Share < 0” and “Share = 0” report the fractions of respondents expecting negative and zero output effects.

Table I.3: Unweighted Policy Elasticity, Strategic Interactions, and Higher-order Errors

|                                                                        | Mean  | S.E. | Share < 0 | Share > 0 | # Obs |
|------------------------------------------------------------------------|-------|------|-----------|-----------|-------|
| <b>Panel A: Policy elasticity and strategic interactions</b>           |       |      |           |           |       |
| Policy elasticity $E_{in}(\sigma_{in})$                                | 2.57  | 0.19 | 0.12      | 0.53      | 1,017 |
| Sectoral-level strategic $\alpha_{in}^s$                               | -0.50 | 0.07 | 0.46      | 0.17      | 645   |
| National-level strategic $\alpha_{in}^a$                               | -0.28 | 0.08 | 0.43      | 0.20      | 609   |
| <b>Panel B: Errors in beliefs about national and sectoral outcomes</b> |       |      |           |           |       |
| National error w. national tax                                         | -0.79 | 0.14 | 0.45      | 0.55      | 1,017 |
| Sectoral error w. national tax                                         | -0.09 | 0.14 | 0.36      | 0.64      | 1,017 |
| Sectoral error w. sectoral tax                                         | -0.05 | 0.16 | 0.35      | 0.65      | 1,017 |

*Notes:* Panel A reports estimates of the policy elasticity and strategic interaction parameters for respondents with non-missing values for each parameter. Panel B reports national and sectoral higher-order errors, defined as the difference between managers' expectations of economy-wide or sectoral output effects and the corresponding effects implied by aggregating respondents' expectations for their own firms (see [Appendix H](#)). Entries report unweighted means and the shares of negative and positive values.

## I.2 Additional heterogeneity results

The two tables below complement [Table 5](#) and [Table 6](#). They present the estimated coefficients for the other controls included in the regression.

Table I.4: Additional Controls — Policy Elasticity and Strategic Interaction Parameters

| Dependent Variable            | $\sigma_{in}$<br>(I) | $\mathbb{1}\{\sigma_{in} > 0\}$<br>(II) | $\alpha_{in}^s$<br>(III) | $\mathbb{1}\{\alpha_{in}^s < 0\}$<br>(IV) | $\alpha_{in}^a$<br>(V) | $\mathbb{1}\{\alpha_{in}^a < 0\}$<br>(VI) |
|-------------------------------|----------------------|-----------------------------------------|--------------------------|-------------------------------------------|------------------------|-------------------------------------------|
| Manufacturing-Machine (bin.)  | -0.046<br>(0.548)    | 0.050<br>(0.041)                        | -0.185<br>(0.263)        | 0.041<br>(0.057)                          | 0.156<br>(0.329)       | 0.082<br>(0.059)                          |
| Share of energy cost          | -0.059<br>(0.059)    | -0.000<br>(0.003)                       | -0.004<br>(0.009)        | 0.004<br>(0.003)                          | -0.025<br>(0.036)      | 0.002<br>(0.004)                          |
| Uses electricity (bin.)       | -1.171<br>(0.870)    | 0.024<br>(0.103)                        | -0.054<br>(0.341)        | 0.098<br>(0.099)                          | 0.560**<br>(0.306)     | -0.125<br>(0.114)                         |
| Uses fossil (bin.)            | -1.202**<br>(0.530)  | 0.132**<br>(0.048)                      | -0.164<br>(0.279)        | 0.091<br>(0.062)                          | -0.262<br>(0.225)      | 0.096<br>(0.069)                          |
| Turnover (log) (millions)     | 0.175<br>(0.110)     | -0.005<br>(0.012)                       | -0.007<br>(0.043)        | 0.015<br>(0.013)                          | -0.058*<br>(0.034)     | 0.034***<br>(0.011)                       |
| Own CO2 more others (bin.)    | -0.770<br>(1.047)    | -0.057<br>(0.101)                       | 0.009<br>(0.252)         | -0.150<br>(0.131)                         | 0.716<br>(0.659)       | 0.025<br>(0.133)                          |
| Sector CO2 more others (bin.) | -0.207<br>(0.746)    | 0.055<br>(0.059)                        | 0.282<br>(0.251)         | 0.044<br>(0.079)                          | -0.399<br>(0.444)      | 0.004<br>(0.079)                          |
| CEO, Director, CFO (bin.)     | -0.730<br>(0.608)    | -0.102**<br>(0.048)                     | 0.375**<br>(0.196)       | -0.051<br>(0.058)                         | 0.087<br>(0.293)       | -0.017<br>(0.064)                         |
| Manager: years in role        | 0.007<br>(0.021)     | 0.001<br>(0.002)                        | 0.003<br>(0.008)         | -0.002<br>(0.002)                         | -0.006<br>(0.008)      | -0.001<br>(0.002)                         |
| Manager: male (bin.)          | -0.421<br>(0.568)    | 0.079<br>(0.058)                        | -0.215<br>(0.239)        | 0.093<br>(0.074)                          | -0.451*<br>(0.287)     | 0.150**<br>(0.075)                        |
| Master or PhD (bin.)          | -0.089<br>(0.525)    | -0.017<br>(0.045)                       | -0.564**<br>(0.267)      | 0.096<br>(0.061)                          | 0.016<br>(0.293)       | 0.032<br>(0.065)                          |
| High tax level (300 CHF)      | 1.870***<br>(0.454)  | 0.017<br>(0.040)                        | 0.177<br>(0.203)         | 0.015<br>(0.052)                          | -0.096<br>(0.248)      | 0.028<br>(0.056)                          |
| Survey duration (min)         | -0.000<br>(0.000)    | 0.000<br>(0.000)                        | 0.000<br>(0.000)         | -0.000<br>(0.000)                         | -0.000<br>(0.000)      | -0.000<br>(0.000)                         |
| Constant                      | -4.360*<br>(2.248)   | 0.802***<br>(0.211)                     | -0.346<br>(0.952)        | 0.281<br>(0.236)                          | 0.484<br>(1.252)       | 0.041<br>(0.272)                          |
| Observations                  | 970                  | 970                                     | 621                      | 621                                       | 588                    | 588                                       |
| Adj. R-squared                | 0.138                | 0.131                                   | 0.020                    | 0.066                                     | 0.032                  | 0.023                                     |

Notes: This table reports coefficient estimates for the additional control variables included in the specifications reported in Table 5 but omitted from that table for brevity. Robust standard errors are reported in parentheses. \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

Table I.5: Additional Controls — National and Sectoral Errors

| Dependent Variable            | National Error<br>(I) | $\mathbb{1}\{\text{Nat. Error} < 0\}$<br>(II) | Sectoral Error<br>(III) | $\mathbb{1}\{\text{Sect. Error} < 0\}$<br>(IV) |
|-------------------------------|-----------------------|-----------------------------------------------|-------------------------|------------------------------------------------|
| Manufacturing-Machine (bin.)  | -0.145<br>(0.411)     | 0.023<br>(0.045)                              | 0.788*<br>(0.437)       | -0.158***<br>(0.043)                           |
| Share of energy cost          | 0.040<br>(0.033)      | 0.002<br>(0.004)                              | -0.023<br>(0.032)       | 0.006*<br>(0.003)                              |
| Uses electricity (bin.)       | -0.569<br>(0.974)     | 0.052<br>(0.094)                              | -0.410<br>(0.919)       | 0.065<br>(0.098)                               |
| Uses fossil (bin.)            | -0.178<br>(0.392)     | 0.099**<br>(0.049)                            | -0.274<br>(0.396)       | 0.069<br>(0.048)                               |
| Turnover (log) (millions)     | 0.085<br>(0.089)      | -0.013<br>(0.011)                             | -0.101<br>(0.130)       | -0.002<br>(0.010)                              |
| Own CO2 more others (bin.)    | 0.105<br>(0.635)      | -0.021<br>(0.099)                             | -0.004<br>(0.652)       | 0.094<br>(0.108)                               |
| Sector CO2 more others (bin.) | -1.076**<br>(0.522)   | 0.011<br>(0.063)                              | -1.117*<br>(0.569)      | 0.003<br>(0.058)                               |
| CEO, Director, CFO (bin.)     | -0.017<br>(0.451)     | -0.006<br>(0.052)                             | -0.212<br>(0.450)       | -0.025<br>(0.049)                              |
| Manager: years in role        | -0.021<br>(0.017)     | 0.001<br>(0.002)                              | -0.021<br>(0.016)       | 0.002<br>(0.002)                               |
| Manager: male (bin.)          | 0.744*<br>(0.448)     | -0.088<br>(0.063)                             | 1.058**<br>(0.534)      | -0.025<br>(0.058)                              |
| Master or PhD (bin.)          | -0.325<br>(0.358)     | 0.090*<br>(0.046)                             | 0.030<br>(0.398)        | -0.012<br>(0.046)                              |
| High tax level (300 CHF)      | 0.885**<br>(0.348)    | 0.001<br>(0.043)                              | 0.701*<br>(0.359)       | -0.038<br>(0.041)                              |
| Survey duration (min)         | -0.000<br>(0.000)     | 0.000<br>(0.000)                              | 0.001**<br>(0.000)      | -0.000***<br>(0.000)                           |
| Constant                      | -2.385<br>(1.667)     | 0.520**<br>(0.215)                            | -0.311<br>(1.817)       | 0.444**<br>(0.207)                             |
| Observations                  | 970                   | 970                                           | 970                     | 970                                            |
| Adj. R-squared                | 0.036                 | 0.065                                         | 0.042                   | 0.076                                          |

*Notes:* This table reports coefficient estimates for the additional control variables included in the specifications reported in [Table 6](#) but omitted from that table for brevity. Robust standard errors are reported in parentheses. \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

### I.3 Robustness for policy elasticity and interaction parameters

We now report additional estimates of the strategic and policy parameters, obtained by excluding different sets of observations and by considering alternative assumptions about potential survey response inconsistencies.

We first examine participants who reported that the carbon tax would raise their real output. In our main sample ( $N = 1,019$ ), 110 managers reported an increase under a tax targeted only at their own firm; of these, 98 also reported increases under the other scenarios, and 82 reported positive values for every scenario and outcome (own firm, other firms, and economy-wide), suggesting they made a sign error in reporting their responses. In ??, Panel B excludes observations with positive values in two or more scenarios, and Panel C instead recodes positive real output impacts as negative. Both adjustments raise the policy elasticity but leave the strategic interaction parameters, and their shares of positive and negative values, largely unchanged. This is intuitive: the strategic parameters are identified by the relative movement of responses across scenarios, and managers reporting positive outcomes thus adjusted much like everyone else, suggesting a simple sign error in their response.

Panel D examines the truncation of expected impacts. The most extreme survey option was a decline of 21% or more, coded as  $-21\%$  in our baseline; recoding these values as  $-25\%$  has little effect on the results.

Our strategic parameters are recovered from a system of equations in which a zero denominator produces a missing value. These zeros at the denominator may arise because responses are integers: a respondent reporting 0 for a competitor's or economy-wide expected change may hold a slightly negative belief that rounds to zero. Rather than dropping such observations, as in our baseline, Panel E replaces the zero denominator with  $-0.3$ , the interior of the rounding interval  $(-0.5, 0)$  from which such a value would be rounded. This has the largest impact among our checks, with the strategic interaction parameters becoming considerably more negative; because dividing by a small number inflates the estimate, however, the check is sensitive to denominators close to zero. Finally, Panel F sets all strategic interaction parameters with missing values to zero, a conservative lower bound that rules out strategic interactions entirely for the managers whose parameters we cannot infer.

Table I.6: Robustness: Policy Elasticity and Strategic Interactions Parameters

|                                                                     | Mean  | S.E. | Share < 0 | Share > 0 | # Obs |
|---------------------------------------------------------------------|-------|------|-----------|-----------|-------|
| <b>Panel A: Main Results</b>                                        |       |      |           |           |       |
| Policy elasticity $E_{in}(\sigma_{in})$                             | 2.68  | 0.25 | 0.11      | 0.52      | 1,017 |
| Sectoral-level strategic $\alpha_{in}^s$                            | -0.46 | 0.10 | 0.45      | 0.20      | 645   |
| National-level strategic $\alpha_{in}^a$                            | -0.25 | 0.13 | 0.42      | 0.20      | 609   |
| <b>Panel B: Managers with Positive Real Output Excluded</b>         |       |      |           |           |       |
| Policy elasticity $E_{in}(\sigma_{in})$                             | 3.56  | 0.25 | 0.00      | 0.58      | 900   |
| Sectoral-level strategic $\alpha_{in}^s$                            | -0.53 | 0.11 | 0.48      | 0.19      | 533   |
| National-level strategic $\alpha_{in}^a$                            | -0.29 | 0.15 | 0.41      | 0.20      | 499   |
| <b>Panel C: Positive Real Output Values Switched to Negative</b>    |       |      |           |           |       |
| Policy elasticity $E_{in}(\sigma_{in})$                             | 3.69  | 0.23 | 0.00      | 0.63      | 1,017 |
| Sectoral-level strategic $\alpha_{in}^s$                            | -0.47 | 0.10 | 0.45      | 0.19      | 645   |
| National-level strategic $\alpha_{in}^a$                            | -0.27 | 0.13 | 0.41      | 0.20      | 609   |
| <b>Panel D: Truncated Values Replaced by -25% instead of -21%</b>   |       |      |           |           |       |
| Policy elasticity $E_{in}(\sigma_{in})$                             | 2.81  | 0.27 | 0.11      | 0.52      | 1,017 |
| Sectoral-level strategic $\alpha_{in}^s$                            | -0.49 | 0.10 | 0.45      | 0.20      | 645   |
| National-level strategic $\alpha_{in}^a$                            | -0.26 | 0.15 | 0.42      | 0.20      | 609   |
| <b>Panel E: Zero-Denominator Replaced with -0.3</b>                 |       |      |           |           |       |
| Policy elasticity $E_{in}(\sigma_{in})$                             | 2.68  | 0.25 | 0.11      | 0.52      | 1,017 |
| Sectoral-level strategic $\alpha_{in}^s$                            | -1.25 | 0.25 | 0.33      | 0.13      | 1,017 |
| National-level strategic $\alpha_{in}^a$                            | -0.75 | 0.32 | 0.30      | 0.17      | 1,017 |
| <b>Panel F: Missing <math>\alpha</math> Replaced by 0</b>           |       |      |           |           |       |
| Policy elasticity $E_{in}(\sigma_{in})$                             | 2.68  | 0.25 | 0.11      | 0.52      | 1,017 |
| Sectoral-level strategic $\alpha_{in}^s$                            | -0.30 | 0.06 | 0.29      | 0.13      | 953   |
| National-level strategic $\alpha_{in}^a$                            | -0.16 | 0.08 | 0.28      | 0.15      | 941   |
| <b>Panel G: Excluding Speeders (Fastest 10% by Survey Duration)</b> |       |      |           |           |       |
| Policy elasticity $E_{in}(\sigma_{in})$                             | 2.77  | 0.27 | 0.10      | 0.53      | 913   |
| Sectoral-level strategic $\alpha_{in}^s$                            | -0.52 | 0.11 | 0.46      | 0.19      | 587   |
| National-level strategic $\alpha_{in}^a$                            | -0.29 | 0.14 | 0.41      | 0.21      | 553   |

Notes: Panel A reports the main results. Only strategic parameters with non-missing values are reported, and no adjustments to managers' responses are done. Panels B to G present results for different samples and adjustments. All means are weighted using the same weighting approach.